



Humanities

Year 7

Cambridge Lower Secondary
Student Manual





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Student Book · Prime Books

ABOUT THIS BOOK

One story, told in order. Part one builds the stage: the Universe, the Sun and its planets, the restless Earth, the five oceans and the seven continents. Part two brings on the players: the first people, Egypt, China and the Inca. Nine units, fifty-two numbered topics, and a short Portugal at this stage panel in every unit, so you always know what was happening on one Atlantic coast while the rest of the world got on with it.

Prime Books is the student book series of Prime School.

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Written in British English. Distances are metric, money is in euros and times are given on the 24 hour clock. Dates before the common era are marked BC, and dates after it are given as plain numbers.

This book was written for the pupils of Prime School by the Prime School Humanities Department. Every fact, date and figure in it was checked against a named source before it was printed, and those sources are listed at the back, where you can check them too.

EDITORIAL BOARD

The Prime Books Subject Series has been developed under the guidance of the Pedagogical Academic Group for each subject and coordinated by the Pedagogical Team of Prime School.

This publication reflects our shared commitment to academic excellence, educational quality, professional integrity, and the continuous advancement of teaching and learning. Through the collective expertise and dedication of our educators, we aim to provide meaningful and engaging learning resources that support students' academic growth.

We extend our sincere appreciation to the Pedagogical Department and the Content Creation Team of Prime School for their invaluable contributions, dedication, and collaborative efforts in making this publication possible.

◆ HOW TO CHECK THIS BOOK

Everything in these pages can be checked, and the book is arranged so that you can check it.

✓ Every **figure, map, graph and table** has a number, a title and a caption

telling you what to look for. A figure numbered 7.2 is the second figure of unit 7.

✓ Every **date and number** was verified against a source before it was printed, and every source is named at the back.

✓ Every **square code** leads to a checked page in English from a museum, a space agency, a national survey or a university. No code appears twice, and the full web address is printed beside it, so the page still works when nobody has a device.

✓ Every **closed question** has an answer at the back. Open questions have

acceptance criteria instead: what a good answer has to contain, rather than the exact words it has to use.

If you find something in this book that is wrong, tell your teacher. That is how the next edition gets better, and it is also the habit this subject is trying to teach you.



Introduction

This book tells one story, and it starts a very long way back.

Most humanities books begin with people. This one begins about thirteen thousand eight hundred million years earlier, because people are late arrivals and it helps to know what they arrived into.

So part one builds the stage. You will start with the whole Universe, come in past our own galaxy to the Sun, then to the planets, then to this one, then inside it, then out across its oceans and continents. By the end of unit 5 you will be standing on a particular piece of land, knowing how it got there, what the weather is like, and who else lives on it.

Part two brings on the players. Our species walks out of Africa, survives an ice age, learns to farm, and then, in four separate corners of the world, builds something remarkable: Egypt on its river, China between two rivers, and the Inca along four thousand kilometres of mountain.

Why the whole world, and not just part of it

Because it is true. Civilisation is not something that happened in one place and spread. It started separately, several times, on different continents, in people who had never heard of each other and who solved the same problems in different ways. A pupil who learns only one of those stories has been given a smaller world than the real one.

You are reading this in a school in Portugal, and many of you were not born here. So Portugal appears in this book the way any country should appear in a world history: as one place among many, on the same line of time as everywhere else. Look for the gold-topped **Portugal at this stage** panels. They are short on purpose. They are there so that when you read about the pyramids going up, you also know what was happening on the Atlantic coast in the same century.

What this book expects of you

Three things, and none of them is memory.

First, **look before you read**. Nearly every page has a picture, a map or a diagram with a caption that tells you what to notice. Look at it first. Reading the words afterwards is much easier.

Second, **ask how we know**. This subject is not a list of facts to be believed. It is a set of conclusions drawn from evidence, and the evidence is always in the room. When this book tells you the Universe had a beginning, it also tells you how anybody could possibly know that.

Third, **be willing to be wrong**. Every unit has a **Careful** panel naming the mistake almost everybody makes. Making it is fine. Making it after you have read the panel is a waste of a good panel.



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Every unit and every numbered topic in this book, with the page it really starts on.

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How to use this book

Every topic is built the same way, so once you have done one you know your way round all fifty-two.

A topic starts with a question, teaches you something, shows you the thinking, lets you practise, and then tells you something you will want to repeat at home. The panels always mean the same thing:

LOOK FIRST	One question about the real world, before any teaching. You are not expected to know the answer. You are expected to have a guess.
THE STORY	The teaching, in ordinary sentences. If a sentence does not make sense, read the figure beside it, then read the sentence again.
WATCH ME WORK IT OUT	One question answered slowly, with the thinking left in. Copy the thinking, not the answer.
LOOK AGAIN	The same evidence read a second way. Two readings of one piece of evidence is what historians and geographers actually do.
YOUR TURN	Numbered practice. Question 1 is always the easiest one. Start there.
ON THE MAP	Where on Earth this happened. Every history topic has one, because history happens somewhere.
PORTUGAL AT THIS STAGE	What was happening in one Atlantic country at the same moment as the rest of the topic. Short, and never the main event.
CAREFUL	The mistake almost everybody makes here, named before you make it.
AMAZING BUT TRUE	One checked fact worth telling somebody at home tonight.
WORDS TO KEEP	The four or five words the topic is really about. They are all in the glossary at the back.
GO FURTHER	For when you have finished and want more. Nothing here is compulsory.

The codes in the margins

Some pages carry a square code. Each one leads to a real page, in English, at a museum, a space agency, a national survey or a university, and each was checked on the day this book was printed. The web address is printed beside every code, so you can also type it, and so the page still works in a room where devices stay in bags.

Ask your teacher before you scan. At Year 7 the adult holds the device. Nothing behind a code is needed to answer any question in this book. It is there because some things are better watched than read.

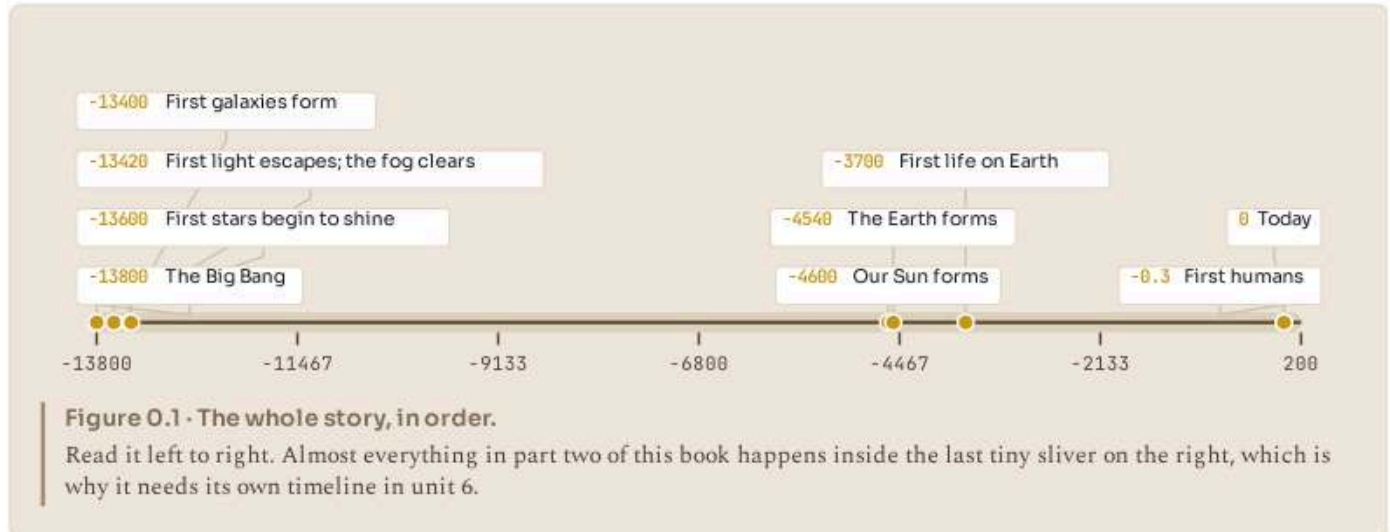
How to use the answers honestly

The answers are at the back, by unit and topic. They are there so that you can find out whether you were right, which is useful, and not so that you can find out what to write, which is not.

The honest order is: try the question, write something down even if you are not sure, then look. If your answer is different from the one at the back, the interesting question is not "who is right" but "where did we take different turnings". Ask that one out loud in class. It is usually the best question of the lesson.

The whole book on one line

Nine units, thirteen thousand eight hundred million years. Come back to this page whenever you lose your place.



Two things are worth noticing before you start.

The first is how late we are. If the whole history of the Universe were squeezed into a single year, with the Big Bang on 1 January, the Earth would form in early September, the first life would appear at the end of that month, the dinosaurs would die out on 30 December, and every single thing in part two of this book, all of it, from the first cave painting to the fall of the Inca, would happen in the last thirty seconds of 31 December.

The second is how much of the story is not about people at all. Units 1 to 5 have almost no humans in them. That is not a delay before the interesting part. The rock, the water, the air and the position of the continents decide where people can live, what they can grow, where they can sail and what they can dig up. Part two makes very little sense without part one.



PART ONE

The stage: our world, and how it was made

Before anybody could do anything, there had to be somewhere to do it. This part builds that somewhere, from the beginning of everything down to the piece of ground you are sitting on: the Universe, the Sun, this planet, its oceans and its seven continents. Part two then puts people on it.





| A galaxy, and one person looking at it. Every atom in that person was made inside a star.

PART ONE • UNIT 1

1 How the Universe began

Everything that exists, and the fourteen thousand million years it has been busy

■ IN THIS UNIT, YOU WILL

- ✓ Explain how we can know about something nobody was there to see.
- ✓ Describe the Big Bang, and say what "expanding" really means.
- ✓ Explain how gravity turned clouds of gas into galaxies.
- ✓ Find the Sun's place inside our own galaxy.
- ✓ Say what a black hole is, and what it is not.

Where we are, and how we know



Figure 1.1 • Five steps out from where you are sitting.

Read it left to right. Each box is far bigger than the one before it, and the last one holds everything there is.

You will start with a question that sounds impossible: how can anybody know what happened before there was anybody? By the end you will be able to answer it, put the Universe's whole history on one line, and point to the Sun's address inside our own galaxy.

THE BIG QUESTION THIS UNIT ANSWERS

How can we possibly know what happened before anybody was there to see it?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

1.1 A story with a beginning

1.2 The Big Bang

1.3 How galaxies were made

1.4 The Milky Way, our home

1.5 Black holes

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Go outside tonight and look up. Every point of light you can see is a sun, and almost all of them are much bigger than ours. Now try to answer three questions without looking anything up. How far away is the nearest one? How long has it been there? And what was there before it?

Write your three guesses down. Keep them. At the end of this unit, come back and see how close you were, and, more interestingly, work out why you guessed what you did.

TOPIC 1.1

A story with a beginning

How can anybody know?

Here is a fair objection, and you should make it. Nobody was there. There were no people, no planets, no atoms. So how can a book tell you what happened thirteen thousand eight hundred million years ago and expect to be believed?

The answer is the same one a detective gives. You do not need a witness if you have enough evidence, and the evidence has to fit together. Nobody watched the crime, but the fingerprints, the footprints and the timing all point one way, and no other story explains all three at once.

Astronomers have three big pieces of evidence, and the useful thing about them is that they were found separately, by different people, looking for different things. Any one of them on its own would be interesting. All three together are hard to argue with.

	What we can see	What it tells us
Galaxies are moving apart	Almost every galaxy is moving away from us, and the far ones move faster.	Run the film backwards and everything was once in one place.
A faint glow, everywhere	A very cold radio glow arrives from every direction in the sky.	It is the light of the young Universe, stretched out and cooled.
What the Universe is made of	Three quarters hydrogen, one quarter helium, almost nothing else.	Exactly the recipe a hot beginning would leave behind.

Figure 1.2 · Three clues, one conclusion.
Read the left column first, then the right. Notice that no single clue proves anything on its own; it is the fact that all three agree that does the work.

Light is a time machine

There is one more thing that makes all this possible, and it is strange enough to be worth a paragraph of its own.

Light is fast, but it is not infinitely fast. It travels about 300 000 kilometres every second. That is quick enough that in a classroom you notice nothing at all. But space is very big, so light takes real time to cross it. Sunlight takes eight minutes and twenty seconds to reach you. The Sun you can see is always the Sun as it was eight minutes ago.

Push that further and it becomes a superpower. A galaxy a thousand million light years away is showing you what it looked like a thousand million years ago, because that is how long its light has been travelling. Astronomers do not have to imagine the young Universe. They can point a telescope at it, because looking far away is looking back in time.

◆ CAREFUL

"A light year" sounds like a length of time. It is not. It is a **distance**: how far light travels in one year, about nine and a half million million kilometres. Saying "it happened four light years ago" is like saying "it happened four kilometres ago".

■ YOUR TURN

- 1 Give one reason why "nobody was there" is not a good reason to reject a scientific explanation.
- 2 The Sun is eight light minutes away. If the Sun stopped shining right now, how long would it take before anyone on Earth noticed? Explain your answer.
- 3 Look at Figure 1.1. Choose the clue you find most convincing, and say why.
- 4 A friend says: "Scientists are only guessing about the Big Bang." Write two sentences replying, using the word *evidence*.

TOPIC 1.2

The Big Bang

What actually happened

About 13 800 million years ago, everything that now exists was packed into a space smaller than a full stop, and unimaginably hot. Then it began to expand, and it has been expanding ever since. That expansion is what we call the Big Bang.

In the first fraction of a second the Universe grew enormously. Within about three minutes it had cooled just enough for the first simple centres of atoms to form: hydrogen, and a quarter as much helium. And then, for a very long time, nothing much happened at all. The Universe was a hot fog, and light could not travel through it any more than a torch beam travels through thick milk.

About 380 000 years in, it cooled enough for that fog to clear. Light escaped in every direction, and it is still travelling. That escaping light is the faint glow astronomers can still detect today from every part of the sky. It is the oldest thing anybody has ever seen.

Not an explosion

Nothing flew outwards into empty space. Space itself got bigger.

Think of a loaf

Currants in rising dough all move apart. No currant is at the centre.

So there is no centre

Every galaxy sees the others moving away. Ours is not special.

And no edge we can find

We can only see as far as light has had time to travel.

Figure 1.3 · Four things the Big Bang was not.

Read all four boxes before you decide what you think the Big Bang was. The loaf in box two is the one to remember.

The bit everyone gets wrong

The Big Bang was not an explosion in space. It was an expansion of space.

That sounds like word games, but it matters, and there is a good way to picture it. Imagine a loaf of raisin bread rising in an oven. Every raisin moves away from every other raisin, not because the raisins are flying about, but because the dough between them is getting bigger. Sit on any raisin you like and you will see all the others moving away from you, with the far ones moving fastest.

That is exactly what astronomers see when they look at galaxies. It is also why there is no centre. No raisin is the middle of the loaf. There is no point in space you could travel to and say, "here is where it started", because it started everywhere at once.



LOOK FURTHER

NASA, *The Universe*. Pictures and short explanations from the people who build the telescopes. Start with the age of the Universe and check it against page one of this unit.

science.nasa.gov/universe/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● AMAZING BUT TRUE

The oldest light in the Universe is arriving at your head right now. A small fraction of the fuzz on an old untuned television was that light, arriving from 380 000 years after the beginning of everything, and being mistaken for a broken aerial.

■ YOUR TURN

- 1 Put these in order, earliest first: the fog clears; the Universe is smaller than a full stop; the first hydrogen forms.
- 2 Explain, in your own words, why the raisin bread is a better picture of the Big Bang than a firework.
- 3 Why is there no centre of the Universe? Answer in one sentence.
- 4 The Universe is about 13 800 million years old. Write that number in figures.

■ GO FURTHER

The word "Big Bang" was invented by an astronomer who did **not** believe the theory. He used it on the radio in 1949 to make the idea sound silly. The name stuck, and the theory turned out to be right. Find out who he was, and what he believed instead.

TOPIC 1.3

How galaxies were made

Gravity, given enough time

After the fog cleared, the Universe was almost, but not quite, smooth. Some regions had very slightly more gas in them than others. Very slightly. A few parts in a hundred thousand.

That was enough. Gravity is a rule with no exceptions: every piece of matter pulls on every other piece. So the slightly denser patches pulled a little harder, gathered a little more gas, and became denser still, which made them pull harder again. Given a hundred million years, small differences become enormous ones.

Where the gas piled up tightly enough, it grew so hot in the middle that it started to shine. That is a star. Where a great many stars formed near each other, held together by their combined gravity, you get a galaxy.

What a galaxy is

A galaxy is not a thing you could ever visit the edge of. It is a vast collection of stars, gas and dust, spinning slowly, held together by gravity. The smallest hold a few million stars. The largest hold thousands of millions of millions.

There are roughly two thousand million galaxies in the part of the Universe we can see, and each one has, on average, something like a hundred thousand million stars. Do not try to feel the size of that. Nobody can. Just notice that when this book says "our galaxy" in the next topic, it means one out of two thousand million.

◆ CAREFUL

Stars are not scattered evenly through space, and neither are galaxies. Gravity clumps things. Stars gather into galaxies, galaxies gather into groups, and groups gather into clusters. The Universe has a structure, a bit like a sponge: long strands of galaxies with huge empty spaces in between.

■ YOUR TURN

- 1 What was needed to turn a smooth cloud of gas into galaxies? Name the one force.
- 2 Explain why a slightly denser patch of gas gets denser still over time.
- 3 What is the difference between a star and a galaxy?
- 4 Roughly how many galaxies are there in the observable Universe? Give the figure this book uses.

TOPIC 1.4

The Milky Way, our home

Finding our own address

Our galaxy is called the Milky Way. It is a spiral: a flat disc with curved arms, a bright bulge in the middle, and something like two hundred thousand million stars.

It is about 100 000 light years across. Light, the fastest thing there is, takes a hundred thousand years to cross it.

Our Sun is not at the centre. It sits about halfway out, in one of the spiral arms, going round the middle once every 225 million years or so. The last time the Sun was in this part of its orbit, dinosaurs had not yet appeared.

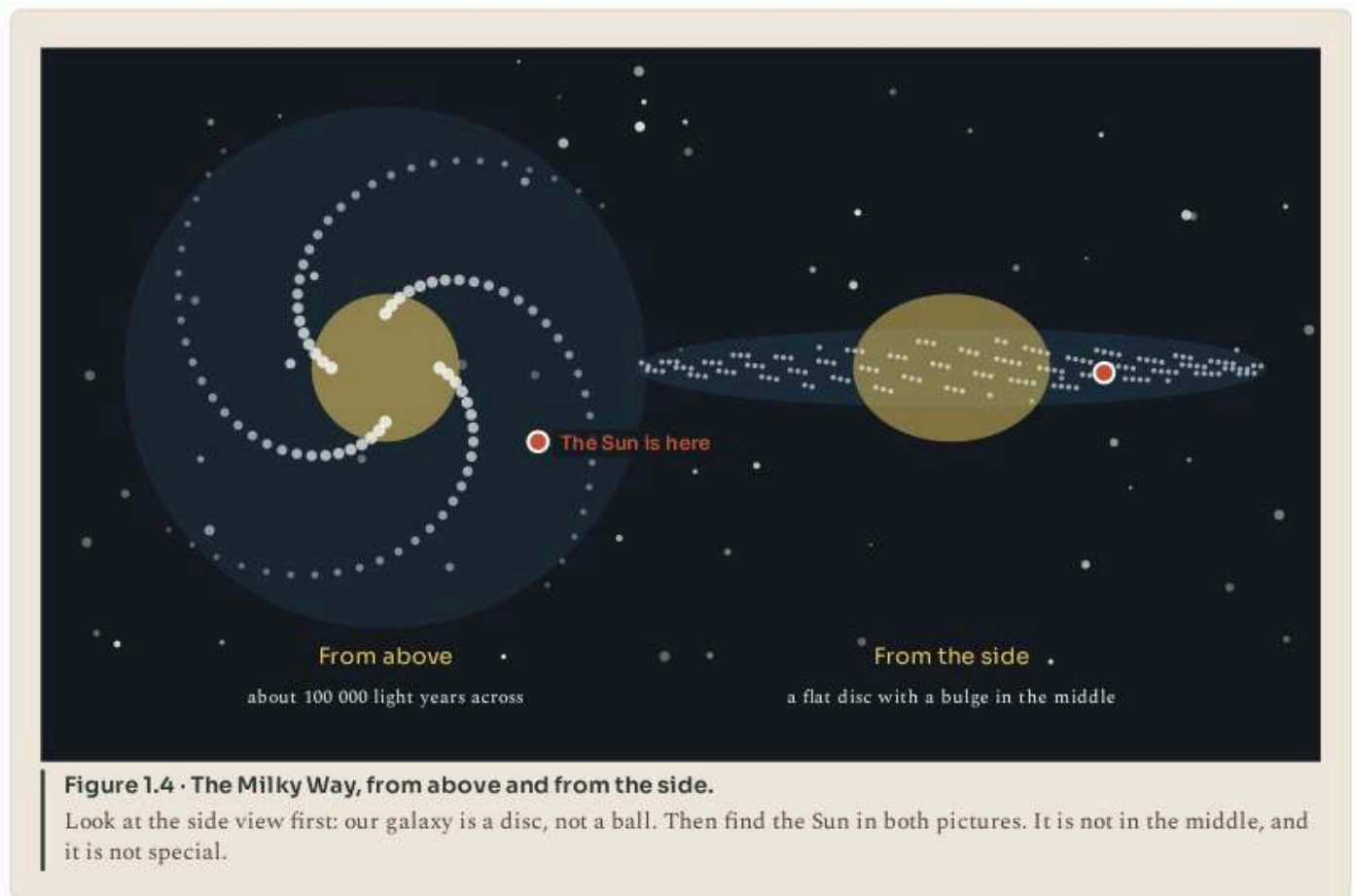


Figure 1.4 · The Milky Way, from above and from the side.

Look at the side view first: our galaxy is a disc, not a ball. Then find the Sun in both pictures. It is not in the middle, and it is not special.

Why it looks like spilt milk

On a genuinely dark night, far from streetlights, you can see a pale band across the sky. That band is our own galaxy, seen edge on, from the inside. We are in the disc, so when we look along the disc we see enormous numbers of distant stars blurred into a faint stripe. When we look up out of the disc, we see far fewer.

Most people alive today have never seen it. Not because it has gone, but because outdoor lighting has made the night sky too bright in nearly every city on Earth.

■ PORTUGAL AT THIS STAGE

Portugal is one of the easier places in Europe to see this. In 2011 the Alqueva region in the Alentejo, an area of very low population around a large reservoir, became the world's first certified Starlight Tourism Destination. Lighting there is deliberately kept low so that the sky stays dark. On a clear night the Milky Way is plainly visible across the whole sky, exactly as it was to every person who lived before electric light. It is worth knowing that this took a decision. Dark sky is now something a country has to choose to keep.



LOOK UP TONIGHT

Dark Sky Alqueva Reserve, Portugal. The first place in the world certified as a starlight tourism destination. Find out how far you have to go to see the Milky Way.

darkskyalqueva.com/en/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- ① What shape is the Milky Way? Describe it in one sentence, using the word *disc*.
- ② Where in the galaxy is the Sun? Say roughly how far out, and whether it is in the centre.
- ③ Explain why the Milky Way looks like a band across the sky rather than a spiral.
- ④ Give one reason why fewer people see the Milky Way today than a hundred years ago.

TOPIC 1.5

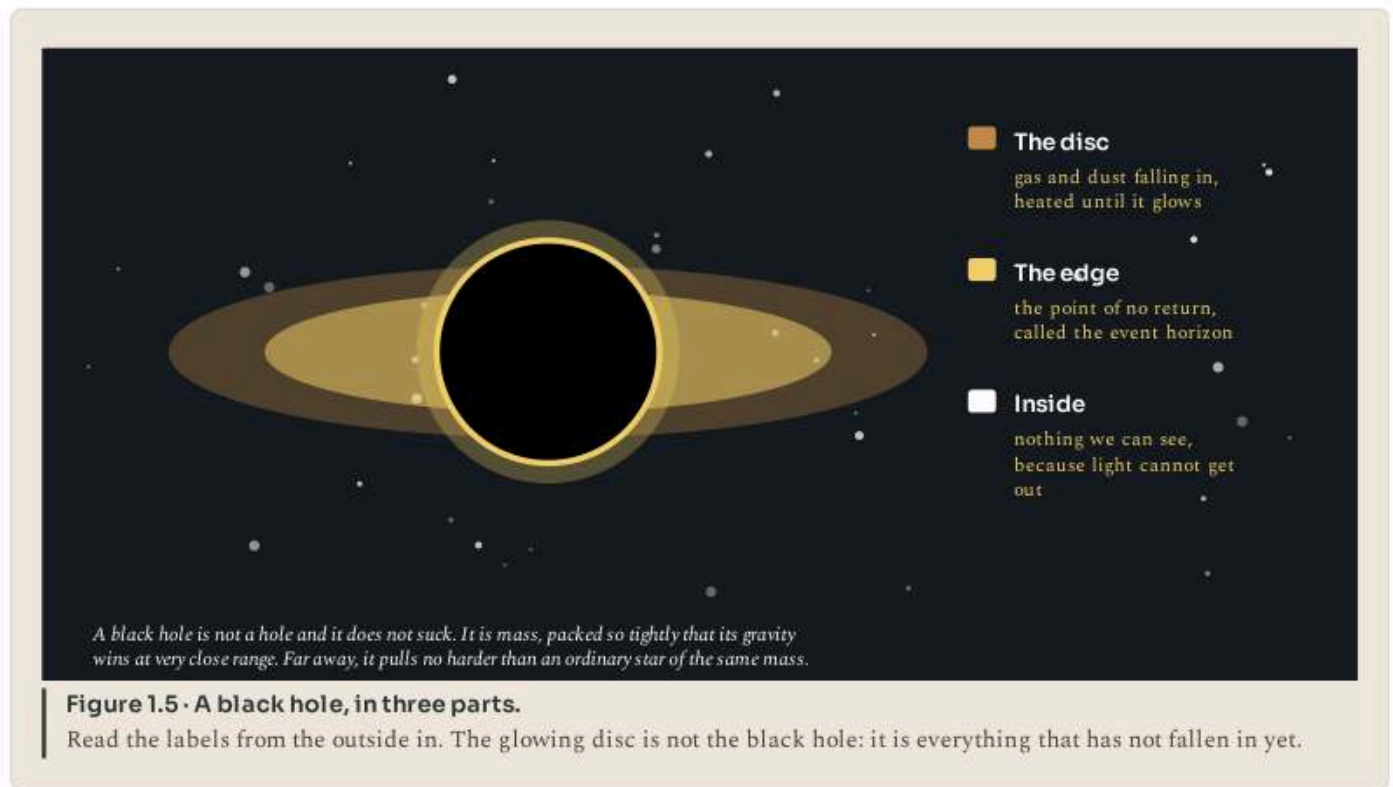
Black holes

What one actually is

A black hole is not a hole. Nothing has been removed. A black hole is the opposite: it is matter, packed so tightly into so small a space that its gravity, close up, becomes overwhelming.

Gravity depends on two things: how much mass there is, and how close you are to it. Squash the same mass into a smaller ball and you can get much closer to its middle, so the pull you feel at the surface becomes far stronger. Squash it enough and you reach a point where nothing that comes too near can get out again. Not a spaceship, not a signal, not even light.

The boundary around it is called the event horizon. It is not a surface you could touch. It is simply the line beyond which leaving is no longer possible.



Where they come from, and what they do not do

Most black holes are what is left when a very large star runs out of fuel. The star can no longer hold itself up, so its core collapses inwards, and the outer layers are blown off in an enormous explosion called a supernova.

There is also a much bigger kind. Almost every large galaxy, including ours, has a supermassive black hole at its centre. The one at the centre of the Milky Way has about four million times the mass of the Sun. It is 26 000 light years away, and it is not going anywhere.

◆ CAREFUL

Black holes do not suck. This is the single most common mistake about them. Gravity does not reach further just because something is dense. If the Sun were replaced today by a black hole of exactly the same mass, the Earth would carry on in exactly the same orbit. It would get very dark and very cold, but we would not be pulled in.

● AMAZING BUT TRUE

In 2019 astronomers published the first photograph of a black hole's shadow, using eight radio telescopes on four continents working together as if they were a single telescope the size of the Earth. The object is 55 million light years away. The photograph took two years to assemble from the data.



LOOK FURTHER

NASA, Black holes. What a black hole really is, from the mission that photographed one. Look for the sentence that says a black hole does not suck.

science.nasa.gov/universe/black-holes/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- 1 Explain why a black hole is not really a hole.
- 2 What is the event horizon?
- 3 A friend says a black hole would swallow the Solar System if it replaced the Sun. Correct them, and explain why they are wrong.
- 4 Where do most black holes come from?
- 5 How far away is the black hole at the centre of our galaxy?

◆ WATCH ME WORK IT OUT

Question. The nearest star after the Sun is Proxima Centauri, 4.2 light years away. A spacecraft travelling at 60 000 km/h would take about 76 000 years to get there. Explain what "4.2 light years" tells you, and why the journey takes so much longer than 4.2 years.

Thinking it through. First, name the quantity. A light year is a distance, not a time, so 4.2 light years means the distance light covers in 4.2 years.

Second, ask who is doing the travelling. Light does that distance in 4.2 years because light is extremely fast: 300 000 km every second.

Third, compare the speeds. Our spacecraft manages 60 000 km per **hour**. Light does 300 000 km per **second**, which is 1 080 000 000 km per hour. So light is about eighteen thousand times faster than the spacecraft.

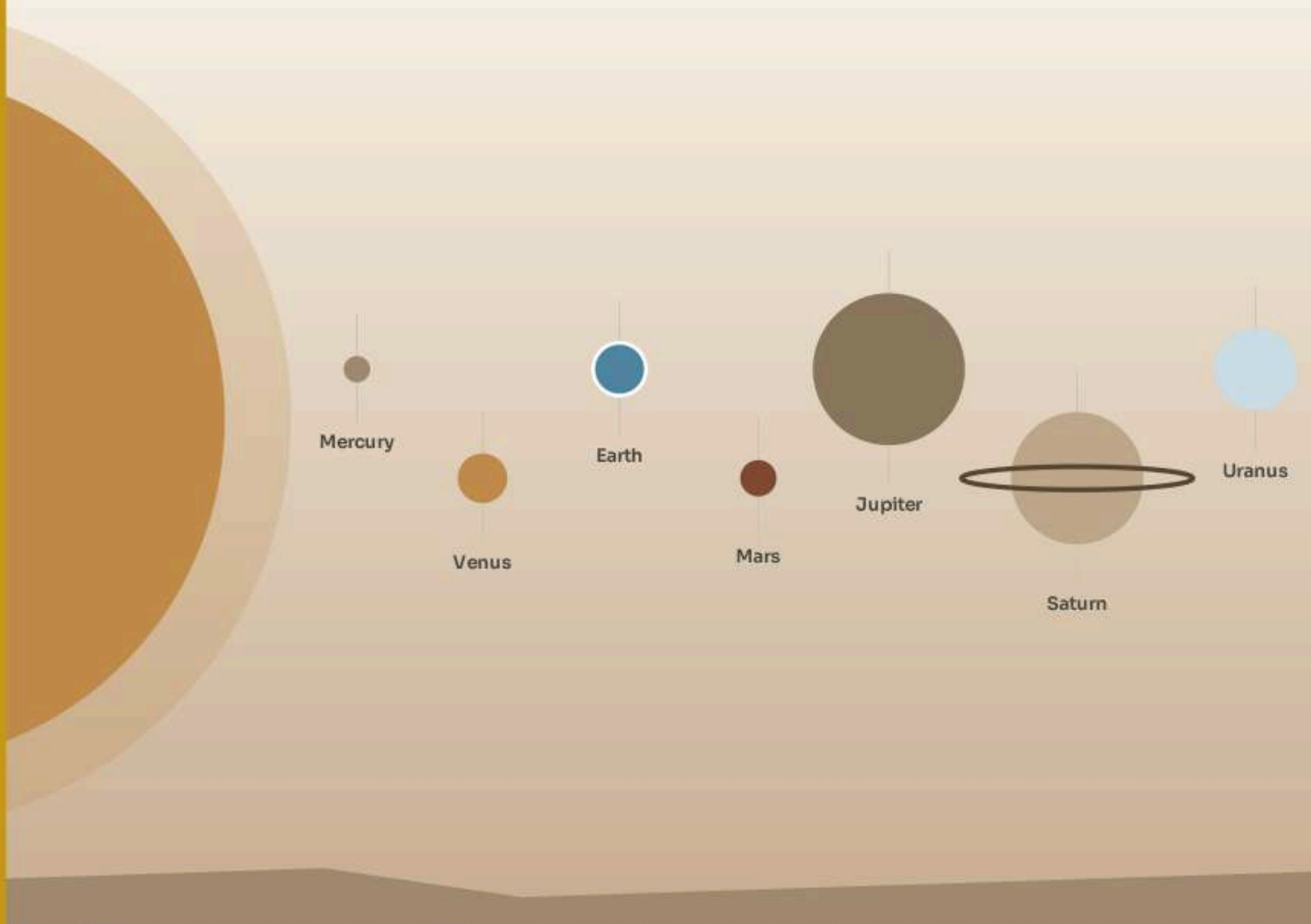
Answer. 4.2 light years is the distance light travels in 4.2 years. The spacecraft is roughly eighteen thousand times slower than light, so it needs roughly eighteen thousand times as long: about 76 000 years. The distance has not changed. Only the speed has.

■ WHAT YOU CAN NOW DO

- ✓ Explain how we can know about events nobody witnessed.
- ✓ Describe the Big Bang, and say what "expanding" really means.
- ✓ Explain how gravity turned smooth gas into galaxies.
- ✓ Describe the Milky Way and find the Sun's place in it.
- ✓ Say what a black hole is, and correct two things people wrongly believe.

— WORDS TO KEEP

galaxy, gravity, light year, event horizon, supernova, expansion, evidence



The Sun and the eight planets, in order. The gaps between them are far bigger than this page can show.

PART ONE • UNIT 2

2 The Sun and its family

One ordinary star, eight planets, and the only place we know of where anything is alive

■ IN THIS UNIT, YOU WILL

- ✓ Explain why one ordinary star matters more to you than all the rest.
- ✓ Compare the four rocky planets.
- ✓ Describe the outer Solar System in order.
- ✓ List what living things need, and say why Earth has it.
- ✓ Explain the Moon's phases and the tides.

UNIT 2 • WHAT YOU WILL LEARN

One star, eight planets, one home

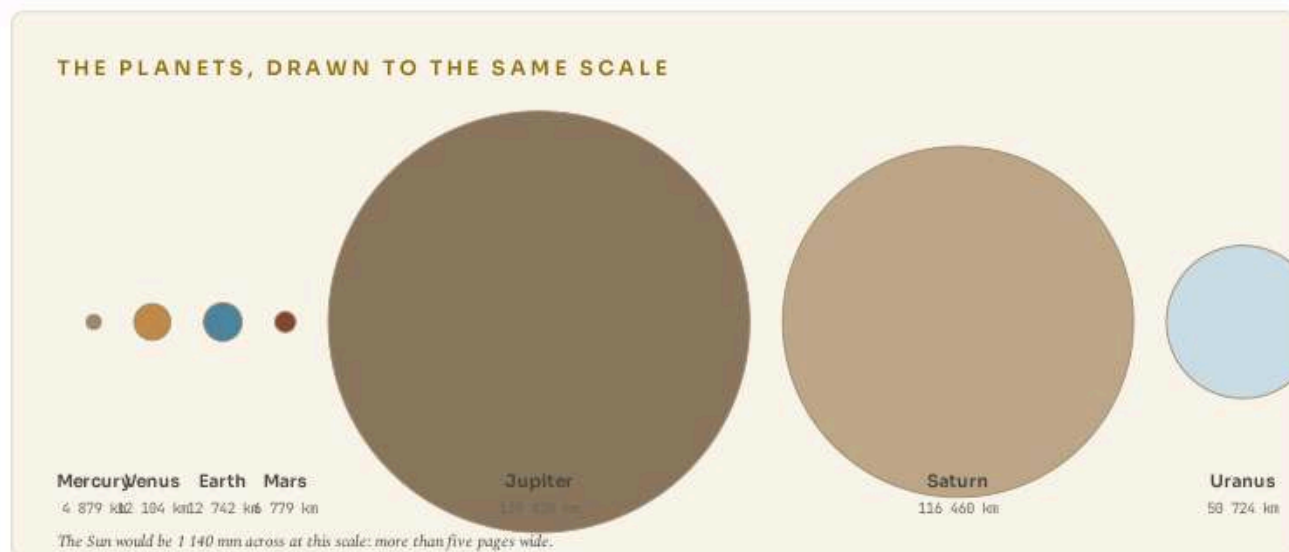


Figure 2.1 • The eight planets, all drawn to the same scale.

Look how small the rocky four are beside the giants, and find Earth without reading the labels.

You will meet the Sun as what it really is, an ordinary star that happens to be ours, then travel outwards past eight planets in order. The unit ends on the two objects that matter most to you: the planet you live on and the Moon that pulls its seas about.

THE BIG QUESTION THIS UNIT ANSWERS

Out of everything in the Solar System, why is this the only place we know of with life on it?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

2.1 The Sun, our own star

2.2 The rocky planets

2.3 The giants, and the edge of the system

2.4 Earth, the planet that got everything right

2.5 The Moon

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

You have almost certainly been told the order of the planets. Try writing it down now, from memory, without help, and put a tick beside any you could describe in one sentence.

Most people can name six or seven and describe two. That is not a failure of memory. It is because a list is a hard thing to remember and a **story** is an easy one. This unit gives you the story: one star, four small rocky worlds close in, four giants further out, and one of the rocky ones that got extraordinarily lucky.

TOPIC 2.1

The Sun, our own star

An ordinary star that happens to be ours

The Sun is a star. It looks different from the others only because it is close: about 150 million kilometres away, while the next nearest star is 270 000 times further off.

It is a ball of hot gas, mostly hydrogen, 1 392 000 kilometres across. You could fit about 1.3 million Earths inside it. It contains more than 99 per cent of all the material in the Solar System; everything else, all eight planets, every moon, every asteroid, is made from the leftovers.

In its core, under enormous pressure, hydrogen is being squeezed together into helium. That releases energy, and that energy is why the Sun shines. It has been doing this for about 4 600 million years and has enough fuel for roughly another 5 000 million.

Why it matters more than anything else in the sky

Almost every source of energy on Earth is the Sun, one step removed.

Sunlight lands on a leaf, and the plant uses it to build itself. An animal eats the plant. You eat the animal, or the plant. Your dinner is sunlight that has changed hands two or three times. Coal, oil and gas are the same thing much older: ancient plants, made of ancient sunlight, buried and squeezed.

The Sun also drives the weather. It heats the tropics more than the poles, which makes air move, which is wind. It evaporates water from the sea, which becomes cloud, which becomes rain. Take the Sun away and within days the Earth would be dark, still and frozen.

◆ CAREFUL

The Sun is not on fire. Fire is burning, which needs oxygen, and there is no oxygen in space. The Sun is doing something quite different: squeezing small atoms together into bigger ones and releasing energy. That is called fusion, and it is why a star can last thousands of millions of years while a fire lasts an evening.

● AMAZING BUT TRUE

Light leaving the Sun's surface reaches you in 8 minutes 20 seconds. But the energy in it took far longer to get out of the Sun in the first place: bouncing around inside the dense interior, it can take tens of thousands of years to reach the surface. The sunlight warming your hand today began its journey before the last ice age ended.

■ YOUR TURN

- 1 Why does the Sun look bigger and brighter than any other star?
- 2 Give two things on Earth that would stop if the Sun stopped shining, and say why for each.
- 3 Explain the difference between burning and fusion.
- 4 About what fraction of the Solar System's material is in the Sun?

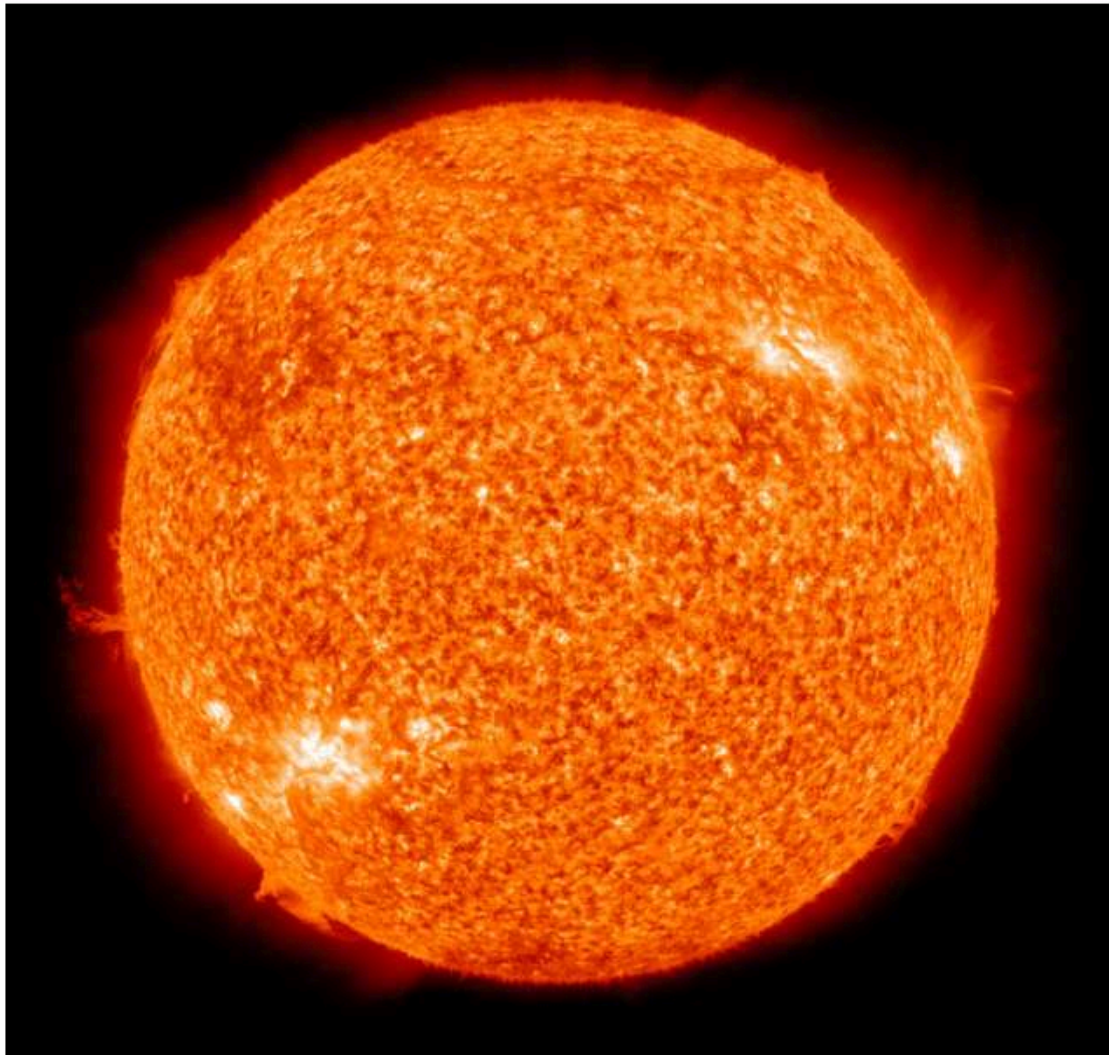


Figure 2.4 - The Sun, photographed by NASA's Solar Dynamics Observatory.

TOPIC 2.2

The rocky planets

The inner four

The four planets closest to the Sun, Mercury, Venus, Earth and Mars, are made of rock and metal. They are small, they have few moons or none, and they are warm because they are close in.

Mercury is the smallest and the fastest, going round the Sun in 88 days. It has almost no atmosphere, so its heat escapes at night: it swings from about 430 degrees C in the day to about minus 180 at night, the biggest temperature range of any planet.

Venus is nearly Earth's twin in size, and nothing like it in every other way. A thick blanket of carbon dioxide traps heat so effectively that its surface sits at about 465 degrees C, day and night, everywhere. It is hotter than Mercury, despite being further from the Sun, because of that blanket.

Earth you already live on. Topic 2.4 is about why it is different.

Mars is about half Earth's size, cold, dry and rusty. Its red colour is iron oxide, which is rust. It has the tallest volcano in the Solar System, Olympus Mons, about 22 kilometres high, and dry valleys that look exactly as though water once ran through them, because it did.

	Venus	Mars
Size	Almost exactly Earth's size: 12 104 km across.	About half Earth's size: 6 779 km across.
Air	A thick blanket of carbon dioxide, ninety times heavier than ours.	A thin wisp of carbon dioxide, less than one hundredth of ours.
Surface	About 465 degrees C, day and night. Hot enough to melt lead.	About minus 63 degrees C on average. Dry, dusty and rusty.
Water	None. It was probably boiled away long ago.	Ice at the poles, and dry river valleys everywhere.

Figure 2.2 • Two planets that could not be more different.
Read across each row. Venus and Mars are both next door to us, and neither is anywhere you could stand.

The lesson Venus teaches

Venus is worth more than a paragraph, because of what it shows.

Venus is roughly the same size as Earth, made of roughly the same stuff, and formed at roughly the same time from the same cloud of material. It is not enormously closer to the Sun. And yet it is a furnace where lead would melt and where a metal spacecraft survived for two hours before being crushed and cooked.

The difference is its atmosphere. Carbon dioxide lets sunlight in and makes it harder for heat to get back out. Venus has an enormous amount of it, so the heat builds up and stays. This is the same effect that keeps Earth comfortable rather than frozen; the difference is one of quantity.

■ YOUR TURN

- 1 Name the four rocky planets in order from the Sun.
- 2 Why is Mercury so hot in the day and so cold at night?
- 3 Venus is further from the Sun than Mercury but hotter. Explain why.
- 4 Give two pieces of evidence that water once flowed on Mars.
- 5 Look at Figure 2.1. Which row do you think is the most important difference between Venus and Mars? Justify your choice.

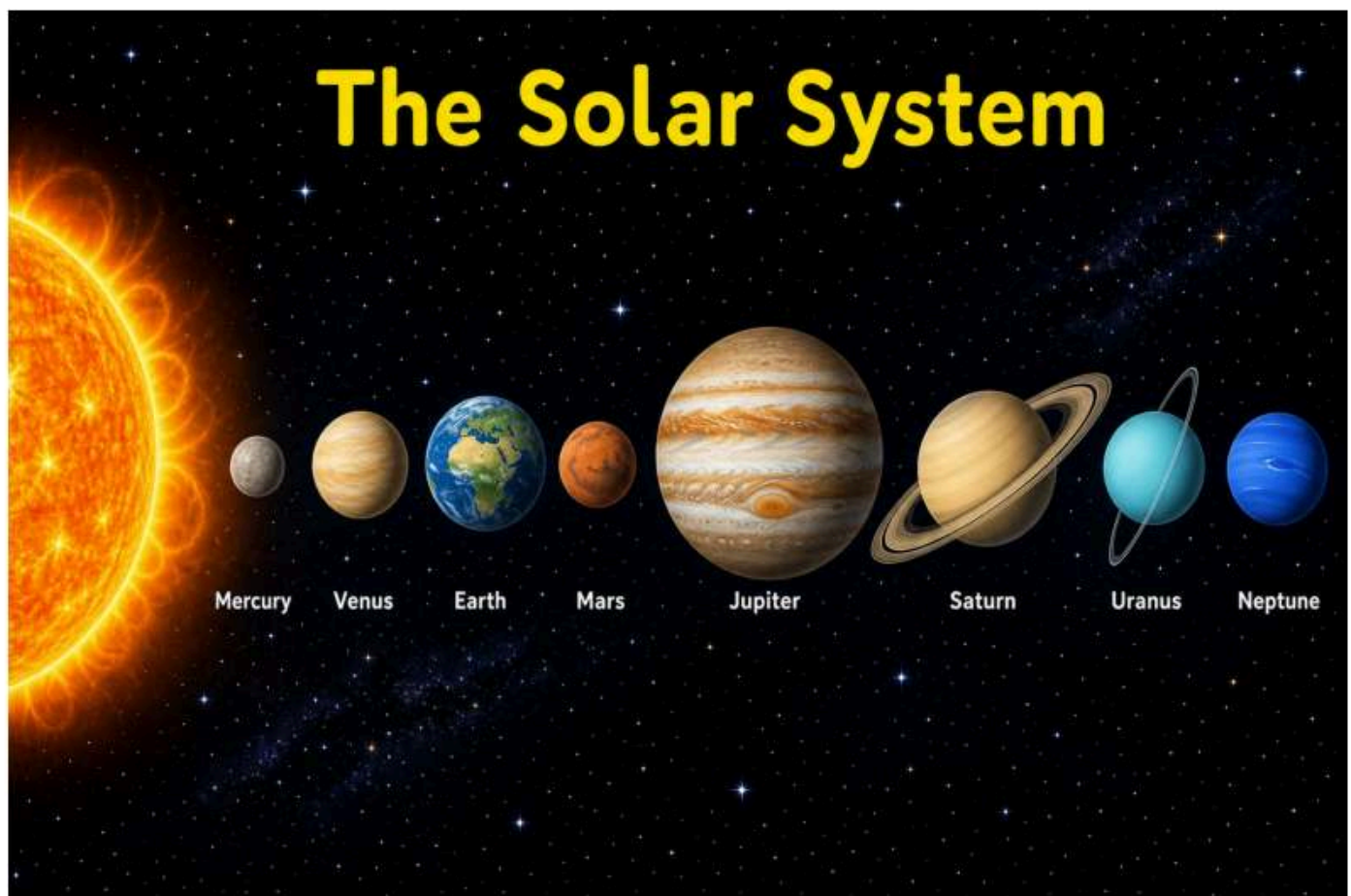


Figure 2.7 • The Solar System, from the Sun outwards.

TOPIC 2.3

The giants, and the edge of the system

Beyond the asteroid belt

Between Mars and Jupiter there is a belt of rock and metal fragments, the asteroid belt. There are perhaps a million objects there bigger than a kilometre across, but they are spread over such a huge volume that a spacecraft can pass straight through without coming near one.

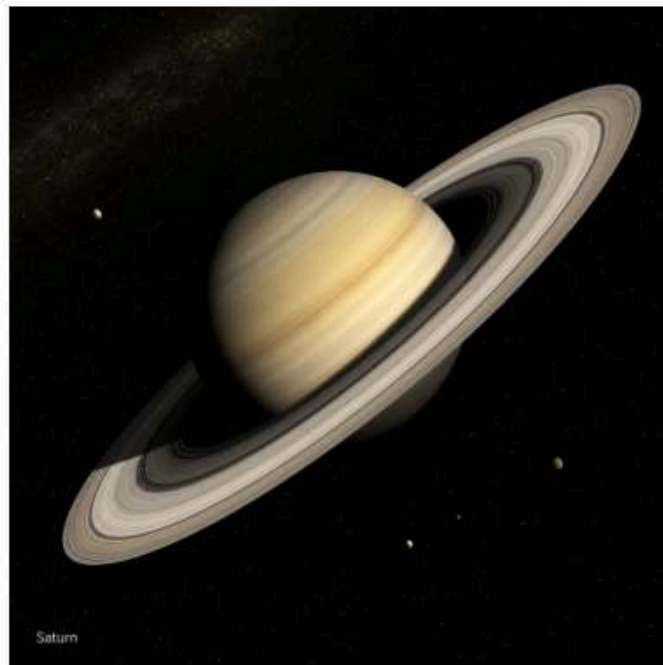
Past the belt, everything changes. The outer four planets are giants: enormous, cold, and made mostly of gas and ice rather than rock. None of them has a surface you could land on.

Jupiter is the largest planet by a long way, 139 820 kilometres across, more than eleven Earths side by side. Its Great Red Spot is a storm that has been running for at least 190 years and is wider than the Earth. It has 95 known moons, one of which, Europa, has a liquid water ocean under an ice crust.

Saturn is only slightly smaller and famously ringed. The rings are not solid: they are billions of chunks of ice and rock, most of them smaller than a house, all orbiting together. They are about 20 metres thick in places, which is remarkably thin for something 280 000 kilometres wide.

Uranus is tipped over on its side, so it rolls round the Sun rather than spinning upright. Each of its poles gets 42 years of continuous daylight followed by 42 years of night.

Neptune is the furthest out and the windiest place known, with winds measured at over 2 000 kilometres per hour. It takes 165 Earth years to go round the Sun once. It was discovered in 1846 by mathematics: astronomers noticed Uranus was not moving quite as predicted, worked out where an unseen planet would have to be, pointed a telescope there, and found it.



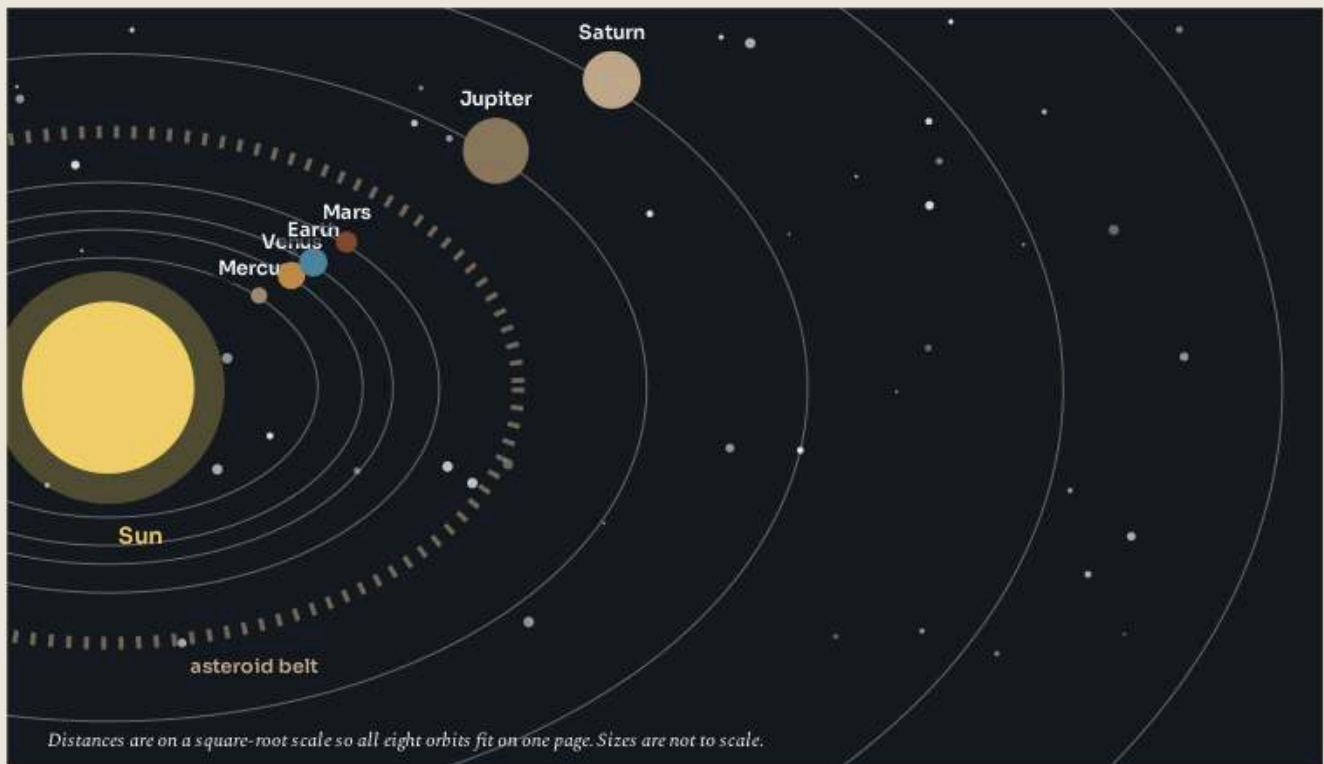


Figure 2.3 · The Solar System from above.

Follow the orbits outwards from the Sun. Notice how much empty space there is past Mars: the four inner planets are crowded together, and the outer four are not.

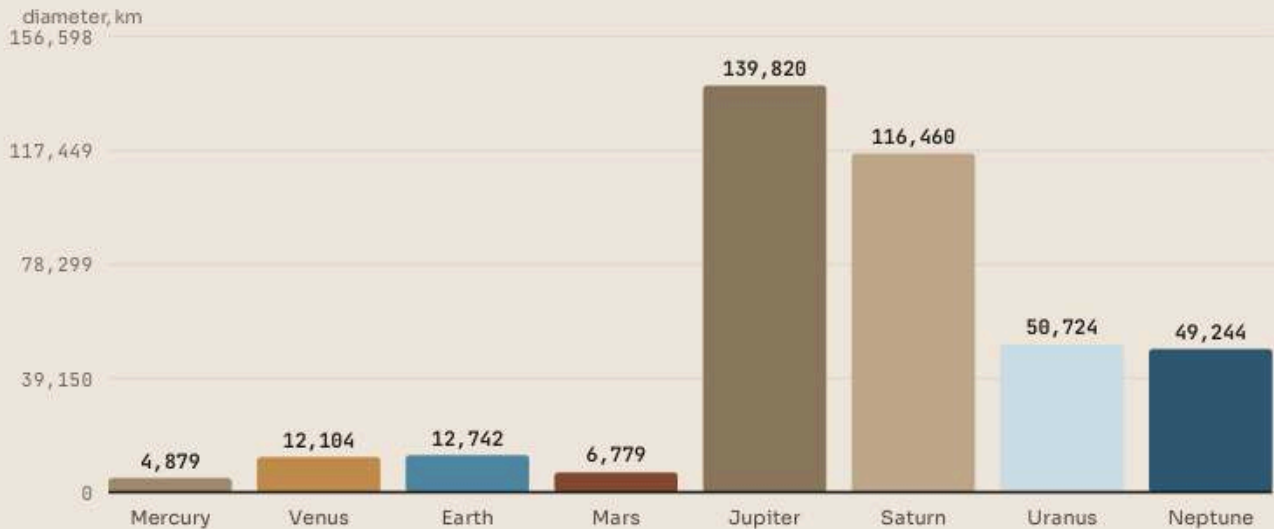


Figure 2.4 · Every planet, drawn to the same scale.

Find Earth. Then look at Jupiter. Then remember that the Sun would be five pages wide at this scale.

And what is not a planet

Pluto was called a planet from 1930 until 2006. Then astronomers found several similar objects out beyond Neptune, one of which is heavier than Pluto, and had to decide: add them all as planets, or tighten the definition.

They tightened it. A planet has to orbit the Sun, be round, and have cleared other objects out of its orbital path. Pluto passes the first two and fails the third: it shares its region with a great many similar icy bodies. So it is now a dwarf planet, and it has plenty of company.

◆ CAREFUL

Pluto was not demoted for doing anything wrong, and it did not change. The **definition** changed, because new evidence arrived. That is not a scientific failure. It is exactly what should happen when you find out more.

■ PORTUGAL AT THIS STAGE

Portugal is a member of the European Space Agency, and Portuguese astronomers work at the European Southern Observatory, whose telescopes sit in the Atacama Desert in Chile because that is the driest, clearest sky on Earth. Portuguese researchers have been part of teams that found planets around other stars, including some of the first Earth-sized ones. It is a reminder that modern science is done by international teams rather than by countries, and that a small country can be part of very large discoveries.



LOOK FURTHER

NASA, *The Solar System*. Every planet, with real photographs and real numbers. Compare its figures with the bar chart in topic 2.2.

science.nasa.gov/solar-system/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

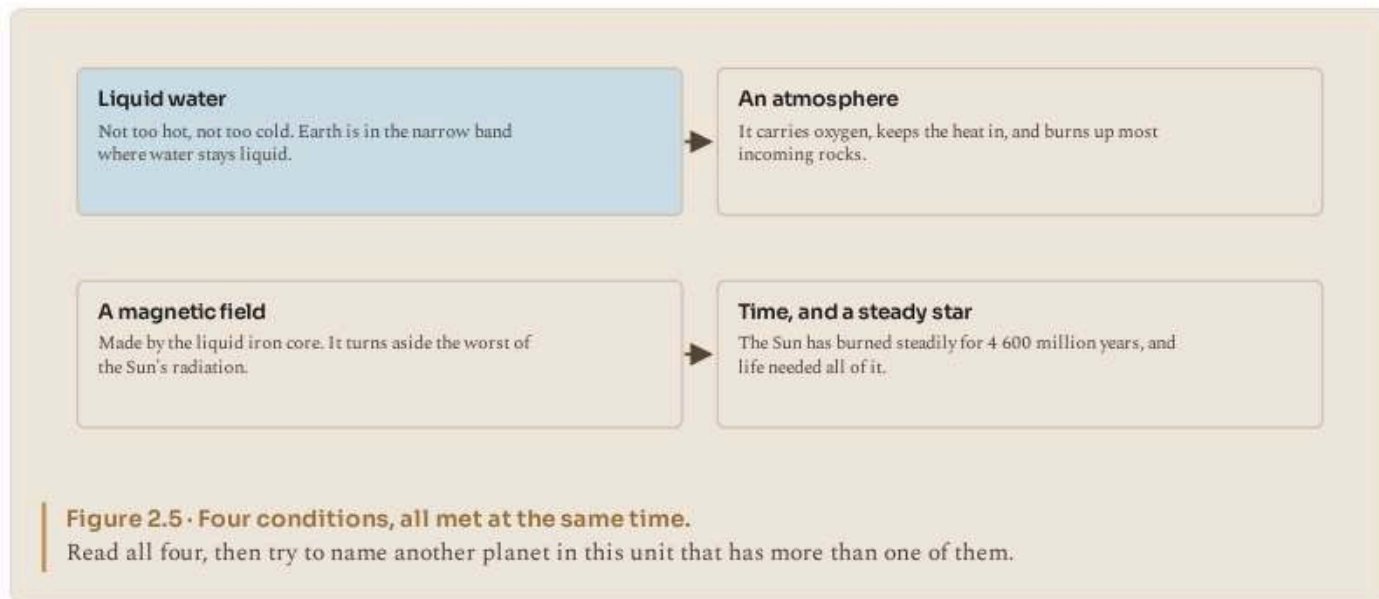
- 1 Name the four giant planets in order from the Sun.
- 2 Give one striking fact about each of Jupiter, Saturn, Uranus and Neptune.
- 3 Neptune was found using mathematics before anyone saw it. Explain how.
- 4 State the three conditions an object must meet to be called a planet.
- 5 Explain why Pluto no longer meets them.

TOPIC 2.4

Earth, the planet that got everything right

What life needs

Life, as far as we know, needs four things, and Earth has all four at once. Take any one away and the story stops.



The first is **liquid water**. Not ice, not steam. Chemistry that can build living things needs a liquid to happen in, and water is very good at it. That requires a narrow temperature range, and Earth's distance from the Sun puts it right inside that range.

The second is an **atmosphere**. Ours is 78 per cent nitrogen and 21 per cent oxygen, which is unusual: oxygen is reactive and would disappear if living things were not constantly producing it. The atmosphere also holds heat in, carries water around the planet, and burns up most of the rocks that come our way.

The third is a **magnetic field**, produced by the swirling liquid iron in the outer core. It deflects the stream of particles the Sun throws out. Mars almost certainly had one once, lost it, and lost most of its atmosphere afterwards.

The fourth is **time**, and a steady star. Life on Earth took perhaps 800 million years to appear at all, and another 3 000 million to become anything more complicated than single cells. That needed a star that neither flared nor faded for thousands of millions of years.

The rare part is not any one of them

Individually, none of these conditions is astonishing. Water is common in the Universe. Atmospheres are common. Magnetic fields are common.

What is unusual is having all four in one place, at the same time, for long enough. Every year astronomers find more planets around other stars, and by now they have confirmed several thousand. Some are roughly Earth-sized and at roughly the right distance from their stars. Whether any of them has the whole set, nobody yet knows.

● AMAZING BUT TRUE

Earth is not a perfect sphere. It bulges at the equator because it spins, so it is about 43 kilometres wider across the equator than from pole to pole. If you stand at sea level on the equator you are further from the centre of the Earth than you would be standing on the summit of Everest.

■ YOUR TURN

- 1 List the four conditions life needs, as this book gives them.
- 2 Why does liquid water need a narrow temperature range?
- 3 Explain the link between Mars losing its magnetic field and Mars losing its atmosphere.
- 4 Oxygen makes up 21 per cent of our air. Why is that surprising, and what keeps it there?



Figure 2.9 • Earth from space: one planet, all four conditions.

TOPIC 2.5

The Moon

One quarter of the way to being a double planet

The Moon is 3 475 kilometres across, about a quarter of Earth's width, and 384 400 kilometres away. Compared with other moons in the Solar System it is enormous relative to its planet.

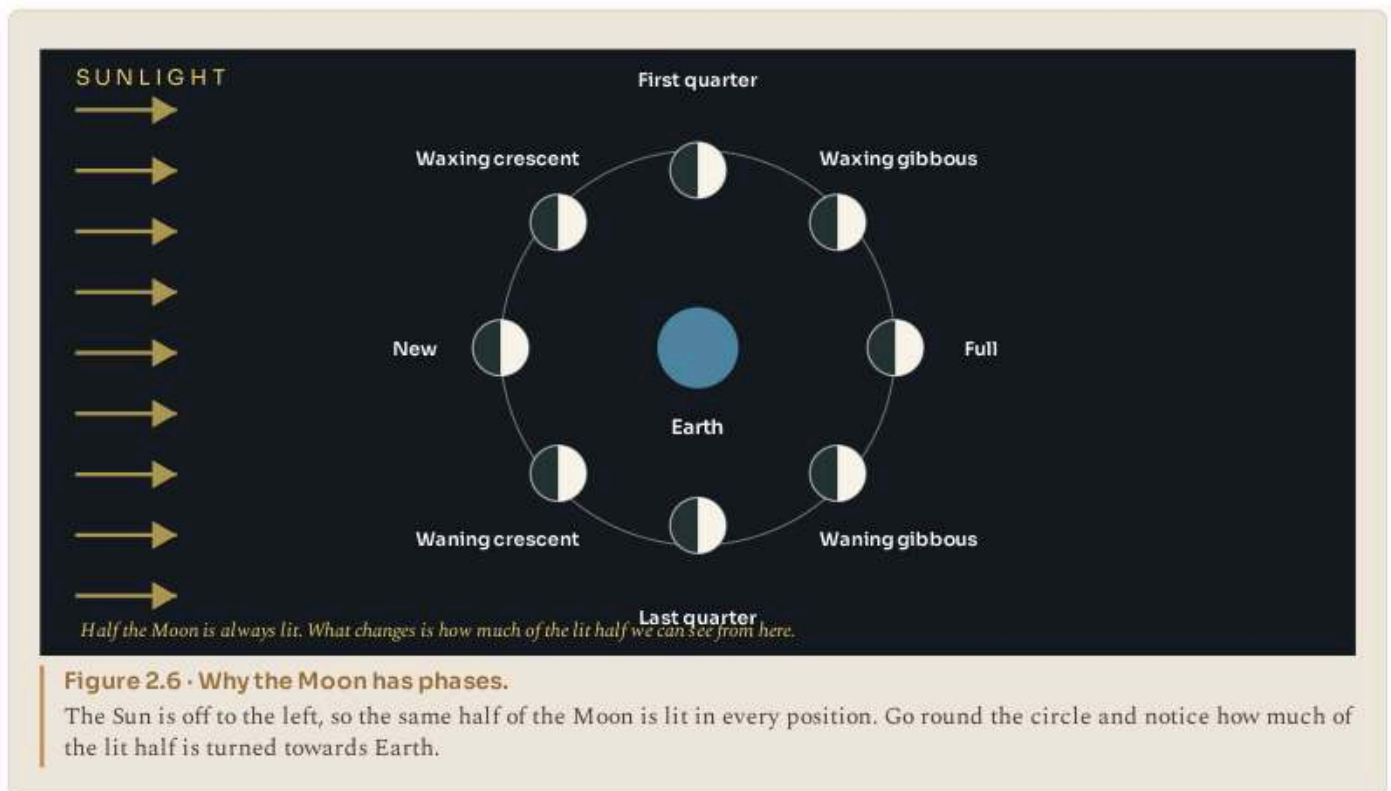
The best explanation for where it came from is dramatic. About 4 500 million years ago, when the Earth was young, something roughly the size of Mars hit it. The collision threw an enormous amount of molten material into orbit, and that material gathered together and cooled into the Moon.

Two pieces of evidence support this. The rocks brought back by the Apollo missions are chemically very like Earth's outer layers and unlike most other bodies. And the Moon has a very small iron core, which is what you would expect if it formed from the outer layers of a planet rather than from scratch.

Phases: the same lit half, seen from different angles

Half of the Moon is always lit by the Sun. Always. What changes over a month is how much of that lit half we can see from where we are standing.

When the Moon is between us and the Sun, the lit side faces away and we see almost nothing: a new moon. When the Earth is between the Sun and the Moon, the lit side faces us fully: a full moon. In between we see slices.



Tides, and why they matter

The Moon's gravity pulls on the Earth, and pulls hardest on the water nearest to it, because gravity weakens with distance. That makes the sea bulge slightly on the side facing the Moon, and, for the same reason in reverse, on the opposite side too. As the Earth turns, coasts pass through those bulges, which is why most places get two high tides a day.

The Sun pulls too, less strongly. When the Sun and Moon line up, the two pulls add together and tides are unusually large. When they are at right angles, the tides are unusually small.

◆ CAREFUL

The Moon does not have a permanent dark side. It has a **far side**, which we never see from Earth because the Moon turns once for each orbit. The far side gets exactly as much sunlight as the near side; we simply never see it lit, because we never see it at all.



LOOK FURTHER

NASA, Earth's Moon. Phases, tides and the far side. Check today's phase against the diagram in topic 2.5.

science.nasa.gov/moon/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

◆ WATCH ME WORK IT OUT

Question. Lisbon has two high tides on most days, roughly 12 hours 25 minutes apart, not 12 hours exactly. Why the extra 25 minutes?

Thinking it through. Start with what causes a high tide: the coast turning into a tidal bulge, and the bulge sits under the Moon.

Now ask what would happen if the Moon stood still. The Earth turns once in 24 hours, so a coast would pass under the bulge, and under the opposite bulge, every 12 hours exactly.

But the Moon does not stand still. It goes round the Earth once in about 27 days, moving in the same direction as the Earth's spin. So by the time a coast has turned all the way round, the Moon has moved on a little, and the coast has to turn slightly further to catch it up.

Answer. The extra 25 minutes is the time the Earth needs to turn a little further to catch up with a Moon that has moved on in its orbit. It is why high tide is roughly 50 minutes later each day.

■ YOUR TURN

- 1 Where did the Moon come from? Give the explanation and one piece of evidence for it.
- 2 How much of the Moon is lit at any moment? Explain what changes to give us phases.
- 3 Why do most coasts get two high tides a day rather than one?
- 4 Correct this sentence: "The dark side of the Moon never gets any sunlight."
- 5 Predict what the tide will be like when the Sun, Earth and Moon are in a straight line, and explain your prediction.

■ WHAT YOU CAN NOW DO

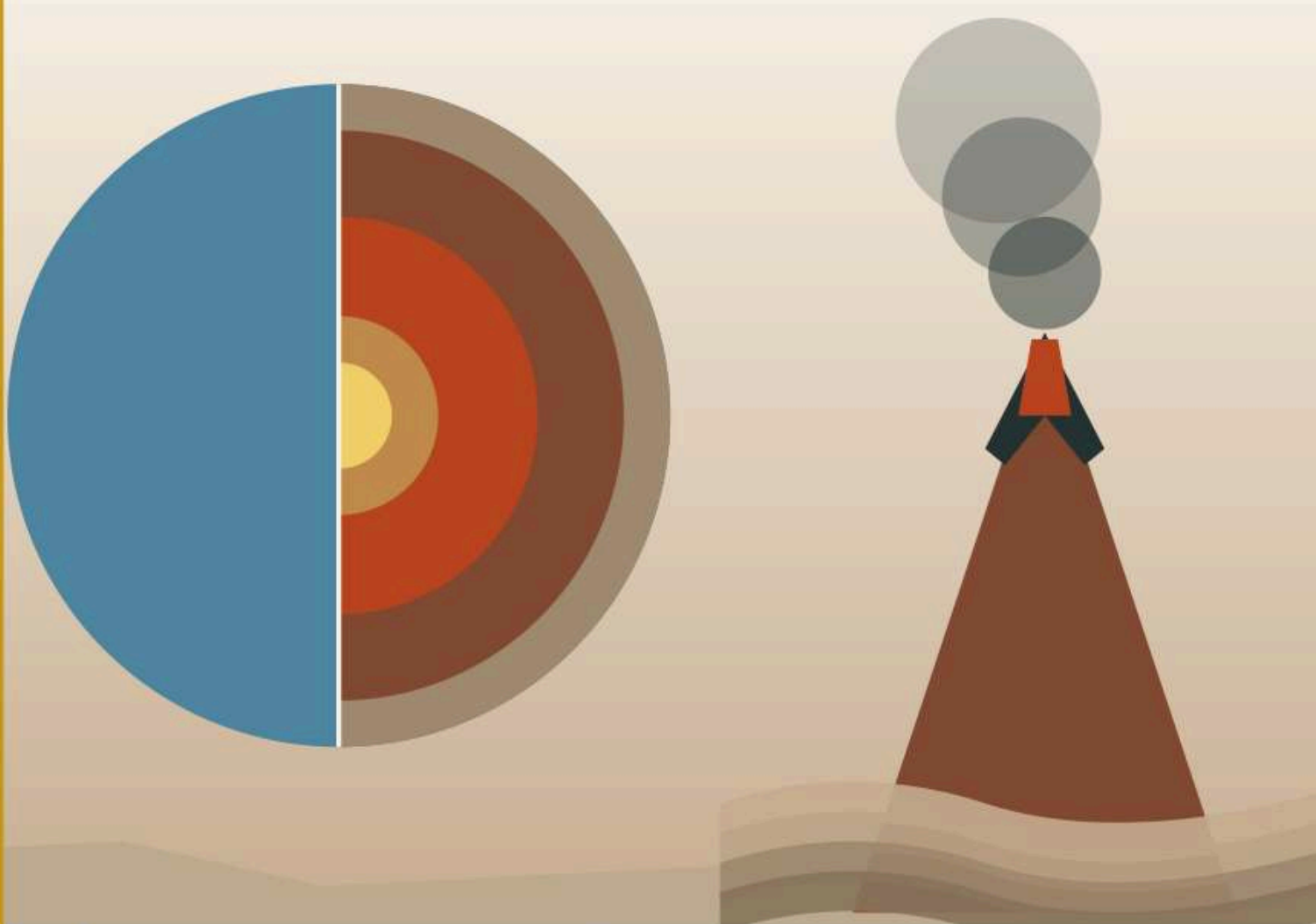
- ✓ Explain what the Sun is, and why almost all our energy comes from it.
- ✓ Compare the four rocky planets, and explain why Venus is hotter than Mercury.
- ✓ Describe the outer Solar System in order, and say why Pluto is not a planet.
- ✓ List the four conditions life needs and explain why Earth has them all.
- ✓ Explain the Moon's phases and the tides, and correct the dark side myth.

■ WORDS TO KEEP

fusion, atmosphere, asteroid belt, dwarf planet, magnetic field, phase, tide



Figure 2.10 · The Moon.



| Inside the Earth, and what happens where the plates meet. The volcano is the mantle finding a way out.

PART ONE • UNIT 3

3 The restless Earth

The ground is moving under your feet. It always has been

■ IN THIS UNIT, YOU WILL

- ✓ Name the Earth's layers, and say how we know they are there.
- ✓ Put the supercontinents in order, and explain what moved them.
- ✓ Match each kind of plate boundary to what it builds.
- ✓ Explain earthquakes and volcanoes, and why people live near them.
- ✓ Explain why today's map is one frame of a very slow film.

UNIT 3 • WHAT YOU WILL LEARN

A planet that will not sit still



Figure 3.1 • Inside the Earth.

Four layers, and only the thin top one is solid rock you could stand on. Nobody has ever drilled past the crust.

You will go inside the Earth, watch the continents move across hundreds of millions of years, and find out why earthquakes and volcanoes happen in lines rather than anywhere. Then you will look at today's world map and see it for what it is: one frame of a very slow film.

THE BIG QUESTION THIS UNIT ANSWERS

If the ground is moving, why does it feel so still?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

- | | |
|--------------------------------------|---|
| 3.1 Inside the Earth | 3.2 Rodinia, Pangaea and the wandering continents |
| 3.3 Plates that push, pull and slide | 3.4 When the Earth moves |
| 3.5 The continents today | |

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Stand still. You feel nothing moving.

In fact you are spinning with the Earth at up to 1 670 kilometres per hour, travelling round the Sun at about 107 000 kilometres per hour, and standing on a piece of the Earth's surface that is sliding sideways at roughly the speed your fingernails grow.

Only the last of those matters for this unit, and it is the one nobody notices. Why can you feel none of it? Write a guess before you read on.

TOPIC 3.1

Inside the Earth

Four layers, and only the top one is solid rock you could stand on

The Earth is 12 742 kilometres across. If you could cut it open you would find four layers, and they are not at all alike.

The **crust** is the thin skin on the outside: between about 5 kilometres thick under the oceans and 70 kilometres thick under big mountain ranges. Everything you have ever touched, every building, every field, every mine, is in the crust. Compared with the whole Earth it is thinner, proportionally, than the skin of an apple.

The **mantle** is by far the biggest layer, nearly 2 900 kilometres deep. It is rock, and it is hot, and here is the part that matters: over very long periods it flows. Not like water. More like extremely stiff treacle, or a glacier. Centimetres a year. But it flows, and that flow is what moves the continents.

The **outer core** is liquid iron and nickel, about 2 200 kilometres thick. Because it is liquid metal and it swirls, it generates the Earth's magnetic field, which is what makes a compass point north and what shields us from much of the Sun's radiation.

The **inner core** is a ball of iron about 2 400 kilometres across, at roughly 5 200 degrees C, which is about the temperature of the Sun's surface. It is solid, despite the heat, because the pressure on it is so enormous.



The deepest hole ever drilled reached 12 km. Everything below that we know from earthquake waves.

Figure 3.2 • A quarter cut out of the Earth.

Start at the outside and work in. Look at how thin the crust is compared with everything under it.

How we know, without ever going there

The deepest hole ever drilled is the Kola Superdeep Borehole in Russia, which reached 12 262 metres in 1989. That is just over 12 kilometres, and it did not get through the crust. Nobody has ever seen the mantle.

So how does anybody know what is down there?

Earthquakes. When the ground breaks, it sends waves through the whole planet, and those waves are recorded by instruments all over the world. Waves travel at different speeds through different materials, bend when they pass from one material into another, and some kinds cannot travel through liquid at all.

By comparing when each wave arrives at hundreds of stations, scientists can work backwards to what it must have passed through. The discovery that the outer core is liquid came from noticing that one type of wave arrives nowhere in a large "shadow" region on the far side of the Earth: it had been stopped by something liquid.

● AMAZING BUT TRUE

The Kola borehole took nineteen years to drill 12 kilometres. Scaled down, if the Earth were an apple, that hole would not have got through the skin. We know more about the far side of the Moon than about the rock 30 kilometres beneath your feet.

■ YOUR TURN

- 1 Name the Earth's four layers in order from the outside in.
- 2 Which layer is thickest, and roughly how thick is it?
- 3 What makes the Earth's magnetic field, and why does it matter?
- 4 Explain how earthquake waves let us map the inside of the Earth.
- 5 The inner core is hotter than the outer core but solid. Explain why.



Lava meeting the sea, Iceland

Figure 3.7 • Lava meeting the sea, Iceland.

TOPIC 3.2

Rodinia, Pangaea and the wandering continents

The map has not always looked like this

Look at a world map and find the eastern coast of South America and the western coast of Africa. They fit. Not roughly: closely, like two pieces of a jigsaw.

People noticed that as soon as decent maps existed, and for centuries it was treated as a coincidence. In 1912 a German scientist named Alfred Wegener said it was not a coincidence, and collected evidence: identical fossils of the same land animals on both coasts, matching rock formations, and glacier scratches in places that are now tropical.

He was laughed at, largely because he could not say what moved the continents. He was right anyway. The mechanism, the slow flow of the mantle, was worked out about fifty years later, long after he had died.

Four maps, hundreds of millions of years apart

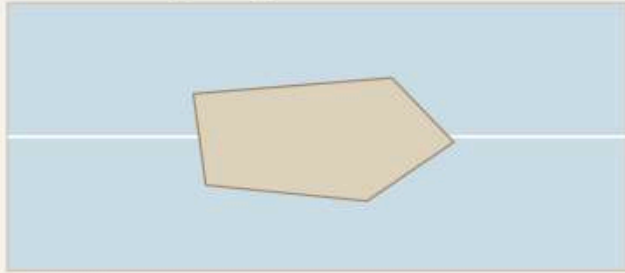
The continents have joined and split several times. Two of those arrangements have names worth knowing.

Rodinia existed about 1 100 million years ago: one great landmass, before there was any life on land at all.

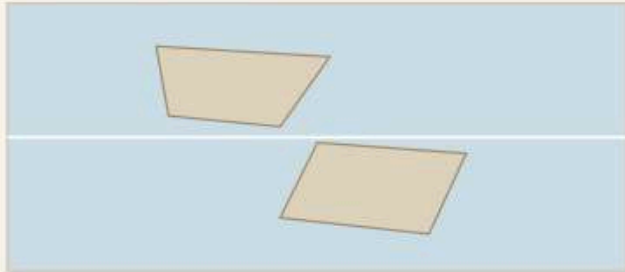
Pangaea formed about 335 million years ago. All of today's continents were joined into a single mass, surrounded by a single ocean. You could have walked from what is now Portugal to what is now Florida without getting your feet wet.

Pangaea began to break up about 175 million years ago, and the pieces have been moving ever since. They have not stopped. The Atlantic is getting wider by about 2.5 centimetres a year, and Australia moves north by about 7 centimetres.

1 100 million years ago

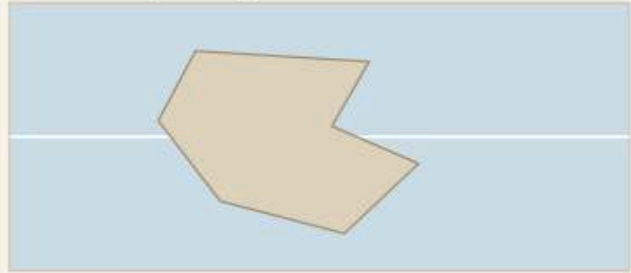
**Rodinia**

one landmass, near the equator
175 million years ago

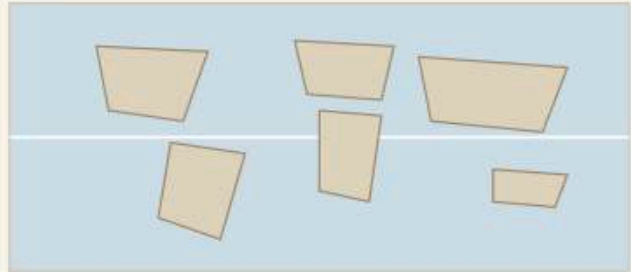
**Breaking up**

The line across each box is the equator. Sketches, not survey maps; they show where the land was.

335 million years ago

**Pangaea**

one landmass again, all of it joined
Today

**Seven continents**

and still moving, about as fast as a nail.

Figure 3.3 • Four snapshots of the same planet.

Read them in the order given, top left to bottom right. The line across each box is the equator, so you can see the land moving north and south as well as sideways.

■ PORTUGAL AT THIS STAGE

Portugal's own coastline is a piece of this story. When Pangaea broke apart and the Atlantic began to open, the crack ran between what became Iberia and what became North America, and the western edge of Portugal is one side of that tear. That is why the rocks of the Algarve and the rocks of Newfoundland have so much in common.

Iberia has also turned. It rotated slowly anticlockwise as the Bay of Biscay opened, and the collision that pushed it against the rest of Europe is what raised the Pyrenees. Portugal, in other words, has not always faced west. Its Atlantic coast is younger than the dinosaurs.

■ YOUR TURN

- 1 What did Wegener notice, and what evidence did he collect?
- 2 Why was his idea rejected at the time?
- 3 Put these in order, oldest first: Pangaea, today, Rodinia, Pangaea breaking up.
- 4 The Atlantic widens by about 2.5 cm a year. How much wider will it be in 1 000 years? Show your working.

Pangaea, about 250 million years ago

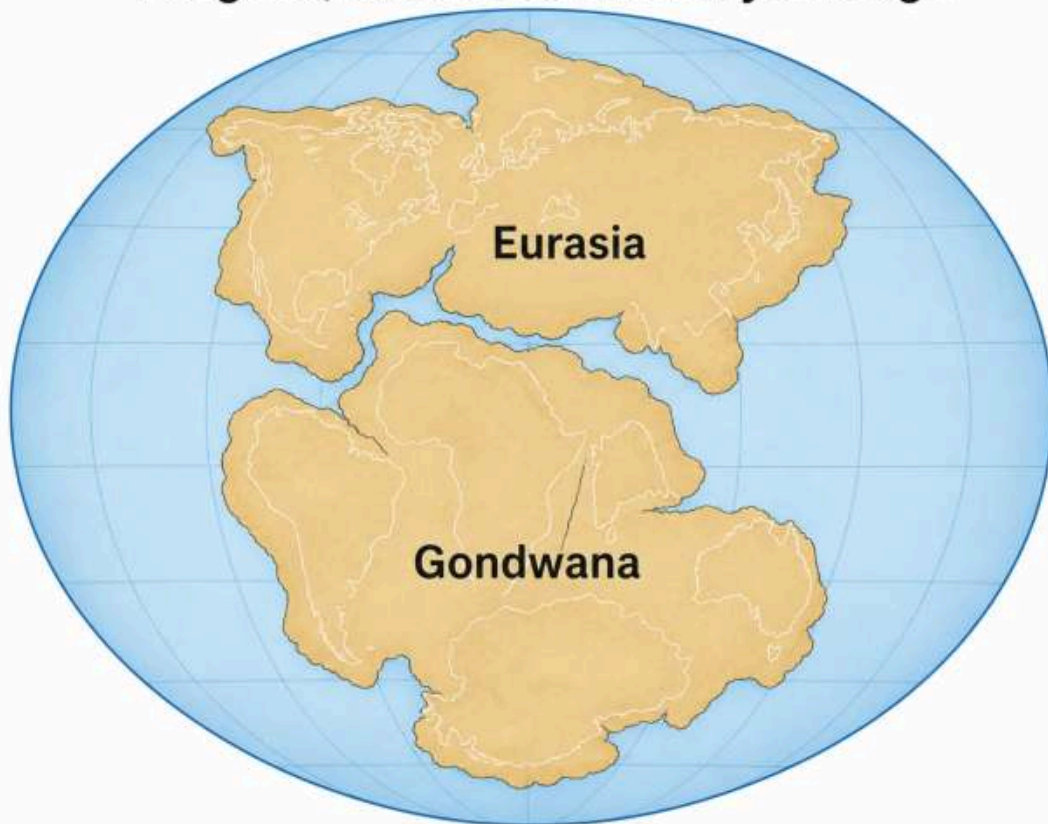


Figure 3.8 · Pangaea, with Eurasia in the north and Gondwana in the south.

TOPIC 3.3

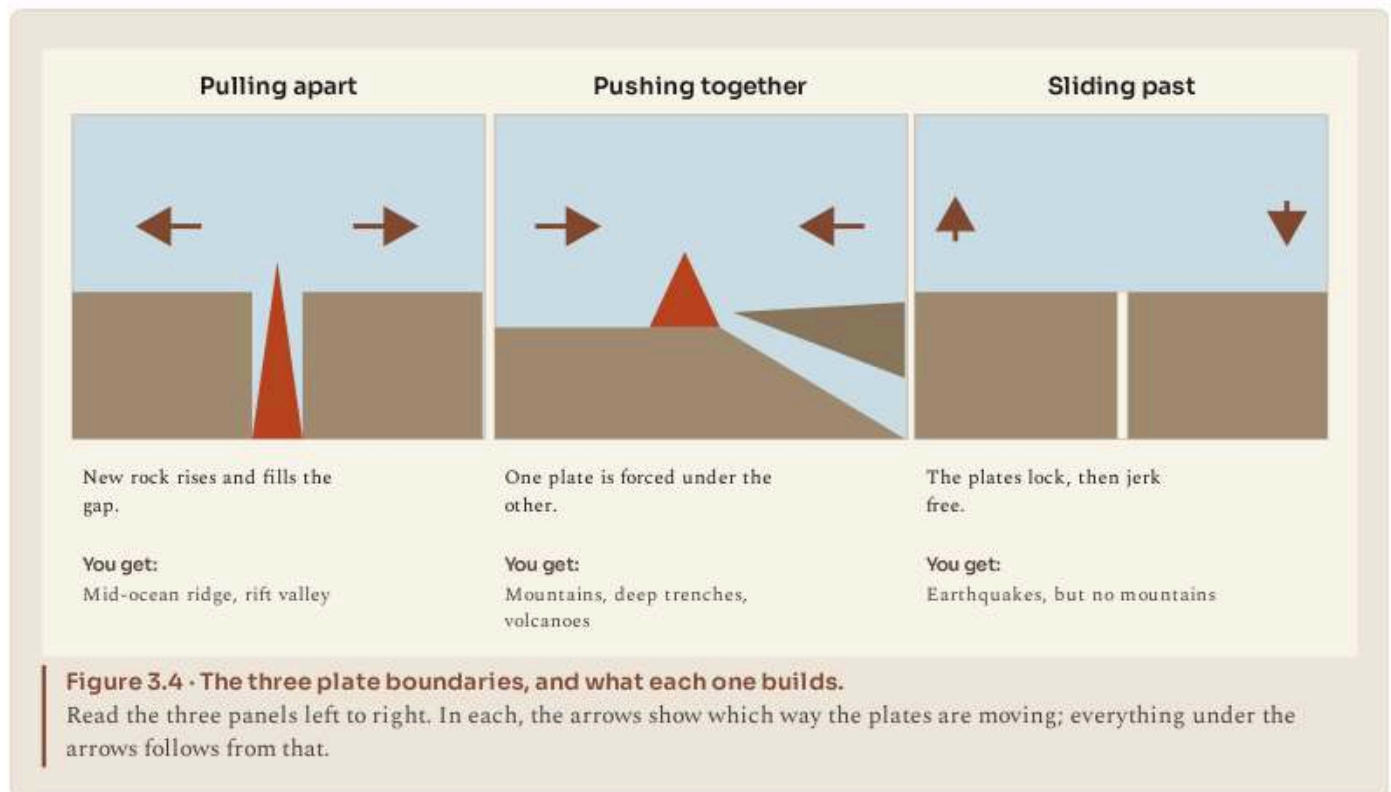
Plates that push, pull and slide

The crust is broken, and the pieces move

The Earth's outer shell is not one continuous skin. It is broken into about fifteen large pieces, called tectonic plates, plus many small ones. They float on the flowing mantle beneath, and they move.

Everything interesting happens at the edges. In the middle of a plate the ground is quiet. At a plate boundary you get mountains, volcanoes, earthquakes, deep ocean trenches and rift valleys, and which of those you get depends entirely on how the two plates are moving relative to each other.

There are only three possibilities, which is a great convenience.



Pulling apart. The plates move away from each other and hot rock rises to fill the gap, cooling into new crust. In the middle of the Atlantic this builds a mountain range 16 000 kilometres long, almost all of it under water. On land it makes rift valleys, like the Great Rift Valley in East Africa.

Pushing together. Something has to give. If one plate is denser it is forced down under the other, which makes a deep trench in the sea floor, melts rock on the way down and sends it back up as volcanoes. If both are equally buoyant continental crust, neither goes down and the rock crumples upwards instead. That is how the Himalayas are being built, as India pushes into Asia at about 5 centimetres a year.

Figure 3.9 · The tectonic plates of the Earth.

Sliding past. The plates grind alongside each other. They do not slide smoothly; they lock, strain builds up for years, and then they jerk free all at once. That gives earthquakes, but no mountains and no volcanoes. The San Andreas fault in California is the famous example.

◆ CAREFUL

Plates are not the same as continents. A plate is a piece of the Earth's outer shell, and most plates carry both continental crust and ocean floor. The South American plate, for example, includes South America **and** half the South Atlantic sea floor. The boundary is not the coastline.

■ YOUR TURN

1 Roughly how many large tectonic plates are there?

2 Copy and complete: pulling apart makes

_____ ; pushing together makes

_____ ; sliding past makes

3 Explain why the Himalayas are still getting taller.

4 Explain the difference between a plate and a continent.

5 Why do sliding boundaries produce earthquakes but not volcanoes?

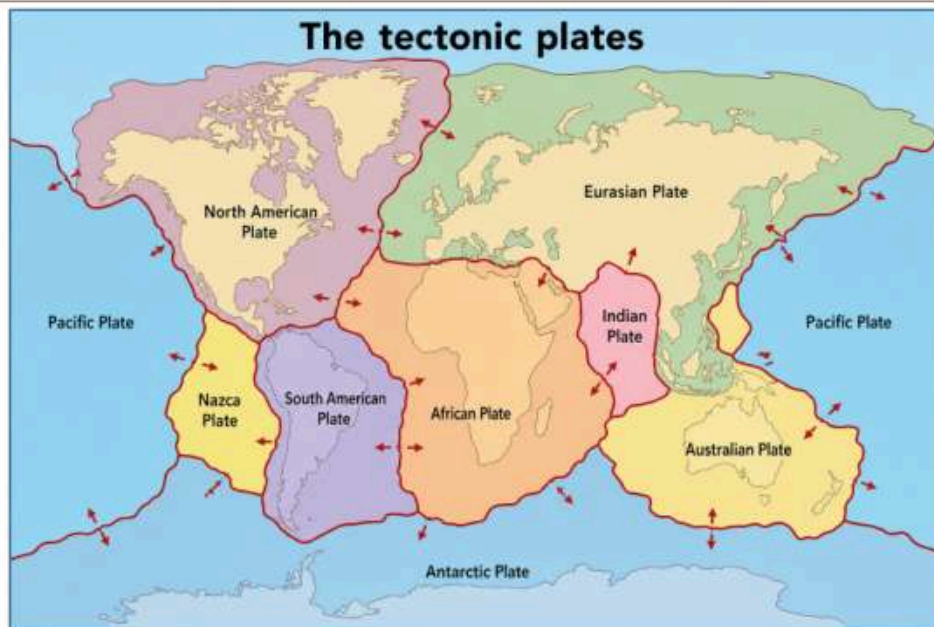


Figure 3.9 · The tectonic plates of the Earth.

TOPIC 3.4

When the Earth moves

Earthquakes

An earthquake is what happens when strain that has been building for years is released in seconds. The rock either side of a fault is locked by friction, the plates keep trying to move, the strain grows, and eventually the rock breaks and both sides lurch into a new position.

The energy travels outward as waves. Their size is measured on a scale where each step up is about thirty times more energy released, so a magnitude 7 is not slightly worse than a magnitude 6; it is roughly thirty times worse.

Almost nobody is killed by the ground shaking. People are killed by buildings. That is why the same magnitude of earthquake can kill a handful of people in one country and tens of thousands in another: the difference is building standards, and building standards cost money.

Volcanoes

A volcano is a place where molten rock from below reaches the surface. Most of them sit exactly where the plate map says they should.

The clearest example is the Ring of Fire, a horseshoe around the edge of the Pacific Ocean where about three quarters of the world's active volcanoes and about 90 per cent of its earthquakes occur. It is not a mystery; it is the outline of a plate.



Figure 3.5 • Where the Earth shakes.

The shaded band is the Ring of Fire. Compare its shape with the edges of the Pacific Ocean, and you have found a plate boundary without being shown one.



LOOK FURTHER

United States Geological Survey, **Volcano Hazards Programme**. Which volcanoes are being watched right now, and why. Find one on the Ring of Fire map in topic 3.4.

volcanoes.usgs.gov/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

So why does anyone live there?

Because volcanic soil is superb. Lava weathers into soil rich in minerals, and farmers get far better harvests on the slopes of a volcano than on ordinary ground nearby. Volcanic areas also give geothermal energy, tourism, and building stone.

People do the same arithmetic everywhere: a certain harvest every year against an eruption that may come once in a lifetime. Put that way, staying is not foolish. It is a bet, and for most people in most years it pays.

■ PORTUGAL AT THIS STAGE

Portugal has this argument at home. The Azores sit on a plate boundary in the middle of the Atlantic, and they are volcanic islands. The Capelinhos eruption on Faial, which began under the sea in 1957 and went on for thirteen months, added new land to the island, buried a lighthouse in ash and made thousands of people leave, many of them for the United States.

The same islands have some of the best soil in Portugal, geothermal power on Sao Miguel, and hot springs people bathe in. Both facts are true at once, which is exactly what this topic is about.



LOOK FURTHER

United States Geological Survey, Earthquake Hazards. A live map of earthquakes in the last day. Compare where they are with the plate boundaries you have just drawn.

www.usgs.gov/programs/earthquake-hazards

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- 1 Explain, in your own words, what causes an earthquake.
- 2 Why do earthquakes of the same size kill very different numbers of people in different countries?
- 3 What is the Ring of Fire, and why is it that shape?
- 4 Give two reasons why people choose to live near an active volcano.
- 5 A magnitude 8 earthquake releases about thirty times the energy of a magnitude 7. Roughly how many times a magnitude 6? Show your working.

TOPIC 3.5

The continents today

The rock cycle: nothing is finished

Rock is not permanent. It is being recycled, all the time, on a timescale so long that it looks permanent to us.

Molten rock cools and hardens into **igneous** rock. Weather breaks that rock into fragments, rivers carry the fragments away, and where they settle and are buried and squeezed they become **sedimentary** rock. Bury rock deeper still, where it is hot and under pressure, and it changes into **metamorphic** rock. Push it deeper again and it melts, and the cycle starts over.

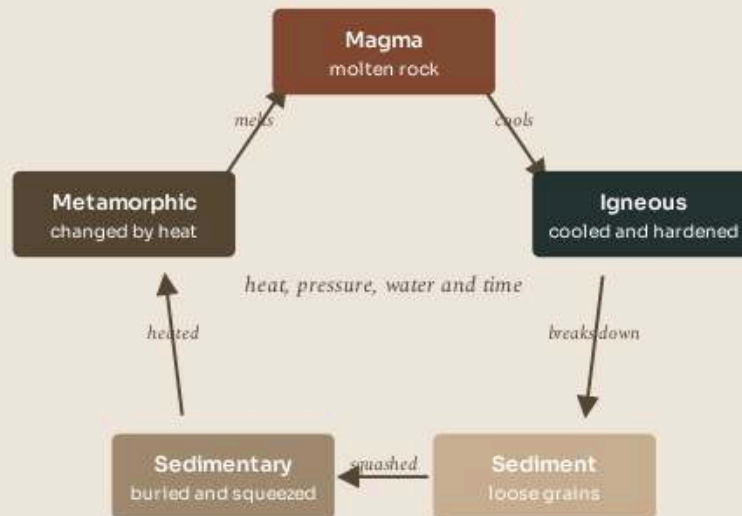


Figure 3.6 · The rock cycle.

Go round the ring once, following the arrows. Notice that it has no beginning and no end, and that every arrow is a process you can name.

Today's map is one frame of a film

Put the three previous topics together and today's world map stops looking like a fact and starts looking like a moment.

The seven continents are where they are because plates have carried them there, and they are still moving. In fifty million years the Mediterranean will have closed as Africa pushes north, and there will be mountains where Italy is. Australia will have collided with south east Asia. The Atlantic will be wider, and the Pacific narrower.

None of that changes anything you will do this week. But it changes how you read a map. The shape of the world is not a design. It is a snapshot of something moving, taken by a species that has only been here for the last 0.002 per cent of the film.

● LOOK AGAIN

Look back at the four maps in Figure 3.2, then at the map of the Ring of Fire in Figure 3.4. Two completely different figures, made for different reasons.

Now read them together. The Ring of Fire is where plates are colliding **now**. The four maps show where plates have carried the continents over hundreds of millions of years. The same process explains both: the same slow flow of the mantle that dragged Pangaea apart is what makes Japan shake this year.

One mechanism, two very different timescales. That is worth remembering, because it is how a lot of geography works.

■ WHAT YOU CAN NOW DO

- ✓ Name the Earth's four layers and explain how we know about them.
- ✓ Put Rodinia, Pangaea and today's continents in order and explain what moved them.
- ✓ Match each of the three plate boundaries to the landforms it builds.
- ✓ Explain earthquakes and volcanoes, and why people live where they happen.
- ✓ Describe the rock cycle, and explain why the world map is a snapshot.

— WORDS TO KEEP

crust, mantle, core, tectonic plate, fault, magnitude, igneous, sedimentary, metamorphic

UNIT 3 AT A GLANCE

The restless Earth

If the ground is moving, why does it feel so still?

THE TOPICS YOU HAVE COVERED

- 3.1 Inside the Earth
- 3.2 Rodinia, Pangaea and the wandering continents
- 3.3 Plates that push, pull and slide
- 3.4 When the Earth moves
- 3.5 The continents today

WHAT YOU CAN NOW DO

- ✓ Name the Earth's four layers and explain how we know about them.
- ✓ Put Rodinia, Pangaea and today's continents in order and explain what moved
- ✓ Match each of the three plate boundaries to the landforms it builds.
- ✓ Explain earthquakes and volcanoes, and why people live where they happen.
- ✓ Describe the rock cycle, and explain why the world map is a snapshot.

THE WORDS THIS UNIT TAUGHT

crust

mantle

core

tectonic plate

fault

magnitude

igneous

sedimentary

metamorphic

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.



| The sea in layers. Sunlight reaches only the top; almost everything below it has never been seen by anybody.

PART ONE • UNIT 4

4 The five oceans

Seven tenths of the planet, and the machine that moves its heat about

■ IN THIS UNIT, YOU WILL

- ✓ Name and find all five oceans.
- ✓ Describe salt, depth, light and temperature in the sea.
- ✓ Explain how currents carry heat around the planet.
- ✓ Link a current to the weather of a real place.
- ✓ Weigh up what people take from the sea, and what they put into it.

UNIT 4 • WHAT YOU WILL LEARN

The blue seven tenths



Figure 4.1 • The five oceans.

Find each name, then trace the coastline that separates it from the next one. Only one of the five has no coast on its northern side.

You will name all five oceans and find them, go down through the layers of the sea into water no one has ever seen, and follow the currents that carry heat from the tropics to the poles. The unit ends with what people take out of the ocean and what they put into it.

THE BIG QUESTION THIS UNIT ANSWERS

The ocean is seven tenths of the planet. What is it doing for us?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

4.1 Meet the five oceans

4.2 What the ocean is really like

4.3 The great conveyor

4.4 The ocean makes the weather

4.5 The ocean and us

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Somebody once said that calling this planet Earth was a mistake, and that it should have been called Ocean.

They had a point. Nearly 71 per cent of the surface is sea. The average depth of that sea is 3 688 metres, which means that if you added up all the space where living things can live, on land, in the air and in the water, well over nine tenths of it is ocean.

Before you read on, write down how many oceans you think there are, and name as many as you can. Then check your list at the end of the next page.

TOPIC 4.1

Meet the five oceans

One ocean, divided into five for convenience

There is really only one ocean. All the water is joined; a bottle dropped off Lisbon can wash up in the Caribbean, and a current leaving the Atlantic can end up in the Pacific. The divisions are made by people, for convenience, and they follow continents and, in one case, a current.

Since 2000 the internationally recognised list has five names on it.

The **Pacific** is the biggest, at 165.2 million square kilometres. It is larger than all the land on Earth put together, and it holds the deepest known point anywhere, the Challenger Deep, at 11 034 metres.

The **Atlantic** is second at 106.5 million square kilometres. It is the one Portugal faces, it is the youngest of the big oceans, and it is still getting wider.

The **Indian** is third at 70.6 million square kilometres, and the warmest. It drives the monsoon that decides whether a fifth of the world's people have a good year.

The **Southern** Ocean, 21.9 million square kilometres, is the one most adults never learned. It rings Antarctica, and it is defined not by land but by a current: the Antarctic Circumpolar Current, which flows all the way round the world without ever meeting a coast.

The **Arctic** is the smallest at 14.1 million square kilometres, the shallowest and the coldest, and much of it is covered by floating sea ice.



Figure 4.2 • The five oceans.

Find each name, then trace the coastline that separates it from the next one. Only one of the five is bounded by a current instead of by land.

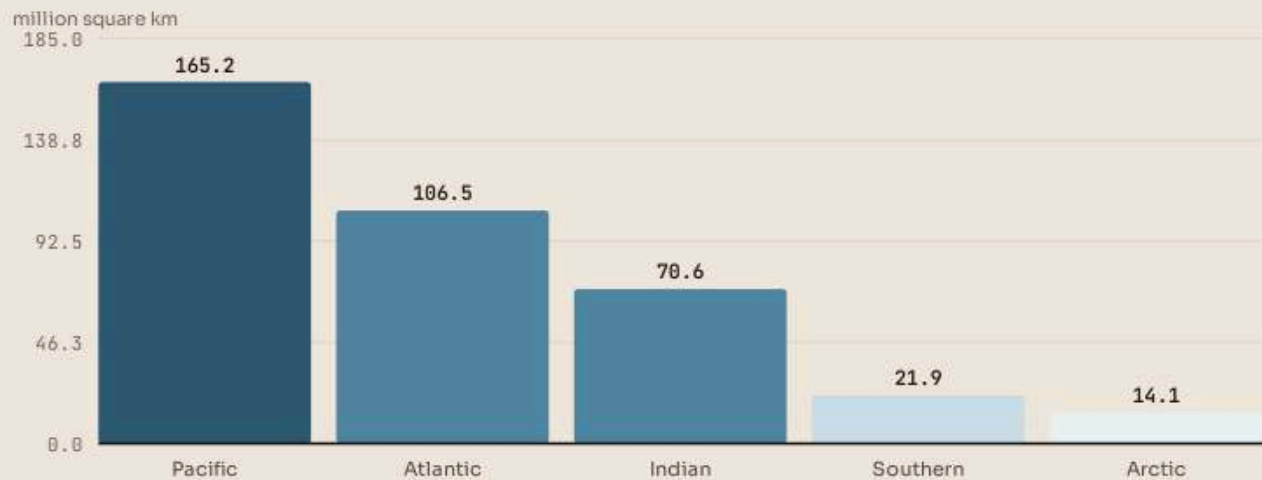


Figure 4.3 • How much bigger the Pacific is.

Read the bars from left to right. The Pacific is larger than the Atlantic and Indian added together.

◆ CAREFUL

"Sea" and "ocean" are not the same word. An ocean is one of the five great bodies of water. A sea is smaller and usually partly enclosed by land: the Mediterranean Sea, the North Sea, the Red Sea. Every sea is part of an ocean.

■ YOUR TURN

- 1 Name the five oceans, largest first.
- 2 Which ocean is defined by a current rather than by land? Name the current.
- 3 What is the difference between a sea and an ocean? Give one example of each.
- 4 Using Figure 4.2, calculate roughly how many times bigger the Pacific is than the Arctic.

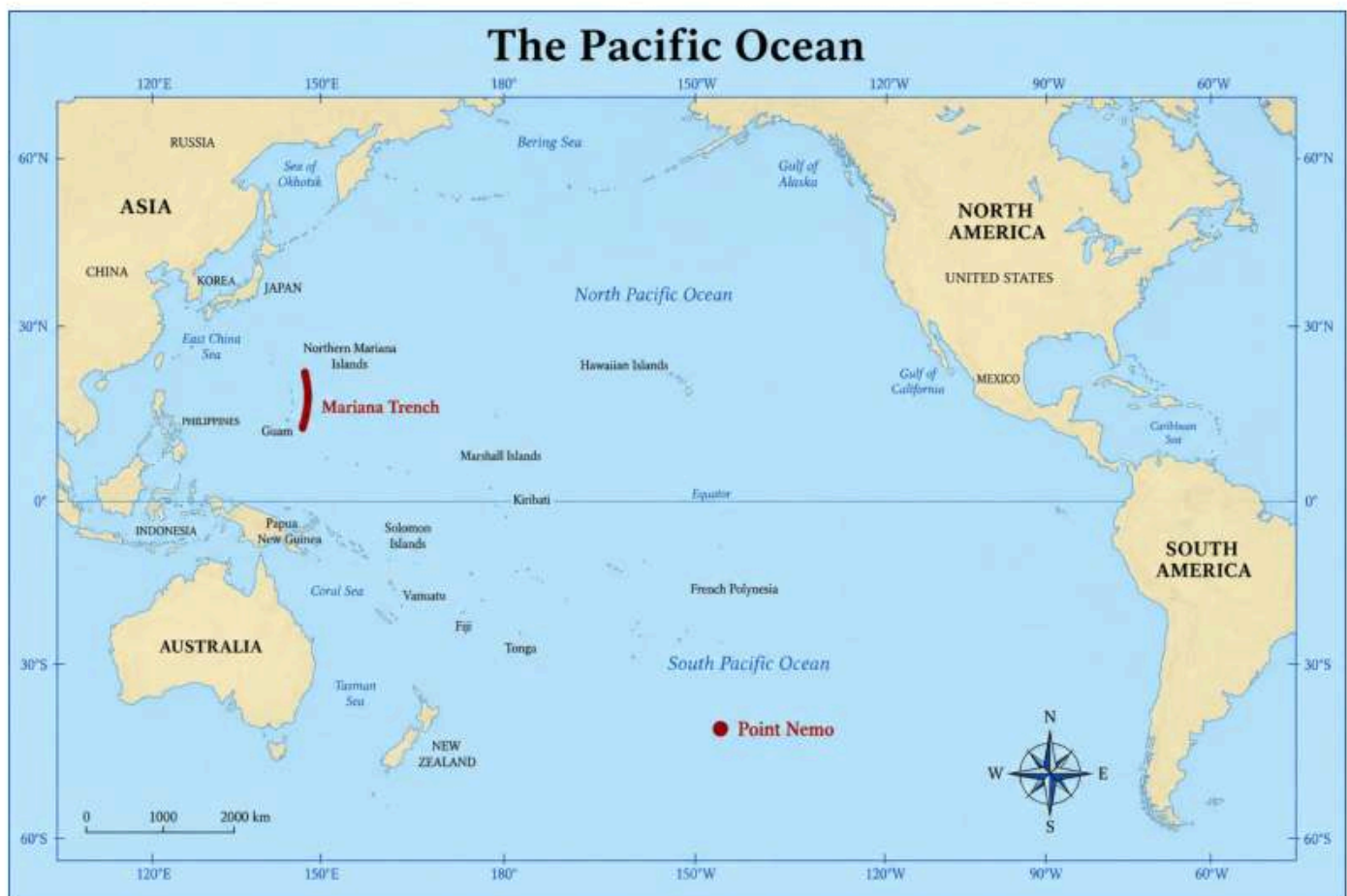


Figure 4.7 · The Pacific Ocean, with the Mariana Trench and Point Nemo.

TOPIC 4.2

What the ocean is really like

Salt

Sea water is about 3.5 per cent salt, which means roughly 35 grams in every litre. If you evaporated all the world's seas you would be left with a layer of salt about 50 metres thick over the whole ocean floor.

The salt comes from the land. Rain is very slightly acidic, it dissolves tiny amounts of minerals out of rock, and rivers carry those minerals to the sea. Water leaves the sea again by evaporating, but the salt does not evaporate with it, so it accumulates.

Depth, and the dark

The sea is not one habitat. It is a stack of them, and the thing that divides them is light.



Sunlight reaches only the top 200 metres or so in useful amounts. That thin layer, less than one twentieth of the average depth, is where nearly all the plant life is, and therefore where nearly all the food comes from.

Below that, light fades fast. By 1 000 metres it is completely, absolutely black. Many animals down there make their own light, which is called bioluminescence, and use it to find food, find mates, or startle predators.

Deeper still it gets colder and the pressure becomes crushing. At the bottom of the Challenger Deep the pressure is over a thousand times what you feel at the surface: the equivalent of about fifty jumbo jets stacked on your head. Fewer people have been to the bottom of the ocean than have walked on the Moon.

● AMAZING BUT TRUE

In 1960 two men descended to the Challenger Deep in a vessel called Trieste. They spent twenty minutes at the bottom, and when they got there they found a flatfish looking at them. Nobody had expected anything with a backbone to be able to live at that pressure.



LOOK FURTHER

NOAA Ocean Exploration. Footage from robot submarines in water no person has ever visited. Match what you see to the depth zones in topic 4.2.

oceanexplorer.noaa.gov/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- ① Roughly what percentage of sea water is salt? Where did the salt come from?
- ② Name the four depth zones this book uses, and give the depth of each.
- ③ Why is almost all the ocean's food produced in the top 200 metres?
- ④ What is bioluminescence, and give two things animals use it for?

TOPIC 4.3

The great conveyor

Water that goes somewhere

Ocean water does not sit still. It moves in currents: enormous, steady rivers within the sea, some of them carrying a hundred times more water than every river on land put together.

Two things drive them. Near the surface, wind. The prevailing winds drag the top of the ocean along with them, and because the Earth is spinning, the water curves as it goes, which is why the big surface currents run in great loops.

Deeper down, density. Cold water is heavier than warm water, and salty water is heavier than fresh. Where surface water gets very cold and very salty, near the poles, it sinks, and other water flows in to replace it. This drives a slow global circulation that takes a thousand years to complete one loop.



Figure 4.5 • Four currents, and which way they carry heat.

Warm currents are drawn in red, cold ones in blue. Follow the Gulf Stream first: it starts in the tropics and ends off Europe.

Why this is the most important machine on the planet

Sunlight does not fall evenly. The tropics get far more of it than the poles do. If nothing moved that heat around, the tropics would be unbearably hot and the poles far colder than they already are.

Something does move it. About half of the heat transport from the equator towards the poles is done by the ocean, and the other half by the atmosphere. Ocean currents are, in effect, a planet-sized central heating system, and they are why the world's climate is as moderate as it is.

■ YOUR TURN

- 1 Name the two things that drive ocean currents, and explain each in one sentence.
- 2 Why does cold, salty water sink?
- 3 Roughly how long does the deep circulation take to complete one loop?
- 4 Explain why the tropics do not get hotter and hotter every year.

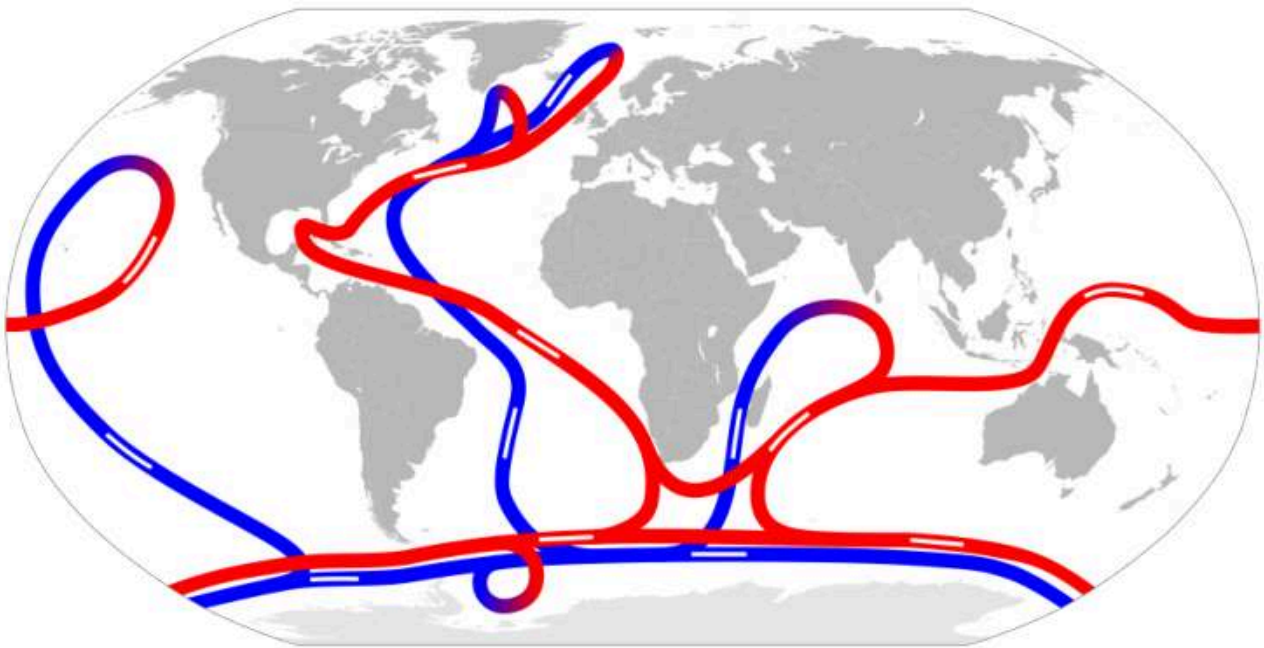


Figure 4.6 - The great conveyor: red at the surface, blue in the deep.

TOPIC 4.4

The ocean makes the weather

Two towns at the same latitude

Here is a fact that ought to be impossible.

Lisbon is at about 38.7 degrees north. So is Washington DC, in the United States, and so, roughly, is Beijing. All three are the same distance from the equator, so all three get almost exactly the same amount of sunlight over a year.

Their winters are not remotely alike. Lisbon's coldest month averages about 11 degrees C. Washington's averages about 2 degrees. Beijing's averages about minus 3.

The difference is the ocean, and it works in two ways.

First, **water heats and cools far more slowly than land**. A coast is therefore milder than an inland place at the same latitude: cooler in summer, warmer in winter, because the sea beside it changes temperature slowly.

Second, **currents carry heat**. The Gulf Stream carries warm water from the Gulf of Mexico across the Atlantic towards Europe. The air passing over it picks up that warmth and carries it onto the land. Western Europe is several degrees warmer in winter than it has any right to be for its latitude.

And the opposite case

Currents can work the other way. The Humboldt Current brings cold water up the west coast of South America, and the air above cold water holds little moisture, so it produces almost no rain. That is a large part of why the Atacama Desert in Chile is the driest place on Earth: some weather stations there have never recorded rainfall at all.

Cold water is also rich in nutrients, so the same current supports one of the world's great fisheries. A cold current can give you a desert on land and a feast in the sea, at the same place, at the same time.

■ PORTUGAL AT THIS STAGE

Portugal is an Atlantic country in a way that shows up in the weather every single day. The Azores High, a large area of high pressure over the mid Atlantic, sits over the country in summer and gives it dry, sunny months. In winter it shifts and lets Atlantic depressions through, which is why almost all Portuguese rain falls between October and March.

The sea also decides where people go. The cool Atlantic keeps Porto and Lisbon milder in August than Madrid, four hundred kilometres inland, and the same Atlantic swell produces the enormous waves at Nazare, where a submarine canyon focuses ocean energy onto one small stretch of coast and where the largest waves ever surfed have been recorded.

◆ WATCH ME WORK IT OUT

Question. Nuuk in Greenland and Oslo in Norway are both at about 60 degrees north. Oslo's coldest month averages about minus 4 degrees C; Nuuk's averages about minus 8. Suggest a reason.

Thinking it through. First, rule out the obvious. Same latitude, so the same sunlight. That cannot be the answer.

Second, look at what is upstream of each one, in the ocean. Oslo is on the European side of the North Atlantic, downstream of the warm current that crosses from the Gulf of Mexico. Nuuk is on the Greenland side, where a cold current comes down from the Arctic.

Third, check the direction of the winds. In these latitudes the prevailing winds blow from the west, so each place gets air that has just crossed the water to its west. For Oslo that water is warm. For Nuuk it is cold.

Answer. The difference is the ocean, not the Sun. Oslo receives westerly air warmed by a warm Atlantic current; Nuuk receives air chilled by a cold Arctic current. Same latitude, different neighbouring water.



LOOK FURTHER

NOAA, Ocean facts. Short answers to real questions about the sea, written for readers who are not scientists.

oceanservice.noaa.gov/facts/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- ① Lisbon and Beijing are at about the same latitude. Explain why Lisbon's winters are so much milder.
- ② Give the two ways in which the ocean affects the climate of a coast.
- ③ Explain how a cold current can cause a desert.
- ④ Why is the same cold current good for fishing?

TOPIC 4.5

The ocean and us

What we take out

Food. Fish and seafood provide a significant part of the protein eaten by around three thousand million people. Some fisheries are managed carefully and are stable. Others have been fished far beyond what the population can replace, and have collapsed.

Trade. Around 80 per cent of world trade by volume travels by sea. Almost everything in the room you are sitting in arrived on a ship. Sea transport is extremely cheap per tonne per kilometre, which is exactly why the world is arranged the way it is.

Energy. Oil and gas from under the sea floor, and increasingly wind farms built offshore, where the wind is stronger and steadier than on land.

What we put in

Plastic. Something like 8 million tonnes of plastic enters the ocean each year. It does not rot. It breaks into smaller and smaller pieces, which are eaten by small animals, which are eaten by bigger ones. Plastic fragments have been found in the deepest trench on Earth and inside animals that have never been near a human being.

Carbon dioxide. The ocean absorbs roughly a quarter of the carbon dioxide people put into the air. That has slowed the warming of the atmosphere, which sounds like good news. But dissolving carbon dioxide in water makes the water slightly more acidic, and that makes it harder for corals, oysters and many small shelled creatures to build their shells.

Food

About 3 000 million people rely on the sea for a large part of the protein they eat.

Trade

Around 80 per cent of world trade by volume travels by ship.

Weather

The ocean holds most of the planet's heat and moves it about. No ocean, no weather as we know it.

And the cost

Roughly 8 million tonnes of plastic enter the sea each year.

Figure 4.6 • Four things the ocean does for people, and one thing people do to it.

Read the first three boxes, then the fourth. The fourth is the price of the first three, and it is not yet paid.

● LOOK AGAIN

Read those two lists again, side by side, and notice something uncomfortable.

Every single thing in the first list is a genuine good: food for a third of humanity, affordable goods, energy without coal. Every single thing in the second list is a consequence of the first. There is no villain in this story and no simple choice.

That is what makes ocean problems hard. They are not a matter of persuading bad people to stop doing something obviously wrong. They are a matter of finding a way to keep the benefits while paying less for them, which is slower, duller and much more useful work.

■ PORTUGAL AT THIS STAGE

Portugal has one of the largest areas of ocean of any country in Europe. Counting the waters around mainland Portugal, Madeira and the Azores, its exclusive economic zone covers about 1.7 million square kilometres, roughly eighteen times the country's land area.

That has consequences. Portugal eats more fish per person than almost any other country in Europe. It has run marine protected areas around the Azores for decades. And when Portugal argues about fishing quotas, pollution or seabed mining, it is arguing about an area far larger than itself.



LOOK FURTHER

Oceanario de Lisboa. One of the largest aquariums in Europe, in Portugal, arranged by ocean. Its four habitats are four of the five oceans.

www.oceanario.pt/en/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- ① Give three things people take from the ocean.
- ② Roughly what proportion of world trade by volume goes by sea?
- ③ Explain why plastic in the ocean is a problem that does not solve itself.
- ④ The ocean absorbing carbon dioxide is both good and bad. Explain both sides.
- ⑤ Portugal's sea area is about eighteen times its land area. Suggest two ways that might change the way the country thinks about itself.

■ WHAT YOU CAN NOW DO

- ✓ Name and locate the five oceans, and explain how they differ.
- ✓ Describe salt, depth, light and pressure in the sea.
- ✓ Explain what drives ocean currents and what they carry.
- ✓ Explain why two places at the same latitude can have very different climates.
- ✓ Weigh up what people take from the ocean against what they put into it.

■ WORDS TO KEEP

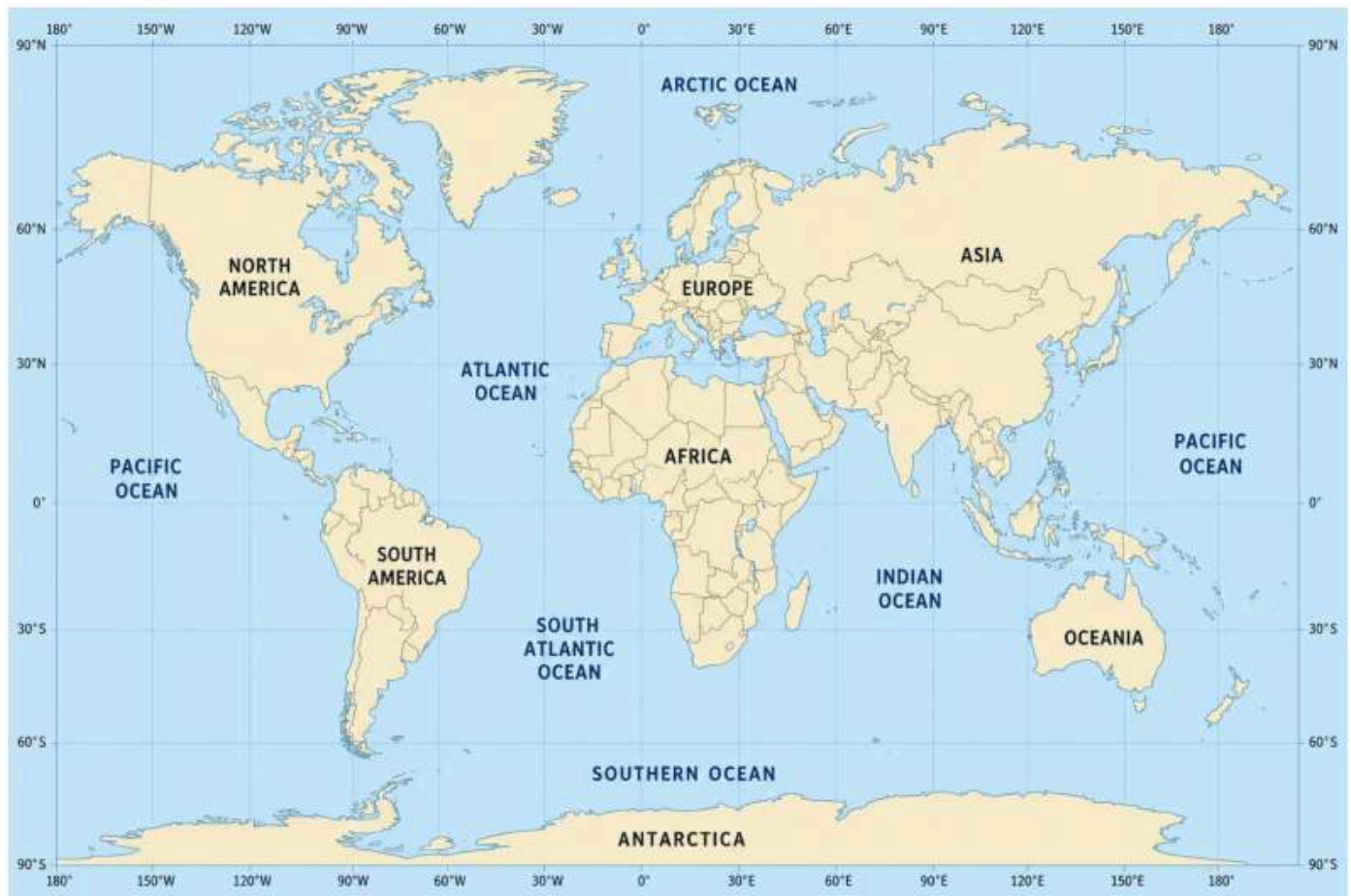
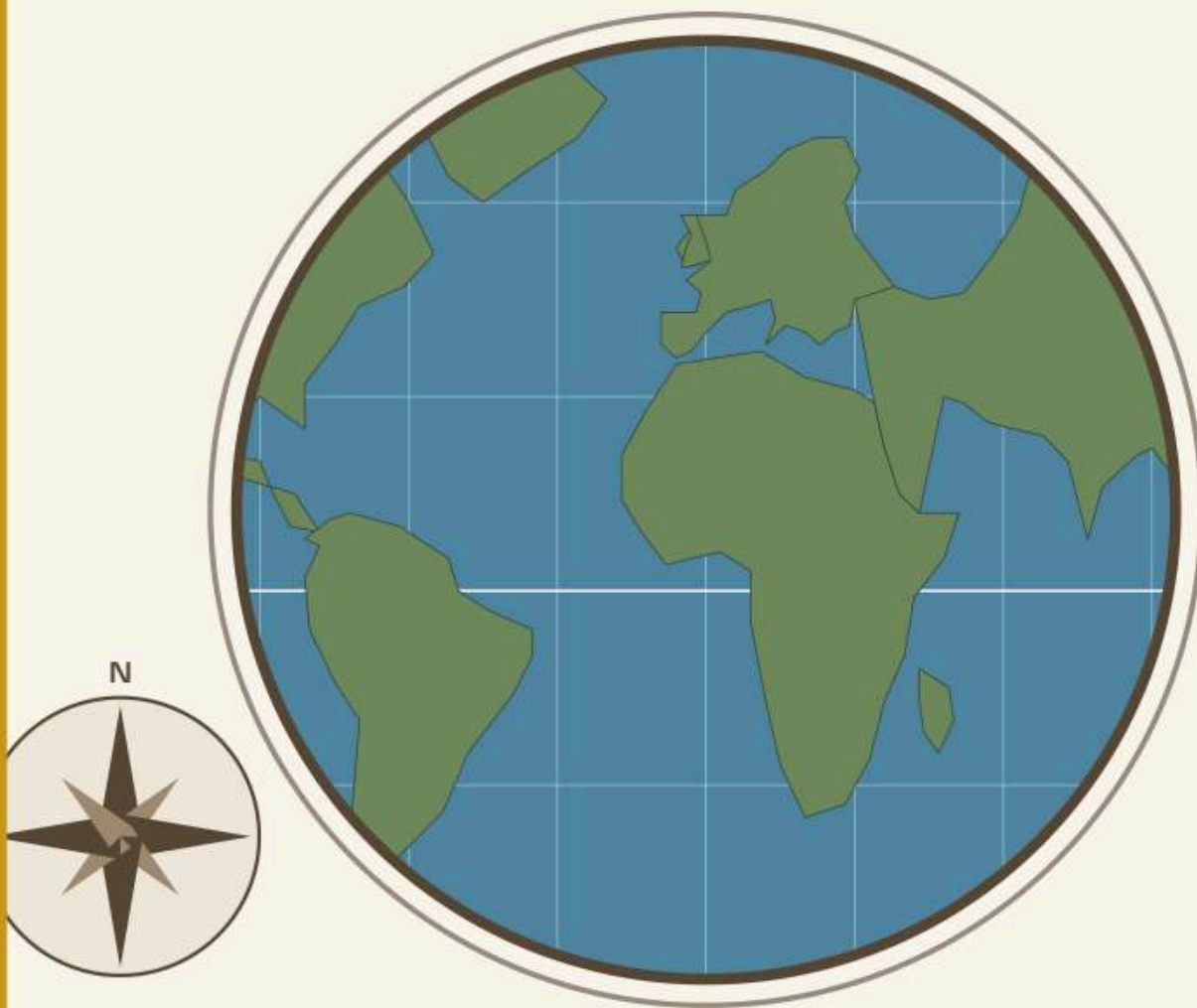


Figure 4.8 · The five oceans and the seven continents.



| The world on a globe. Every continent in this unit is on this picture, and so is the country you were born in.

PART ONE • UNIT 5

5 Seven continents

One planet, seven pieces of land, and a way of reading any of them

■ IN THIS UNIT, YOU WILL

- ✓ Use a map's title, key, scale, direction and coordinates.
- ✓ Describe a continent's relief, climate, biomes and wildlife.
- ✓ Explain where a continent's people live, and why there.
- ✓ Name the main countries and capitals of each continent.
- ✓ Explain why Antarctica matters to everybody, including you.

UNIT 5 • WHAT YOU WILL LEARN

Seven pieces of land



Figure 5.1 • The seven continents.

One of them has no permanent residents at all, and one of them is where every human family ultimately comes from.

You will learn to read any map properly, then take each continent in turn: its shape, its mountains, its climate, the living things that belong there, the people, and the countries. By the end you should be able to describe a continent you have never visited.

THE BIG QUESTION THIS UNIT ANSWERS

How would you describe a continent you have never visited, and be right?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

5.1 How to read a map

5.2 Africa

5.3 Asia

5.4 Europe

5.5 North America

5.6 South America

5.7 Oceania

5.8 Antarctica

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Name the seven continents. Then, without looking, put them in order of size. Then put them in order of population.

Almost everybody gets the two orders wrong in the same way: we all think the places we hear about most are the biggest. Asia holds more people than the other six put together, and one continent has no permanent residents at all.

TOPIC 5.1

How to read a map

Five things every map has

A map is not a picture of the ground. It is an argument about the ground: a person decided what to include, what to leave out and what to make important. Reading one properly starts with five things that are always there.

1. The **TITLE** says what the map shows



2. The **KEY** says what the colours mean

3. The **SCALE** says how far it really is

4. The **ARROW** says which way is north

5. The **GRID** gives every place an address

Every map you meet has these five. Find them before you read anything else.

Figure 5.2 · The five things to find before you read any map.

Look for them in the order they are numbered. It takes about ten seconds and it changes what you see.

The title says what the map is showing. A map of Africa's rainfall and a map of Africa's population look completely different, and the title is how you know which you are holding.

The key explains the colours and symbols. Never guess. Blue usually means water, but on a rainfall map blue means wet, and on a temperature map blue means cold.

The scale tells you how far it really is. It is usually a line marked with a distance. Two maps on the same page can be at wildly different scales.

The direction arrow tells you which way is north. It is normally up, but not always, and on a map of Antarctica "up" is meaningless.

The grid gives every place an address. On a world map that is latitude and longitude: latitude measures how far north or south of the equator you are, from 0 at the equator to 90 at the poles; longitude measures how far east or west, from 0 at Greenwich in London to 180 in the Pacific.

Every flat map of a round world is wrong

You cannot flatten a sphere without stretching something. Try it with an orange peel and you will see the problem in ten seconds.

So every world map has to choose what to get wrong. The common one in classrooms keeps the shapes of countries roughly right, which makes it good for navigation, but it badly stretches areas far from the equator. On it, Greenland looks about the size of Africa. Africa is in fact about fourteen times bigger.

That is not a conspiracy. It is geometry. But it is worth knowing, because a map that quietly makes some parts of the world look bigger than they are will quietly change what you think is important.

◆ CAREFUL

The maps in this book are equirectangular: straight lines of latitude and longitude. They keep directions simple, which is right for a first world map, and they stretch areas near the poles, which is why Antarctica looks enormous. No map escapes this. Always ask what a map has chosen to get wrong.

■ YOUR TURN

- 1 List the five things every map should have, and say what each is for.
- 2 Give the latitude and longitude, roughly, of the place you are sitting.
- 3 Explain why every flat map of the world must distort something.
- 4 On many classroom maps Greenland looks the size of Africa. Explain why this happens, and give the real comparison.

TOPIC 5.2

Africa

The shape of it

Africa is the second largest continent, 30.4 million square kilometres, and the only one crossed by the equator roughly in the middle. That single fact explains most of its geography, because climate is arranged in bands either side of the equator, and Africa gets the full set twice, mirrored north and south.

Working outwards from the equator: rainforest at the middle, then savannah grassland, then desert, then a narrow strip of mild Mediterranean climate at each far end.

The **relief** is unusual. Much of Africa is a high plateau rather than lowland plain, so many rivers reach the coast by dropping off an edge, which gives spectacular waterfalls and, historically, made it very hard to sail inland. In the east, the Great Rift Valley is tearing the continent apart along a line 6 000 kilometres long, and one day the eastern strip will separate.



Figure 5.3 • Africa.

Find the equator first, then the Sahara north of it and the Kalahari south of it. Notice how the four cities marked are all on or near the coast.

Area	30.4 million square kilometres
People	About 1 481 million, and the fastest growing population of any continent
Highest point	Kilimanjaro, 5 895 m, a volcano near the equator with ice on top
Longest river	The Nile, about 6 650 km
Biomes	Rainforest, savannah, desert, Mediterranean scrub, mountain
Resources	Gold, diamonds, cobalt, oil, farmland, and the world's youngest workforce
Countries	54, more than any other continent
Some capitals	Abuja (Nigeria), Cairo (Egypt), Nairobi (Kenya), Pretoria (South Africa)

Table 5.1 • Africa at a glance.

Read the rows in pairs: the physical row above and the human row it explains below.

The living world

The savannah carries the largest land animals left on Earth: elephant, giraffe, lion, rhinoceros, wildebeest. The Serengeti migration, in which nearly two million wildebeest and zebra move in an annual circle following the rain, is the largest movement of land animals anywhere.

The Sahara, at 9.2 million square kilometres, is the largest hot desert in the world, and it is bigger than the whole of Europe. It is not empty: it has mountains, oases, salt mines and trade routes thousands of years old.

● AMAZING BUT TRUE

Africa is so much larger than it looks on most classroom maps that you could fit the United States, China, India, Japan and most of Europe inside its outline, all at the same time, with room left over.

■ YOUR TURN

- 1 Why does Africa have the same climate bands twice?
- 2 Explain how Africa's plateau shape affected people trying to travel inland.
- 3 Name three biomes found in Africa and give one animal from each.
- 4 How many countries are there in Africa? Which continent has more?

TOPIC 5.3

Asia

The largest of everything

Asia is the biggest continent, 44.6 million square kilometres, and it holds about 4 753 million people, which is roughly six people in every ten alive.

It also holds the extremes. The highest point on Earth, Everest at 8 849 metres. The lowest land, the shore of the Dead Sea at 430 metres below sea level. The coldest inhabited places, in Siberia. Some of the wettest, in north east India. The largest country by area, Russia, and the two largest by population, India and China.



Figure 5.4 • Asia.

Find the Himalayas between India and China, then trace the great rivers running off them. Almost every early Asian civilisation sits on one of those rivers.

The Himalayas are the key to a great deal of it. They are still rising as India pushes north into the rest of the continent, and they do two enormous jobs. They block the cold air coming south, which is why northern India is far warmer in winter than places at the same latitude in China. And they feed the great rivers, the Indus, Ganges, Brahmaputra, Mekong, Yangtze and Yellow, on which perhaps one and a half thousand million people depend.

Area	44.6 million square kilometres
People	About 4 753 million
Highest point	Everest, 8 849 m
Longest river	The Yangtze, about 6 300 km
Biomes	Tundra, boreal forest, steppe, desert, monsoon forest, tropical rainforest
Resources	Oil and gas, rice and wheat, manufacturing, rare metals
Countries	49
Some capitals	Beijing (China), New Delhi (India), Tokyo (Japan), Riyadh (Saudi Arabia)

Table 5.2 · Asia at a glance.

The population row is the one to dwell on: it is more than the other six continents added together.

The monsoon

Much of southern and eastern Asia lives by the monsoon: a seasonal reversal of the wind. In summer the land heats faster than the sea, air rises over the land, and moist air is drawn in off the ocean, bringing months of heavy rain. In winter the flow reverses and the air is dry.

Whole harvests depend on it arriving on time. A monsoon that comes late or brings too little rain is a national emergency for several countries at once.

■ YOUR TURN

- 1 Roughly what share of the world's people live in Asia?
- 2 Give three ways the Himalayas affect the rest of the continent.
- 3 Explain what the monsoon is and why it reverses.
- 4 Name the two most populated countries in the world, and their capitals.

TOPIC 5.4

Europe

Small, crowded, and cut about by sea

Europe is the second smallest continent, 10.2 million square kilometres, but it holds about 745 million people, so it is densely populated. It has no obvious eastern edge, and the line usually drawn, along the Ural Mountains, is a convention rather than a natural break: Europe and Asia are one landmass.

Its most important feature is its coastline. Europe is a peninsula covered in smaller peninsulas: Iberia, Italy, Scandinavia, the Balkans, Brittany, Jutland, plus large islands. Nowhere in western Europe is very far from the sea, and that made trade, fishing and, later, overseas expansion far easier than in a continental interior.



Figure 5.5 • Europe.

Find Lisbon at the far west, then follow the coastline round. Count the seas: the Mediterranean, the Atlantic, the North Sea, the Baltic, the Black Sea.

Area	10.2 million square kilometres
People	About 745 million
Highest point	Elbrus, 5 642 m, in the Caucasus
Longest river	The Volga, about 3 530 km
Biomes	Tundra in the far north, boreal and deciduous forest, grassland, Mediterranean scrub
Resources	Farmland, coal and iron historically, North Sea oil and gas, manufacturing
Countries	44 to 50, depending where the boundary is drawn
Some capitals	Lisbon (Portugal), Paris (France), Berlin (Germany), Warsaw (Poland)

Table 5.3 · Europe at a glance.

Compare the area row with Africa's, then compare the population rows.

Climate, and the reason for it

Europe is much warmer than its latitude deserves, and unit 4 explained why: the Atlantic, and the warm current crossing it. Lisbon and New York are at similar latitudes, and Lisbon's winter is far milder.

That warmth thins out as you go east, away from the ocean. Western Europe has mild wet winters and cool summers; central and eastern Europe have colder winters and hotter summers. The change is gradual and it is entirely about distance from the Atlantic.

■ PORTUGAL AT THIS STAGE

Portugal is the far south western corner of the whole continent, and being at the edge has shaped everything about it. It faces the Atlantic rather than the Mediterranean, which set the direction of its history: south and west, not east. Its land border, with Spain, has been almost unchanged since 1297, which makes it one of the oldest borders in the world.

The country is also more varied than its size suggests. The north is green, granite and wet, with more than 1 200 mm of rain a year in places. The south is dry, hot and open, and parts of the Alentejo get less than 500 mm. Add Madeira and the Azores, hundreds of kilometres out in the Atlantic, and one small country reaches from mainland Europe into the middle of an ocean.

■ YOUR TURN

- 1 Why is the eastern boundary of Europe a matter of convention?
- 2 Explain how Europe's long coastline affected its history.
- 3 Why does eastern Europe have colder winters than western Europe?
- 4 Give the capital cities of Portugal, France, Germany and Poland.

TOPIC 5.5

North America

A continent arranged in stripes

North America runs from the Arctic to the tropics, 24.7 million square kilometres, with about 604 million people. Its main features run roughly north to south, which has a large effect on its weather.

In the west, the Rocky Mountains. In the east, the much older and lower Appalachians. Between them, an enormous plain drained by the Mississippi, which is some of the best farmland on the planet.

Because the mountains run north to south, there is nothing to stop cold Arctic air sweeping straight down the middle of the continent, or warm moist air coming north from the Gulf of Mexico. Where those two meet, you get violent weather: this region has more tornadoes than the rest of the world combined.

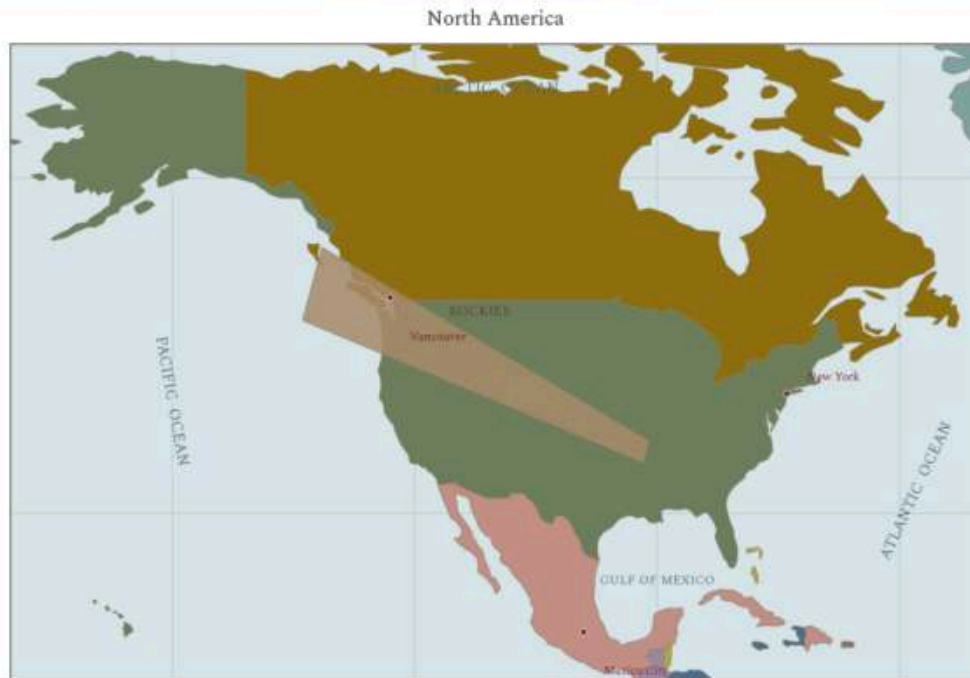


Figure 5.6 • North America.

Find the mountain chains on the west and east, then the plain between them. Notice that the three cities marked are all a long way apart.

Area	24.7 million square kilometres
People	About 604 million
Highest point	Denali, 6 190 m, in Alaska
Longest river system	Mississippi and Missouri, about 6 275 km
Biomes	Arctic tundra, boreal forest, prairie grassland, desert, temperate forest, tropical forest in the south
Resources	Grain, timber, oil and gas, fresh water, minerals
Countries	23
Some capitals	Ottawa (Canada), Washington DC (United States), Mexico City (Mexico)

Table 5.4 · North America at a glance.

The biome row runs almost the full range: only true tropical rainforest and true desert at both extremes are missing.

The Great Lakes on the Canada and United States border hold about a fifth of all the fresh surface water on Earth. Further south, Central America narrows to a strip only 50 kilometres wide at its thinnest, which is where the Panama Canal cuts through and saves ships a 13 000 kilometre journey round South America.

■ YOUR TURN

- 1 Name the two great mountain chains of North America and say where each is.
- 2 Explain why the middle of the continent has such extreme weather.
- 3 What proportion of the world's fresh surface water is in the Great Lakes?
- 4 Why was the Panama Canal built where it was?

TOPIC 5.6

South America

The wettest, the driest, and the highest outside Asia

South America is 17.8 million square kilometres with about 442 million people. Almost the whole continent is dominated by two things: a mountain range down one side, and the largest river system in the world beside it.

The **Andes** run 7 000 kilometres down the western edge, the longest mountain range on any continent, thrown up where the Pacific sea floor is being forced under South America. They are high enough to have permanent ice at the equator, and you will meet them again properly in unit 10.

The **Amazon** carries more water than the next seven largest rivers on Earth put together, and drains a rainforest of about 5.5 million square kilometres, holding perhaps a tenth of all known species.

And squeezed between the Andes and the Pacific is the **Atacama**, the driest desert in the world, for the reason unit 4 gave: a cold current offshore.



Figure 5.7 · South America.

Follow the Andes down the western edge, and notice how close they are to the coast. Everything east of them drains into the Atlantic.

Area	17.8 million square kilometres
People	About 442 million
Highest point	Aconcagua, 6 961 m
Longest river	The Amazon, about 6 400 km, and the largest by volume anywhere
Biomes	Rainforest, savannah, grassland pampas, desert, high mountain
Resources	Copper, lithium, iron, soya, coffee, beef, timber
Countries	12
Some capitals	Brasilia (Brazil), Buenos Aires (Argentina), Lima (Peru), Bogota (Colombia)

Table 5.5 • South America at a glance.

Look at the highest point and the driest place: they are about 300 km apart.

● AMAZING BUT TRUE

The Amazon river is so large that fresh water can be detected in the Atlantic Ocean more than 150 kilometres out from its mouth. Early sailors filled their water barrels while still out of sight of land, and could not work out where the fresh water was coming from.

■ YOUR TURN

- 1 Name the longest mountain range on any continent, and explain how it formed.
- 2 Give two records held by the Amazon.
- 3 Why is the Atacama so dry, given that it is beside an ocean?
- 4 Name four South American countries and their capitals.

TOPIC 5.7

Oceania

Mostly water

Oceania is the smallest continent by land, about 8.5 million square kilometres, with only about 45 million people. But it is spread across an area of ocean larger than any other continent's land, which makes it a very unusual place to describe.

Australia is most of the land: an old, flat, dry continent, with a large desert interior called the outback and most of its people living on a narrow coastal strip in the south east. It has been separated from other landmasses for tens of millions of years, which is why so much of its wildlife exists nowhere else: kangaroos, koalas, platypuses, wombats.

New Zealand is quite different: mountainous, wet, green, and sitting on a plate boundary, which gives it volcanoes and earthquakes.

The Pacific islands are thousands of small islands across an enormous area, some volcanic and mountainous, many of them low coral atolls only a couple of metres above sea level. Countries such as Tuvalu and Kiribati are among the most threatened on Earth by rising seas.



Figure 5.8 • Oceania.

Notice how much of this map is water. The distance from Perth to Auckland is greater than from Lisbon to Moscow.

Area	About 8.5 million square kilometres of land
People	About 45 million
Highest point	Puncak Jaya, 4 884 m, in New Guinea
Longest river	The Murray, about 2 508 km
Biomes	Desert, grassland, eucalyptus forest, tropical rainforest, coral reef
Resources	Iron ore, coal, gold, wool, wheat, fisheries, tourism
Countries	14
Some capitals	Canberra (Australia), Wellington (New Zealand), Suva (Fiji)

Table 5.6 · Oceania at a glance.

The population row is the smallest of any inhabited continent.

The Great Barrier Reef, off north east Australia, is about 2 300 kilometres long and is the largest structure on Earth built by living things.

■ YOUR TURN

- 1 Why is Australian wildlife so unlike wildlife anywhere else?
- 2 Contrast Australia and New Zealand in two ways.
- 3 Why are low coral islands particularly at risk from rising sea levels?
- 4 What is the Great Barrier Reef, and what makes it remarkable?

TOPIC 5.8

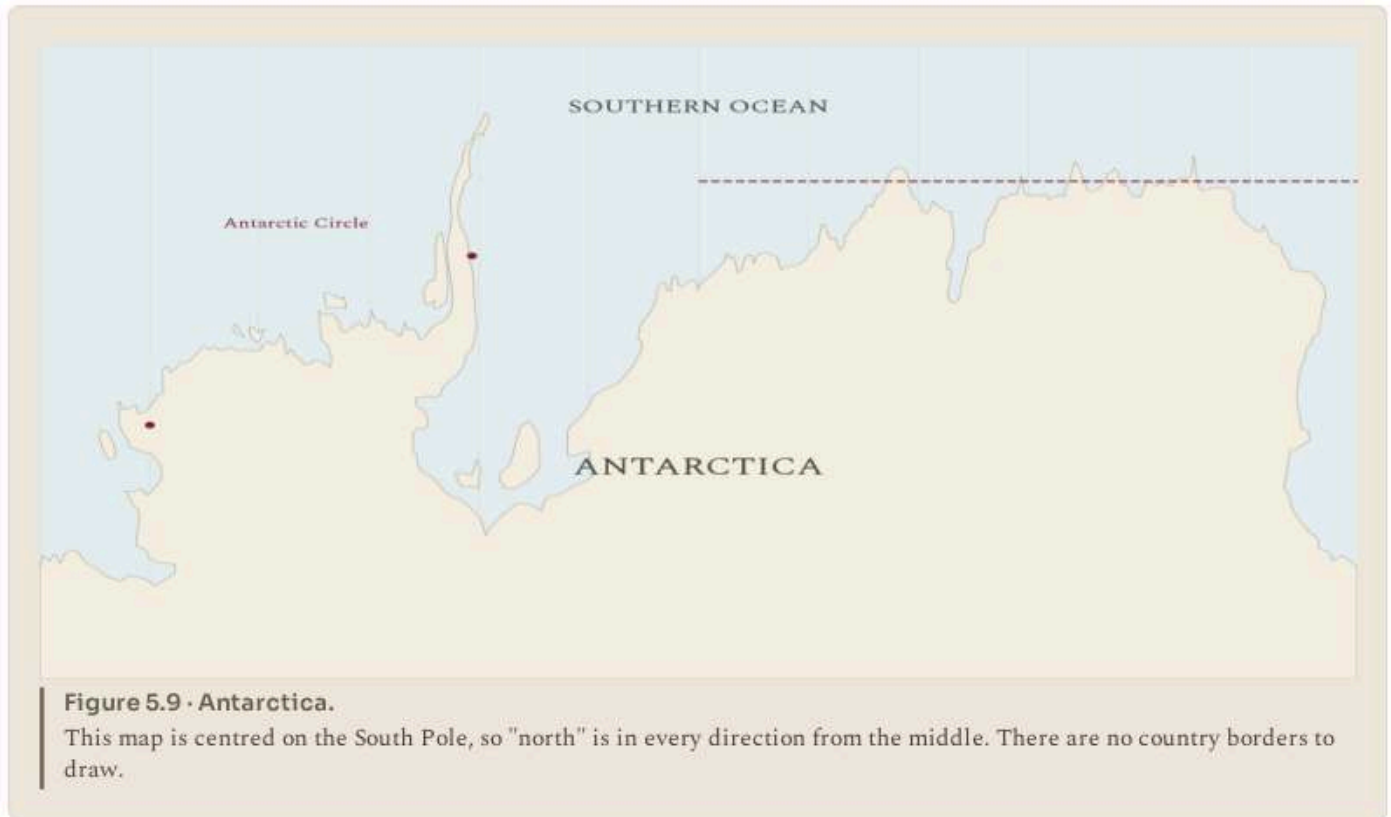
Antarctica

A continent with no countries

Antarctica is 14.2 million square kilometres, larger than Europe, and it has no permanent population at all. Between about 1 000 and 5 000 scientists and support staff live there at any time, depending on the season, and everyone goes home eventually.

It is covered by an ice sheet averaging about 1.9 kilometres thick, which holds roughly 90 per cent of all the ice on Earth and about 70 per cent of all its fresh water. Under the ice there is land, and in places that land is pressed down below sea level by the sheer weight above it.

It is also the coldest, windiest and driest continent. The lowest natural air temperature ever recorded on Earth, minus 89.2 degrees C, was measured at Vostok Station in 1983. And despite all that ice, most of Antarctica receives less precipitation than the Sahara: it is technically a desert.



Why it matters to everybody

Three reasons, and none of them is scenery.

The ice is a record. Snow falls, is buried, and traps bubbles of the air of that year. Drill a core down through the ice and you can sample the atmosphere going back more than 800 000 years, which is how we know what the air used to be like.

The ice is a threat and a promise. If the whole sheet melted, sea level would rise by about 58 metres. That is not going to happen soon, but even small losses raise sea level everywhere, including at every coastal city in the world.

It is governed differently. The Antarctic Treaty, signed in 1959 and now agreed by more than fifty countries, sets the whole continent aside for peaceful scientific use, bans military activity and suspends all territorial claims. There is no other place on Earth run this way, and it has held for over sixty years.



LOOK FURTHER

British Antarctic Survey. What it is actually like to live and work on a continent with no countries. Look up the current temperature at the research station.

www.bas.ac.uk/about/antarctica/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- 1 Give three reasons Antarctica counts as a desert, a freezer and a library at the same time.
- 2 What proportion of the world's fresh water is in the Antarctic ice sheet?
- 3 Explain how ice cores tell us about ancient air.
- 4 What does the Antarctic Treaty do, and why is it unusual?

◆ WATCH ME WORK IT OUT

Question. Asia has about 4 753 million people and 44.6 million square kilometres. Africa has about 1 481 million people and 30.4 million square kilometres. Which is more crowded, and by how much?

Thinking it through. Crowded means people per unit of area, so divide population by area for each.

Asia: 4 753 divided by 44.6, which is about 107 people per square kilometre.

Africa: 1 481 divided by 30.4, which is about 49 people per square kilometre.

Then compare: 107 divided by 49 is about 2.2.

Answer. Asia is more crowded, at roughly 107 people per square kilometre against Africa's 49, so about 2.2 times as densely populated. Be careful with the word "crowded", though: both figures are averages over enormous areas, and both continents contain empty deserts and packed cities.

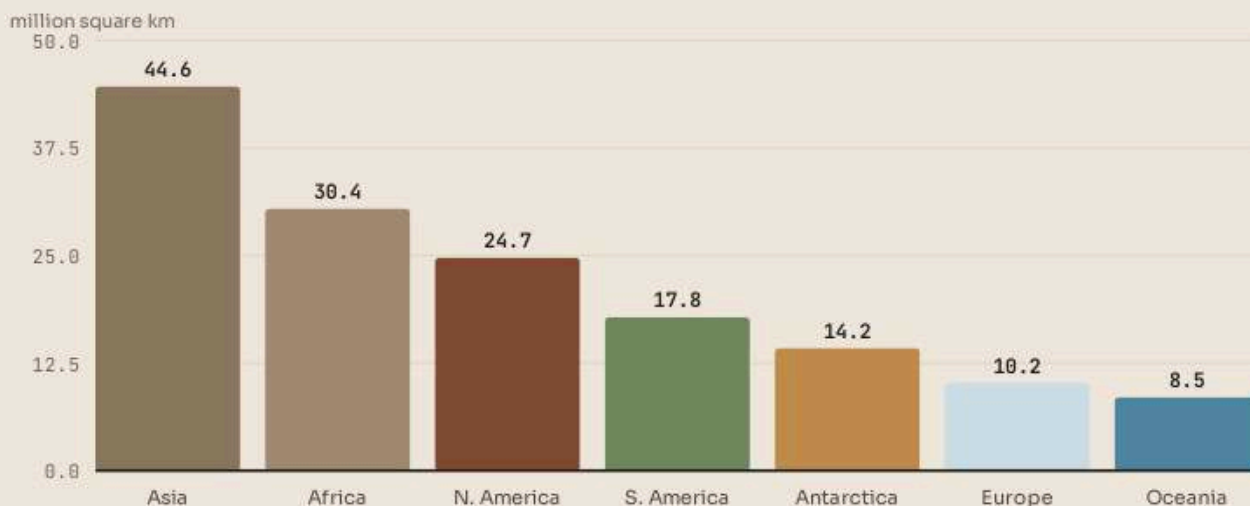


Figure 5.10 - The seven continents by size.

Read the bars, then compare with the next figure. The order is not the same.

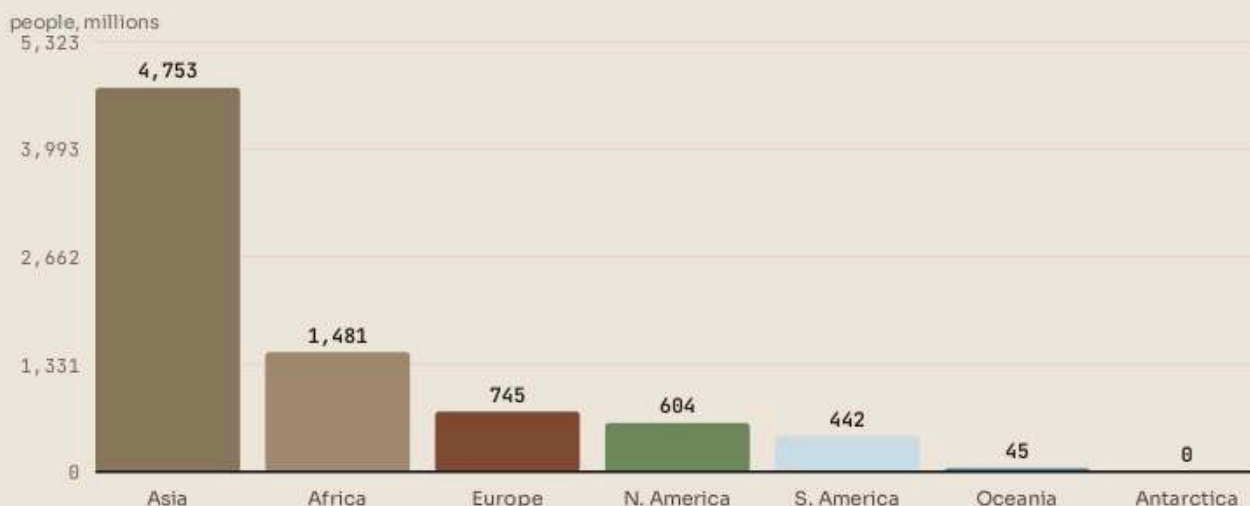


Figure 5.11 - The seven continents by population.

Now compare the two orders. Asia is first in both; Antarctica is fifth by area and last by population, with nobody living there permanently.



LOOK FURTHER

NASA Earth Observatory. The whole planet, photographed from orbit and explained one image at a time. Find your own continent from space.

earthobservatory.nasa.gov/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ WHAT YOU CAN NOW DO

- ✓ Use a map's title, key, scale, direction and grid, and say what it distorts.
- ✓ Describe any continent's relief, climate, biomes and wildlife.
- ✓ Explain where a continent's people live and why.
- ✓ Name the main countries and capitals of each continent.
- ✓ Explain why Antarctica matters to people who will never go there.

■ WORDS TO KEEP

continent, plateau, savannah, monsoon, biome, latitude, longitude, ice sheet, treaty

PART TWO

The players: the human story, in order

Now the people arrive. This part runs forwards in time without jumping about: out of Africa, through an ice age, into farming and the first cities, then to Egypt on its river, China between two rivers, and the Inca along four thousand kilometres of mountain. Five units, one timeline, four continents.

The stage: our world, and how it was made

The units in this part, and the question each one answers. Come back to this page when you want to know where you are.

Unit 1

How the Universe began

How can we possibly know what happened before anybody was there to see it?

Unit 2

The Sun and its family

Out of everything in the Solar System, why is this the only place we know of with life on it?

Unit 3

The restless Earth

If the ground is moving, why does it feel so still?

Unit 4

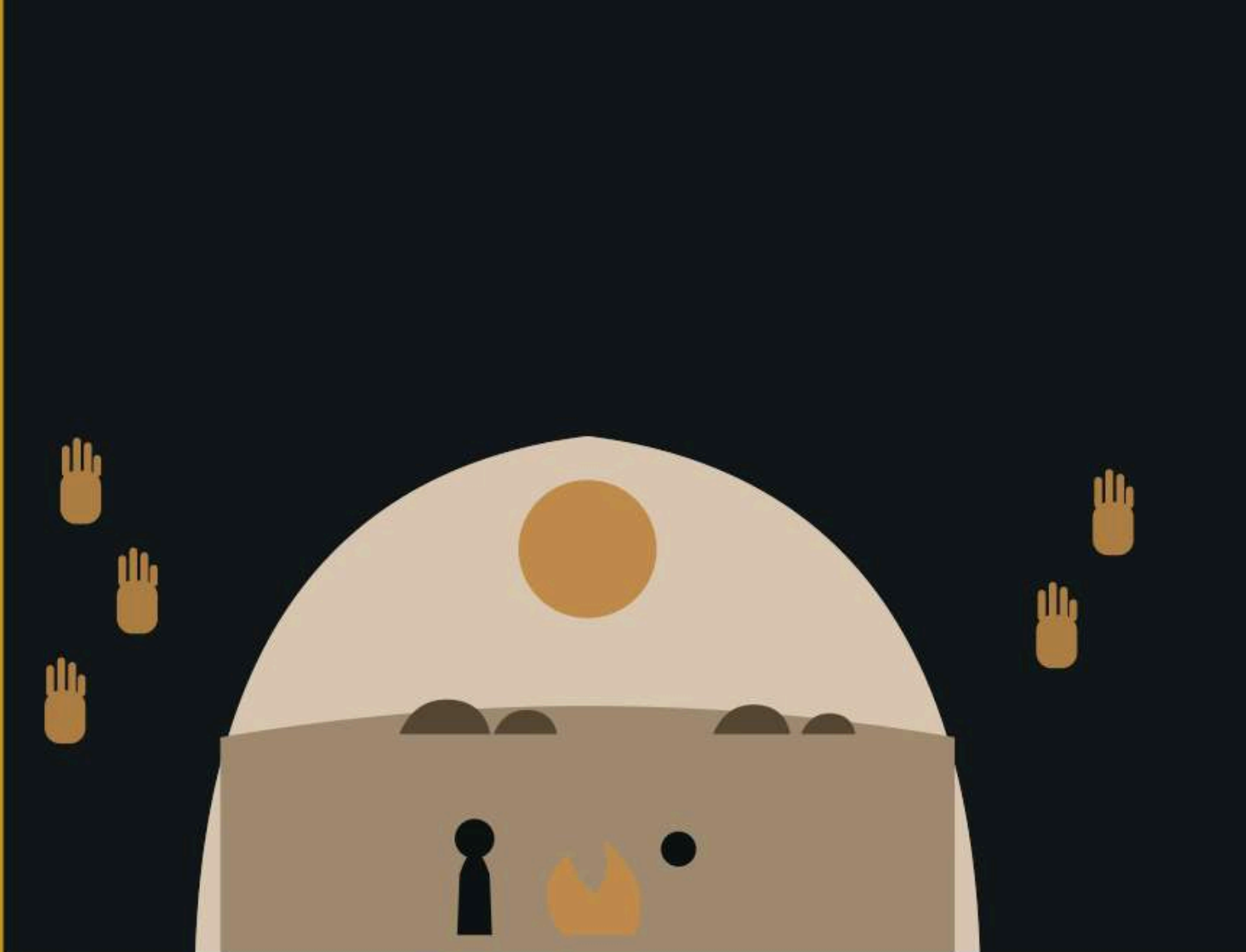
The five oceans

The ocean is seven tenths of the planet. What is it doing for us?

Unit 5

Seven continents

How would you describe a continent you have never visited, and be right?



| A cave mouth, a fire and a painted hand. The hand is a signature: somebody wanted to be remembered.

PART TWO • UNIT 6

6 The first people

Out of Africa, through an ice age, and into the first cities

■ IN THIS UNIT, YOU WILL

- ✓ Trace how our species spread out of Africa across the world.
- ✓ Explain how hunters and gatherers lived through the Ice Age.
- ✓ Explain why farming changed everything, for better and for worse.
- ✓ Describe the first cities, and explain why writing was invented.

UNIT 6 • WHAT YOU WILL LEARN

Where everybody came from



Figure 6.1 • Out of Africa.

Arrows show roughly when people reached each region. The numbers are years ago, and they are approximate: this is a map of a very slow journey.

You will follow our own species out of Africa and across every continent, survive an ice age, and then meet the single change that made everything afterwards possible: farming. The unit ends in the first cities, where somebody invented writing to keep track of grain.

THE BIG QUESTION THIS UNIT ANSWERS

Farming gave people more food and harder lives. Was it worth it?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

6.1 Out of Africa

6.2 Living through the Ice Age

6.3 The great change

6.4 The first cities

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Every single person you have ever met has ancestors who lived in Africa. So do you. So does everybody in your class, whatever they look like and wherever they were born.

That is not a slogan. It is the plain reading of the fossils and of the DNA in living people, and this topic explains how we know it.

TOPIC 6.1

Out of Africa

Where we start

Our species, *Homo sapiens*, appears in the African fossil record around 300 000 years ago. For most of that time, everybody alive was in Africa.

Then, in stages, small groups moved out. The main expansion that leads to everybody living outside Africa today seems to have begun around 70 000 to 60 000 years ago, crossing from north east Africa into south west Asia, and spreading from there.

The dates are approximate, and they shift as new sites are dug up. What does not shift is the direction: out of Africa, and outwards from there.

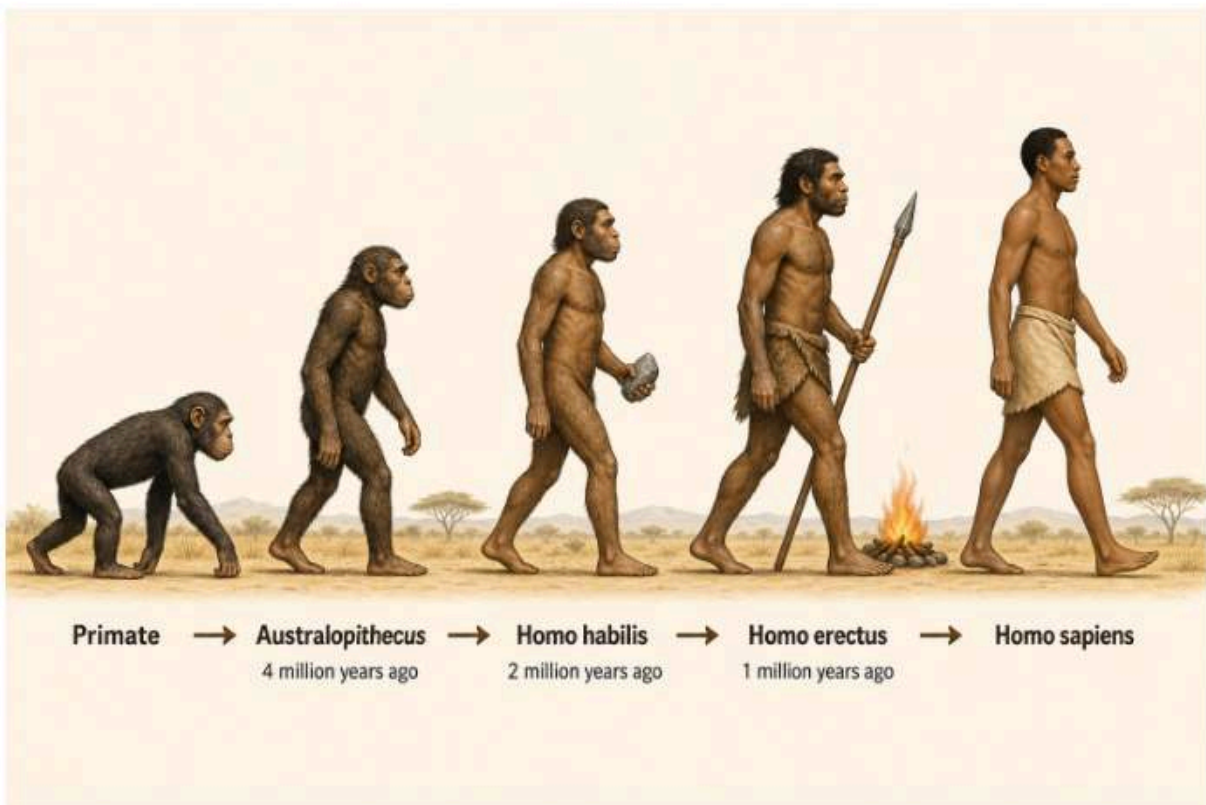


Figure 6.2 · From primate to *Homo sapiens*: the long road to us.

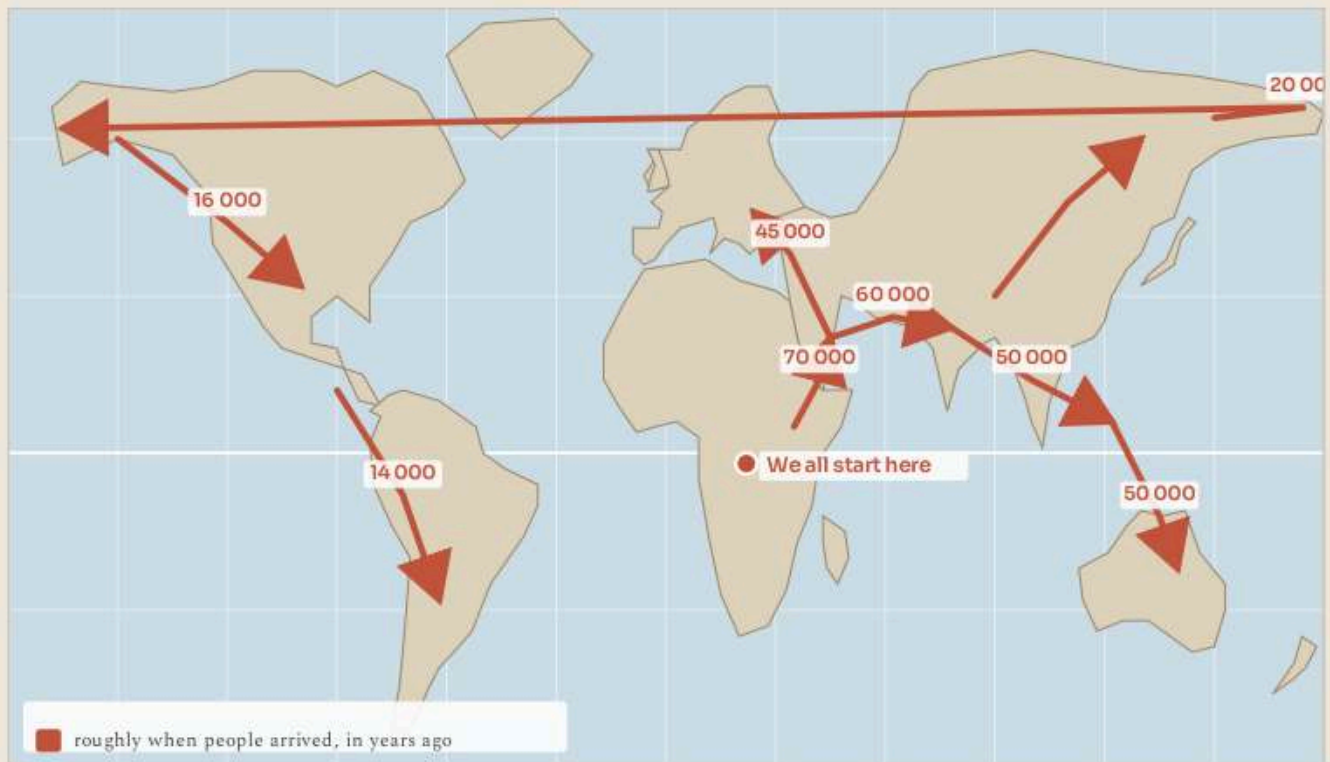


Figure 6.3 - Out of Africa.

Start at the red pin and follow the arrows outwards. The numbers are years ago, and they are approximate: this is a map of a journey that took fifty thousand years.

Three kinds of evidence, again

This is the same detective work as unit 1, and it is worth noticing the pattern.

Fossils. The oldest *Homo sapiens* remains are African. The further from Africa you look, the younger the earliest remains get.

DNA. Genetic variety is greatest in Africa and decreases the further a population is from it. That is exactly what you would expect if every population outside Africa descends from a relatively small group that left.

Tools and objects. Stone tool styles, shell beads and pigments appear in Africa first, then in the places people moved to.

Three separate lines of evidence, gathered by different specialists, pointing the same way. That is what makes a conclusion strong.

◆ CAREFUL

"Out of Africa" does not mean everyone left. It means the people who ended up outside Africa came from there. Africa was never emptied; it has been continuously inhabited by our species for the whole 300 000 years, longer than anywhere else on Earth.

**LOOK FURTHER**

Smithsonian Institution, Human Origins. The fossils themselves, and what each one tells us. Follow the family tree back past the point where it leaves Africa.

humanorigins.si.edu/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- 1 Roughly how long has our species existed, and where did it appear?
- 2 Name the three kinds of evidence for the spread out of Africa.
- 3 Genetic variety is greatest in Africa. Explain why that supports the story.
- 4 Correct this sentence: "About 60 000 years ago, humans left Africa."



Figure 6.4 • Daily life in a cave: fire, tools and the first art.

TOPIC 6.2

Living through the Ice Age

A colder world

For most of the time people were spreading across the planet, the world was in an ice age. The last cold peak was about 20 000 years ago, and it changed the map.

Enormous ice sheets covered northern Europe, northern Asia and much of North America. So much water was locked up as ice that sea level was about 120 metres lower than today. Land that is now sea floor was dry: Britain was joined to Europe, Asia was joined to North America at the Bering land bridge, and it was possible to walk between islands that are now separate.

That lower sea level is a large part of how people reached the Americas and Australia at all.

How people lived

They hunted and they gathered. Groups were small, perhaps twenty to fifty people, and they moved with the seasons, following herds and ripening plants.

They were extremely good at it. They knew which hundred plants in their area were edible and which were poison, when each one fruited, how to track animals, how to make fire, how to sew fitted clothing from hides using bone needles, and how to make stone tools with edges sharper than surgical steel.

They also made art. The cave paintings of western Europe, at Lascaux, Altamira and Chauvet, are between 17 000 and 36 000 years old: horses, bison, lions and rhinoceros drawn with real skill by people who had never seen writing.

■ PORTUGAL AT THIS STAGE

Portugal holds one of the most important rock art sites in the world, and it was very nearly drowned.

In the Coa Valley in the north east, thousands of animal figures were carved into open-air rock faces, many of them between about 22 000 and 10 000 years old: horses, aurochs, ibex and fish. They are outdoors, not in a cave, which is rare and which makes them fragile.

They were found in 1994, during work on a dam that would have flooded the whole valley. After a long public argument the dam was stopped, the valley was protected, and in 1998 it became a UNESCO World Heritage site. It is a useful reminder that heritage is not automatically saved. Somebody has to decide.

**LOOK FURTHER**

Coa Valley Archaeological Park, Portugal. Open-air rock carvings made about 20 000 years ago, on the same river where they still are. A UNESCO World Heritage site.

arte-coa.pt/en/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● **AMAZING BUT TRUE**

Some of the pigment used in European cave paintings was carried tens of kilometres from where it was mined. People were transporting art materials across a landscape, on foot, in an ice age, before anybody farmed anything.

■ **YOUR TURN**

- 1 How much lower was sea level at the peak of the last ice age?
- 2 Give two places that were joined by land then and are separated by sea now.
- 3 List four skills a hunter-gatherer needed.
- 4 Why is the Coa Valley art unusual compared with Lascaux or Altamira?



Figure 6.5 · Humanity during the Ice Age: hunting mammoths in furs and hides.

TOPIC 6.3

The great change

Farming

Around 11 500 years ago, as the climate warmed, something happened in south west Asia that had never happened before: people began deliberately planting the seeds of the plants they wanted, and keeping the animals they ate.

It happened separately in at least eleven different places around the world, with different crops each time: wheat and barley in south west Asia, rice and millet in China, maize, beans and squash in Central America, potatoes in the Andes, sorghum in Africa, taro in New Guinea.

That "separately" matters. Nobody taught anybody. Groups of people who had never heard of each other reached the same idea within a few thousand years, because the same warming climate made it possible in each place.

It made life better, and it made life worse

This is where the topic gets interesting, and where most books stop too early.

	Hunting and gathering	Farming
Food	Whatever the season gives. Varied, and usually enough.	A few crops, in large amounts. More food, less variety.
Moving	The group follows the food through the year.	You stay. The field cannot be carried.
Work	Studies of similar groups suggest a few hours a day.	Ploughing, weeding, watering, storing, all year round.
What it made possible	Small groups, few possessions, no rulers.	Villages, then towns, then kings, taxes, armies and writing.

Figure 6.5 • Before and after.
Read the rows across, not down. Notice that farming wins the first row and loses the third.

Farming produced far more food from the same area of land, so populations grew, and they grew fast. But the food was less varied, so the diet got worse. Skeletons from early farming villages are shorter than the hunter-gatherer skeletons before them, and show more tooth decay and more signs of disease.

Living close together, with animals, also made it far easier for diseases to spread. Most of the epidemic diseases of history came originally from farm animals.

And farming created something new: surplus. If you grow more than you eat, the extra can be stored, which means it can be taken, which means somebody has to decide who takes it. Stored grain is the beginning of taxes, of rulers, of soldiers to defend it and of writing to count it.

● LOOK AGAIN

Was farming a good thing? Read the last two paragraphs again before you answer.

The honest answer is that it depends what you are measuring. If you measure the number of humans alive, farming is the most successful decision our species has ever made: it is why there are eight thousand million of us instead of a few million. If you measure the health, height and working hours of the average individual, early farming made things clearly worse for a very long time.

Both of those are true, and they do not cancel out. In this subject you will often be asked to judge a change, and the first question is always the same: better for whom?

■ YOUR TURN

- 1 When and where did farming begin, according to this book?
- 2 Give three other places where farming began independently, with a crop for each.
- 3 Give two ways farming improved life and two ways it made it worse.
- 4 Explain the link between stored grain and the first rulers.



Figure 6.6 · A lakeside village: houses on stilts, canoes and nets.

TOPIC 6.4

The first cities

Mesopotamia

Between the Tigris and the Euphrates, in what is now Iraq, farming villages grew into something new. By about 3500 BC there were cities: Uruk, Ur, Lagash and others, some with tens of thousands of people.

The land there is fertile but dry, so the farming depended on irrigation: digging and maintaining canals to bring river water to the fields. Irrigation is not something a family can do alone. It needs many people to work together, agree who gets water when, and keep the canals clear. Organisation on that scale needs somebody in charge, and that is how cities acquire governments.



Figure 6.7 - Where the first cities appeared.

Four separate places, four different dates. None of them taught the others.

And then somebody invented writing

Around 3300 BC, in Uruk, scribes began pressing marks into wet clay tablets with a cut reed. We call the script cuneiform, which means wedge-shaped.

Here is the part pupils rarely expect. The earliest writing is not poetry, prayer or history. It is accounts. Lists of grain, sheep, beer and oil, and who owed what to the temple. Writing was invented to keep track of things, and only later did anybody use it to tell a story.

That single invention changes what history even means. Before writing we have objects, and we guess. After writing we have people telling us, in their own words, what they thought was happening. Everything before is called prehistory, and everything after is history, and the line between them is a clay receipt for some barley.

● AMAZING BUT TRUE

One of the oldest complaint letters ever found is a cuneiform tablet from about 1750 BC, in which a customer named Nanni tells a copper merchant named Ea-nasir that the copper he delivered was poor quality, that his messenger was treated rudely, and that he wants his money back. It is nearly four thousand years old and it reads exactly like an angry email.

■ PORTUGAL AT THIS STAGE

While the first cities were rising in Mesopotamia, the people living on the Atlantic coast of Iberia were doing something quite different and equally remarkable: they were moving enormous stones.

Near Evora stands the Almendres Cromlech, a great oval of standing stones begun around 6000 BC and rearranged over thousands of years. It is older than Stonehenge and older than the pyramids. The region has hundreds of megalithic tombs, including the Anta Grande do Zambujeiro, one of the largest in Europe.

These were farming communities without cities, without writing and without kings, and they were still able to organise hundreds of people to quarry, move and raise stones weighing many tonnes. Cities are one way of being organised. They are not the only one.

■ YOUR TURN

- 1 Where and roughly when did the first cities appear?
- 2 Explain the link between irrigation and organised government.
- 3 What was the earliest writing used for? Why is that surprising?
- 4 Explain the difference between prehistory and history.
- 5 The Almendres Cromlech was built by people with no cities and no writing. What does that tell you about what a society needs in order to achieve something big?

■ WHAT YOU CAN NOW DO

- ✓ Trace how our species spread out of Africa, and give the evidence.
- ✓ Explain how people lived through the Ice Age, and what the lower sea level made possible.
- ✓ Explain why farming began, and judge it fairly on both sides.
- ✓ Describe the first cities and explain why writing was invented.

■ WORDS TO KEEP

hunter-gatherer, ice age, land bridge, irrigation, surplus, cuneiform, prehistory, megalith



Pyramids, palms and the Nile. Everything green in this picture is there because of the river; everything beyond it is desert.

PART TWO • UNIT 7

7 Egypt, the gift of the Nile

Three thousand years of one civilisation, all of it within walking distance of one river

■ IN THIS UNIT, YOU WILL

- ✓ Explain the flood, the black land and the red land.
- ✓ Describe how Egypt was governed, from pharaoh down to farmer.
- ✓ Explain what Egyptians believed, and why it shows in their tombs.
- ✓ Read a cartouche, and explain how the code was cracked.
- ✓ Explain how the pyramids were built, and reject two popular myths.

UNIT 7 • WHAT YOU WILL LEARN

One river, three thousand years

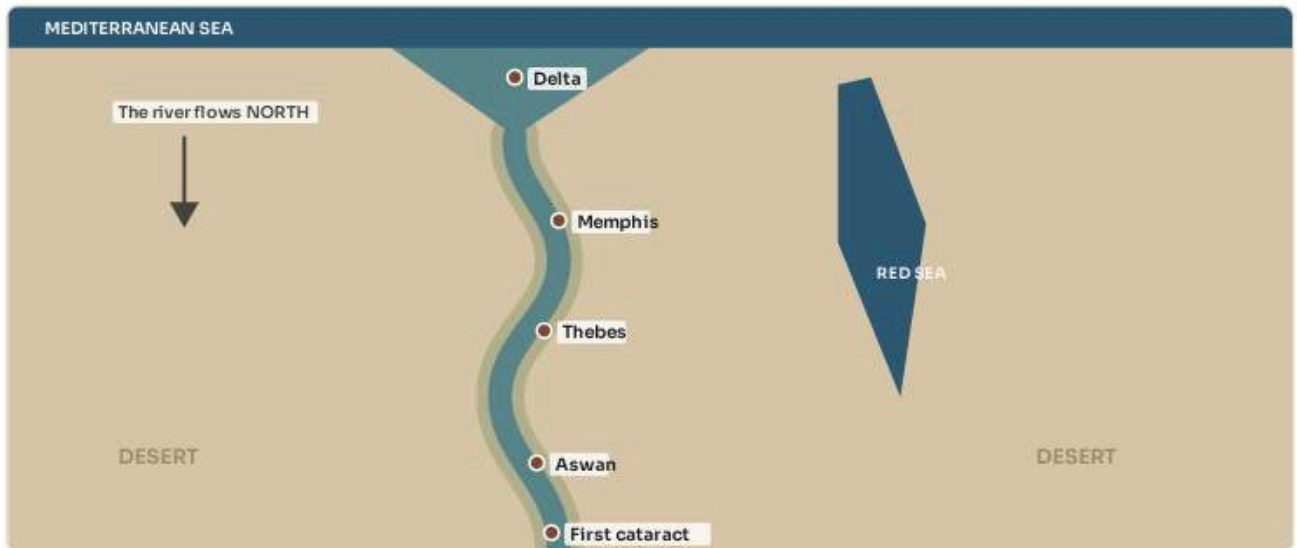


Figure 7.1 • Egypt is the Nile.

Almost every Egyptian lived within sight of this river. Notice which way it flows, and where it fans out at the end.

You will find out why a country could grow in a desert, meet the pharaoh and everybody underneath him, work out what the Egyptians believed and why their tombs prove it, learn to read a name in hieroglyphs, and finish by building a pyramid without a single machine.

THE BIG QUESTION THIS UNIT ANSWERS

How did a country in a desert last for three thousand years?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

7.1 The river that made a country

7.2 Pharaoh and the pyramid of people

7.3 Gods, temples and the afterlife

7.4 Hieroglyphs and the scribe

7.5 Building the pyramids

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Ancient Egypt lasted about three thousand years as a recognisable civilisation.

To feel how long that is, count backwards. Three thousand years before today is 1000 BC, which is before Rome was a city worth mentioning. Cleopatra, the last ruler of ancient Egypt, lived closer in time to the first Moon landing than to the building of the Great Pyramid.

How does anything last that long? That is the question this unit answers, and the answer is mostly a river.

TOPIC 7.1

The river that made a country

A green line through a desert

Egypt is almost entirely desert. Yet for three thousand years it was one of the richest places on Earth, and the reason runs straight through the middle of it.

The Nile is about 6 650 kilometres long and flows **north**, from the highlands of east Africa to the Mediterranean. That surprises people, because north is up on a map, but rivers flow downhill, not down the page.

Every year, summer rains far to the south swelled the river until it overflowed its banks in Egypt, from about June to September. When the water went down it left behind a layer of dark, fertile silt.

The Egyptians called that strip **kemet**, the black land: a ribbon of superb farmland, in places only a few kilometres wide. Beyond it was **deshret**, the red land, which was desert and where nobody could live. Almost every Egyptian in three thousand years lived within sight of the river.



Figure 7.2 · Hapi, god of the Nile.



The year began with the flood, not with a date somebody chose. The river set the calendar.

Figure 7.3 · The Egyptian year, set by the river.

Read the three seasons round the circle, starting at the top. The year did not begin on a chosen date; it began when the water rose.

What a river does for a country

The Nile did four jobs at once, and it is worth separating them.

It **fed** people: the silt renewed the soil every year, so Egyptian fields did not wear out the way fields elsewhere did.

It **watered** them, in a country with almost no rain.

It **carried** them. The current runs north, and the prevailing wind blows south, so you could drift downstream and sail back up. A boat could go either way without much effort, which made Egypt easy to hold together as one country.

And it **protected** them. Desert on both sides, sea to the north, cataracts (rocky rapids) to the south. Egypt was hard to invade, which is a large part of why it lasted.

◆ CAREFUL

Upper Egypt is in the **south** and Lower Egypt is in the **north**. That is not a mistake. They are named for the river: upstream and downstream. Get this right and Egyptian history makes sense; get it wrong and every map confuses you.

■ YOUR TURN

- 1 Which way does the Nile flow? Explain why that is not strange.
- 2 What were kemet and deshret, and where was each?
- 3 Give the four jobs the Nile did for Egypt, and explain each in a sentence.
- 4 Why is Upper Egypt in the south?

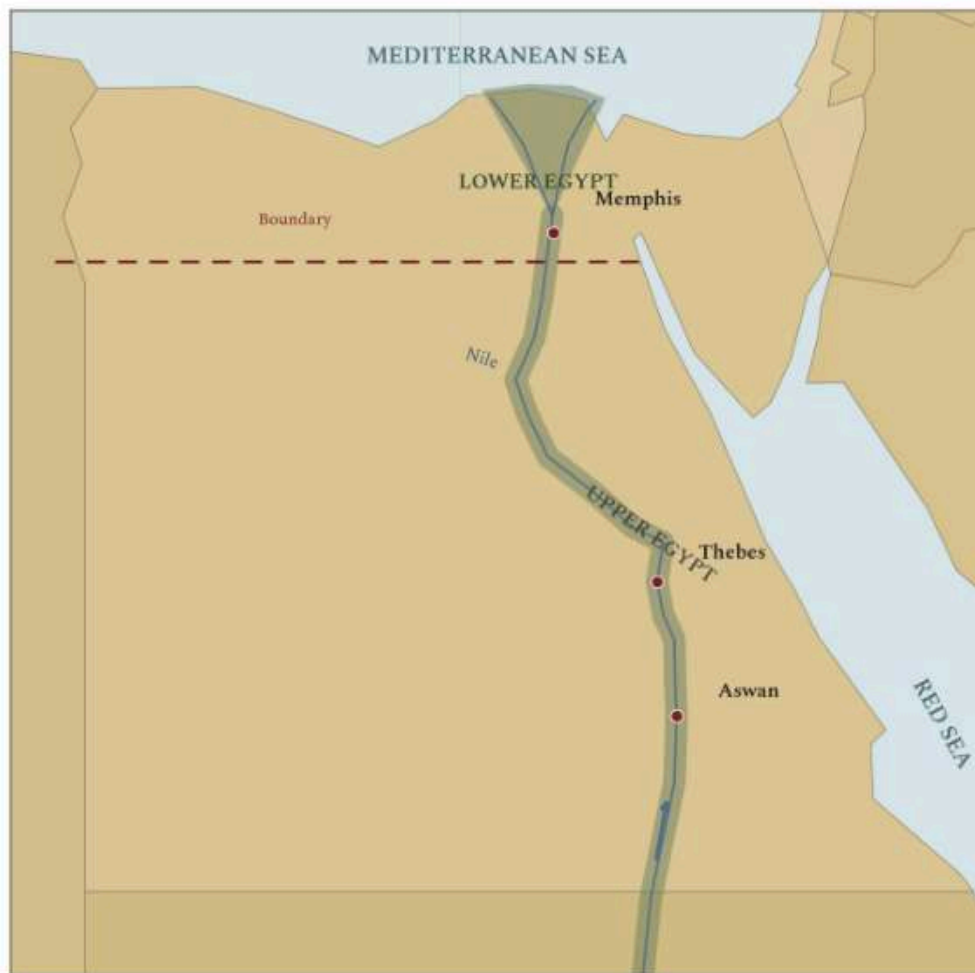


Figure 7.4 • Upper and Lower Egypt, and the river that joins them.

TOPIC 7.2

Pharaoh and the pyramid of people

One person at the top

Egypt was ruled by a pharaoh, and a pharaoh was not simply a king. He, or occasionally she, was believed to be a god on Earth, the link between the gods and the people, personally responsible for keeping the world in order.

That belief did a great deal of practical work. If the harvest was good, the pharaoh was doing the job. If the country was invaded or the flood failed, people concluded the pharaoh had failed, and dynasties ended.

Egypt was united in about 3100 BC, and it was ruled by a long series of dynasties after that. Not all of them were men: **Hatshepsut** ruled for about twenty years from 1479 BC, and Egypt prospered under her.

Pharaoh

King, and also a god on Earth. Owned the land in theory.

Officials and priests

Ran the country, the temples and the granaries.

Scribes

Could read and write, so they counted everything.

Craftworkers and soldiers

Built, carved, made and fought.

Farmers

Nine people in ten. They fed everybody above them.

Figure 7.5 · **Egyptian society, from the top down.**

Read every box, then count them: nine people in ten are in the bottom one.

Who did what

Below the pharaoh were officials and priests, who ran the country and the temples. Below them were scribes, who could read and write and therefore counted everything: harvests, taxes, workers, cattle.

Below them came craftworkers and soldiers. And at the bottom, about nine people in every ten, were farmers, who grew all the food that everybody above them ate.

For part of the year, during the flood, the fields were under water and farmers could not farm. That is when they worked on the pharaoh's building projects. This is important, and it corrects a very common belief: the pyramids were not built by slaves. They were built by paid, fed, housed Egyptian workers, many of them farmers with nothing else to do that season. Archaeologists have found their villages, their bakeries, their medical care and their tombs.

● AMAZING BUT TRUE

The workers' town beside the Giza pyramids has been excavated, and the bones show something remarkable: many workers had healed broken limbs, set properly by someone who knew what they were doing. These were valued workers, treated well enough to be nursed back to health.

■ YOUR TURN

- 1 What did Egyptians believe a pharaoh was?
- 2 Explain how that belief could bring a dynasty to an end.
- 3 List the five levels of Egyptian society, top to bottom.
- 4 Why could farmers work on building projects for part of each year?
- 5 Give one piece of archaeological evidence that pyramid workers were not slaves.



Figure 7.6 · The funerary mask of Tutankhamun.

TOPIC 7.3

Gods, temples and the afterlife

Many gods, and one big idea

The Egyptians had hundreds of gods. **Ra** the sun god, who crossed the sky each day. **Osiris**, god of the dead and of rebirth. **Isis**, his wife, a goddess of magic and protection. **Anubis**, with the head of a jackal, who looked after the dead. **Thoth**, with the head of an ibis, who invented writing.

Behind them all stood one idea: **maat**. It is usually translated as truth, balance, order, justice, and it means all of those at once. Maat was the way the world was supposed to be: the river rising, the sun returning, the strong not robbing the weak. The pharaoh's central duty was to uphold maat, and every Egyptian was expected to live by it.

Why they built for the dead

Egyptians believed that death was a passage rather than an ending, and that a person needed their body, their name and their possessions to continue.

So they preserved bodies by mummification: removing the organs, drying the body with natron salt for about forty days, then wrapping it in linen. They buried the dead with food, furniture, tools and jewellery. And they carved the person's name, because a name that was spoken was a person who continued.

At the judgement, the Egyptians believed, the dead person's heart was weighed against the feather of maat. If the heart was heavy with wrongdoing, it failed. If it balanced, the person went on.

◆ CAREFUL

It is easy to describe Egypt as a civilisation obsessed with death. That gets it backwards. Almost everything that survives from Egypt is from tombs and temples, because those were built of stone in the dry desert, while houses, including the pharaoh's, were built of mud brick beside the river and have washed away. We are not seeing what Egyptians cared about most. We are seeing what happened to last.

■ YOUR TURN

- 1 Name four Egyptian gods and say what each was responsible for.
- 2 Explain what maat meant, in your own words.
- 3 Why did Egyptians preserve bodies and record names?
- 4 Explain why so much of what survives from Egypt comes from tombs.

TOPIC 7.4

Hieroglyphs and the scribe

A writing system that does three jobs at once

Egyptian hieroglyphs look like pictures, and some of them are. But the system is cleverer than that, and it works in three ways at the same time.

Some signs stand for a **sound**, rather like our letters. Some stand for a whole **word** or idea. And some are silent **determinatives**, added at the end of a word to tell you which kind of thing is meant, which is useful in a script that does not write vowels.

A royal name was written inside an oval called a **cartouche**, meaning "everything the sun goes round". That oval turned out to be the key that unlocked the whole language.

Cleopatra, written in hieroglyphs

K	L	E	O	P	A
a bowl seen from the side	a lion lying down	a reed	a loop of rope	a mat	an eagle

The oval means "everything the sun goes round". Only a royal name is written inside one.

Figure 7.7 • A royal name in a cartouche.
Read the signs left to right, then read what each one is underneath. Notice that the pictures are being used for their sounds, not their meanings.

How the code was cracked

By about 400 AD nobody could read hieroglyphs any more, and the knowledge was lost for fourteen hundred years.

In 1799 French soldiers in Egypt found a slab of stone at Rosetta carrying the same decree in three scripts: hieroglyphs, a later Egyptian script called demotic, and Greek. Scholars could read Greek. So they had, in effect, an answer key.

Even so it took more than twenty years. The breakthrough came in 1822 from Jean-Francois Champollion, who realised something others had missed: hieroglyphs were not purely picture-writing. Many signs recorded **sounds**. He tested the idea on the cartouches, where he could guess the royal names

from the Greek, and it worked.

● AMAZING BUT TRUE

Champollion had an advantage nobody else had. He had learned Coptic, the language used in Egyptian churches, which is the direct descendant of ancient Egyptian written in Greek letters. When he sounded out a cartouche, he recognised words. The language had never really died; it had only changed its alphabet.



LOOK FURTHER

Institute for the Study of Ancient Cultures, University of Chicago. A working Egyptology institute, with its own photographs of tombs, temples and hieroglyphs. Try writing your name from its sign list.

isac.uchicago.edu/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- 1 Give the three jobs that hieroglyphic signs can do.
- 2 What is a cartouche, and why was it useful to scholars?
- 3 Explain what the Rosetta Stone is and why it mattered.
- 4 What was Champollion's key insight?
- 5 Why did knowing Coptic help him?



Scarab amulet

Figure 7.8 · A scarab amulet.

TOPIC 7.5

Building the pyramids

The Great Pyramid, by the numbers

The Great Pyramid at Giza was built for the pharaoh Khufu around 2560 BC. It was 146.6 metres tall when finished, and it remained the tallest building anywhere on Earth for about 3 800 years.

It contains roughly 2.3 million blocks of stone, most weighing between 2 and 15 tonnes. Its base is level to within a couple of centimetres across more than five acres, and its sides are aligned to true north to within a fraction of a degree. It was built in about twenty years.

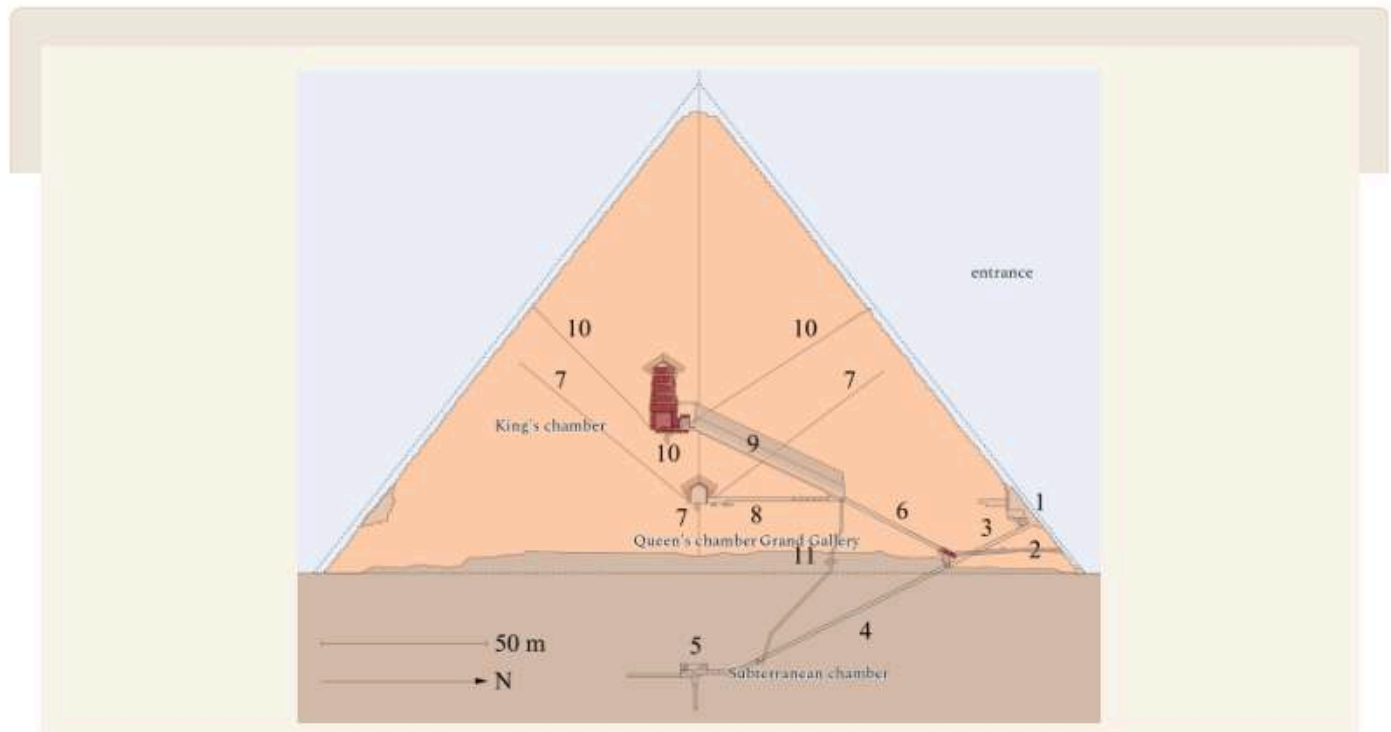


Figure 7.9 · Inside the Great Pyramid.

Follow the passage from the entrance on the left. There are three chambers, and only one of them was ever used.

How, without machines

Nobody has a complete answer, but the main pieces are well supported.

Quarrying. Most of the stone came from a quarry a few hundred metres away. The fine white limestone facing came from across the river, and the granite from Aswan, 900 kilometres upstream, floated down on barges during the flood.

Moving. Blocks were dragged on wooden sledges. A wall painting from a tomb shows a huge statue being pulled on a sledge with a man pouring water onto the sand in front of it. That was long thought to be a religious gesture. Physics experiments have since shown that wetting sand to the right dampness

roughly halves the friction, so it was engineering.

Lifting. Some kind of ramp system, almost certainly, though whether it was one long straight ramp, a spiral around the outside, or ramps inside the structure is still argued about.

Organising. This is the part that impresses engineers most. Feeding, housing, paying and rotating a workforce of tens of thousands for twenty years, and delivering the right stone to the right place on time, is a management achievement as much as a building one.

◆ CAREFUL

Two things you may have heard, and why both are wrong.

"Slaves built the pyramids." No. The workers' town, bakeries, breweries, medical care and tombs have been excavated. They were Egyptian workers, fed and paid in bread and beer, and proud enough of the work to be buried in its shadow.

"It could not have been done without help from elsewhere." This claim always appears about monuments built by people the claimant does not expect to be capable. It is not a theory about pyramids; it is an insult to Egyptians. We have the quarries, the tools, the ramps, the workers' town and the pay records.

■ PORTUGAL AT THIS STAGE

While Khufu's pyramid was rising, the people of the Iberian peninsula were building too, though on a smaller scale and in a different style.

This was the time of the great megalithic tombs of the Alentejo, and of the first fortified settlements. At Los Millares in the south east of the peninsula and, a few centuries later, at Zambujal near Torres Vedras, communities built walled settlements with towers, working in copper. Zambujal's walls, begun around 2600 BC, are almost exactly the same age as the Great Pyramid.

Nobody in Iberia was building anything the size of Giza. But the same century that raised the pyramid saw organised communities on the Atlantic coast quarry stone, raise walls and work metal. Both are on the same line of time.

◆ WATCH ME WORK IT OUT

Question. The Great Pyramid contains about 2.3 million blocks and took about twenty years. How many blocks had to be placed per day, if work went on every day?

Thinking it through. First, get everything into the same units. Twenty years, at 365 days each, is 7 300 days.

Then divide: 2 300 000 divided by 7 300.

That is about 315.

Then sanity check it. Is one block every four and a half minutes, all day, every day for twenty years, believable? Only if a great many teams were working at once on different parts of the structure, which is what the archaeology shows.

Answer. About 315 blocks a day, which is roughly one every five minutes in a twenty-four hour day. The number is only possible with many teams working in parallel, which tells you that the hard part was organisation, not lifting.

■ YOUR TURN

- 1 When was the Great Pyramid built, and how tall was it?
- 2 For how long was it the tallest building on Earth?
- 3 Explain how wetting the sand helped move blocks.
- 4 Give two claims about the pyramids that this book says are wrong, and the evidence against each.
- 5 What does this book say was the hardest part of building a pyramid?

■ WHAT YOU CAN NOW DO

- ✓ Explain the Nile flood, the black land and the red land.
- ✓ Describe how Egypt was governed, from pharaoh to farmer.
- ✓ Explain what Egyptians believed, and why so much of it survives in tombs.
- ✓ Read a cartouche and explain how hieroglyphs were deciphered.
- ✓ Explain how the pyramids were built, and reject two popular myths.

■ WORDS TO KEEP

pharaoh, silt, cataract, maat, mummification, hieroglyph, cartouche, determinative, dynasty

UNIT 7 AT A GLANCE

Egypt, the gift of the Nile

How did a country in a desert last for three thousand years?

THE TOPICS YOU HAVE COVERED

- 7.1 The river that made a country
- 7.2 Pharaoh and the pyramid of people
- 7.3 Gods, temples and the afterlife
- 7.4 Hieroglyphs and the scribe
- 7.5 Building the pyramids

WHAT YOU CAN NOW DO

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THE WORDS THIS UNIT TAUGHT

pharaoh

silt

cataract

maat

mummification

hieroglyph

cartouche

determinative

dynasty



Ancient Egypt in one emblem: ankh, Eye of Horus, scarab, lotus, pyramids and the Nile.

able to explain how, from pharaoh to farmer, and why so much of it survives in tombs on the right, go back to the topic that taught it, not to the answers.



| The Great Wall along a ridge. It was never one wall, and it was never only about keeping people out.

PART TWO • UNIT 8

8 Ancient China

The oldest state still on the map, and four inventions that changed everybody else

■ IN THIS UNIT, YOU WILL

- ✓ Explain why Chinese civilisation began between two rivers.
- ✓ Put the main dynasties in order on one timeline.
- ✓ Compare two Chinese ideas about how to run a country.
- ✓ Explain four inventions, and what each one changed.
- ✓ Explain what the Silk Road and the Great Wall were really for.

UNIT 8 • WHAT YOU WILL LEARN

Two rivers, many dynasties



Figure 8.1 • China, its two great rivers and the Silk Road.

The dashed line is not a road: it is a chain of routes along which goods were passed from trader to trader.

You will find out why Chinese civilisation began where it did, put three thousand years of dynasties on one timeline, compare two very different ideas about how to run a country, and meet four inventions that eventually changed the whole world.

THE BIG QUESTION THIS UNIT ANSWERS

Why did four inventions from one country change the whole world?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

- | | |
|--|---------------------------------------|
| 8.1 Two great rivers | 8.2 Three thousand years of dynasties |
| 8.3 Confucius and how to run a country | 8.4 Four inventions |
| 8.5 The Silk Road and the Great Wall | |

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before you scan.

● LOOK FIRST

Look at the four things below. Paper. Printing. The compass. Gunpowder.

Now imagine removing all four from the world. No books, no newspapers, no cheap records, no sailing out of sight of land, no fireworks, no guns.

All four were invented in China, and China had them for centuries before anybody else. This unit is about the country that produced them, and about why it took so long for the rest of the world to catch up.

TOPIC 8.1

Two great rivers

Where and why

Chinese civilisation began along two rivers that run west to east across the country.

The **Huang He**, or Yellow River, is the northern one. It gets its name and its colour from loess: a fine yellow soil, blown there by wind over thousands of years, easy to dig and very fertile. The first Chinese states grew up along it, farming millet and wheat.

The **Chang Jiang**, or Yangtze, is the southern one and the longest river in Asia at about 6 300 kilometres. It is warmer and wetter there, which suits rice.



Figure 8.1 · The rivers of China: the Huang He in the north, the Chang Jiang in the south.



Figure 8.2 · The giant panda, China's most famous animal, eating bamboo.

China's Sorrow

The Yellow River carries so much silt that it drops it on the river bed, which slowly rises. People build banks to hold the river in; the bed rises further; the banks are raised again. In places the river now runs above the level of the surrounding fields, held in by its own banks.

When those banks fail, the flood is catastrophic and the river can shift its whole course by hundreds of kilometres. It has done so many times, and it has killed enormous numbers of people. It was known as China's Sorrow.

This produced something politically important. Controlling that river needed work on a scale no village could manage: dykes, dredging and canals, organised across whole regions. A strong central government that could organise flood control was not a luxury in China; it was the difference between eating and starving. The very earliest Chinese legends are about a hero who tamed the floods, and they were remembered for a reason.

■ YOUR TURN

- ① Name China's two great rivers, and the crop associated with each.
- ② What is loess, and why does it matter?
- ③ Explain why the Yellow River's bed rises over time.
- ④ Explain the link between flood control and strong central government in China.



Figure 8.2 · Left: the Huang He in the loess gorges. Right: the Chang Jiang among green mountains.

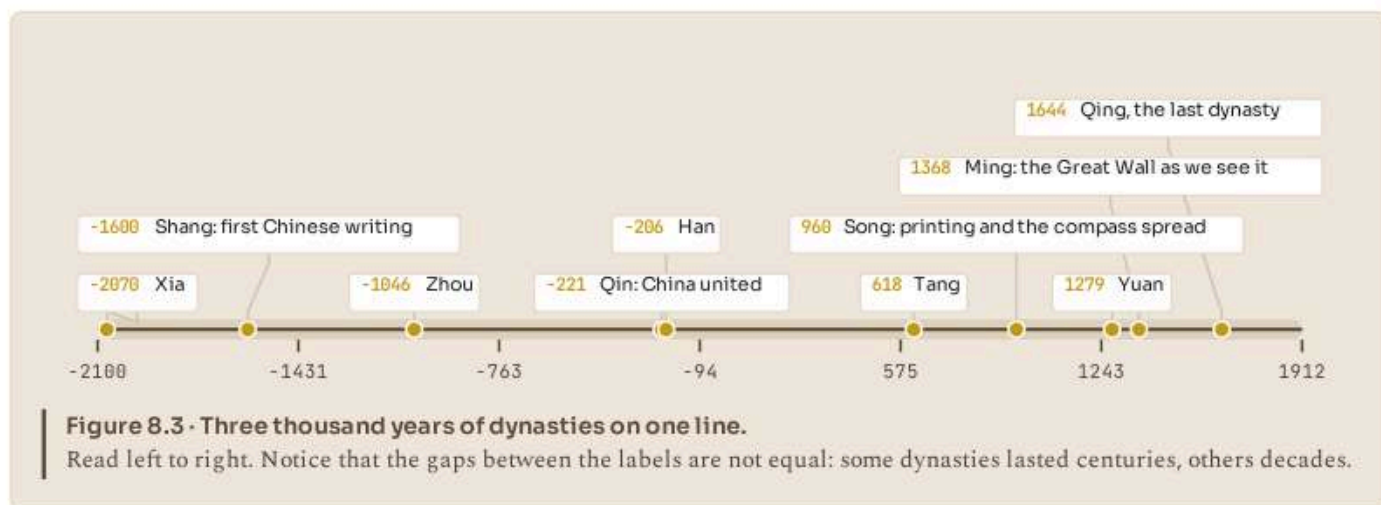
TOPIC 8.2

Three thousand years of dynasties

One country, many families

Chinese history is usually told in dynasties: periods when one family ruled. The pattern repeats often enough that Chinese historians gave it a name.

A dynasty takes power, restores order, and the country prospers. Over generations the rulers grow distant and the government grows expensive. Taxes rise, floods or famines are handled badly, rebellions start, and eventually the dynasty falls and a new one takes over, restoring order.



Four are worth remembering now.

The **Shang**, from about 1600 BC, gave us the first Chinese writing, scratched onto ox bones and turtle shells used for fortune telling. That script is the direct ancestor of the Chinese writing used today, which makes it the longest-lived writing system on Earth.

The **Qin**, from 221 BC, united China for the first time under one emperor, Qin Shi Huang. He standardised the writing, the coins, the laws and even the axle width of carts, so that roads would fit all vehicles. China gets its name from the Qin. He is also the emperor buried with the Terracotta Army.

The **Han**, from 206 BC, lasted four hundred years, opened the Silk Road, and was so successful that the majority ethnic group in China still calls itself Han today.

The **Song**, from 960, was a period of extraordinary invention: printing, paper money, gunpowder weapons and the magnetic compass all spread widely under the Song.

The Mandate of Heaven

Chinese rulers explained their right to rule with an idea called the Mandate of Heaven. Heaven gave a just ruler the right to govern; if a ruler became cruel or incompetent, Heaven withdrew it, and rebellion became legitimate.

It is worth comparing with Egypt. Both said the ruler's authority came from beyond the human world. But the Egyptian pharaoh was a god, while the Chinese emperor was a man holding a licence that could be revoked. That built a right of rebellion into Chinese political thinking, which is unusual and which lasted for two thousand years.

■ PORTUGAL AT THIS STAGE

While the Han dynasty was ruling China, the north west of the Iberian peninsula was covered in hill forts.

The Castro culture built fortified settlements of round stone houses on hilltops, ringed by walls: places like Citania de Briteiros and Sanfins, in what is now northern Portugal. Several hundred are known. They were farming and metal-working communities, trading tin and gold, and they were still there when Rome arrived.

So while one end of the same landmass had a unified empire of perhaps sixty million people, a professional civil service and paper, the other end had independent hill communities of a few hundred people each. Both are the same century. Neither had any idea the other existed.



LOOK FURTHER

Columbia University, Asia for Educators. Chinese history explained period by period, with sources you can read yourself.

afe.easia.columbia.edu/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

■ YOUR TURN

- 1 Describe the dynastic cycle in your own words.
- 2 Give one important thing about each of the Shang, Qin, Han and Song dynasties.
- 3 What was the Mandate of Heaven?
- 4 Compare the Chinese emperor with the Egyptian pharaoh. Give one similarity and one difference.

TOPIC 8.3

Confucius and how to run a country

Two answers to one question

Around 500 BC China was divided and at war, and thinkers asked a practical question: how do you make people behave well enough for a country to function?

Two very different answers competed, and the argument between them shaped Chinese government for two thousand years.

	Confucius	The Legalists
People are	Good, or can be made good by example.	Selfish, and will only behave if it pays to.
So a ruler should	Set an example, and be worth following.	Write clear laws, and enforce them hard.
Education	Essential. Anyone able should be able to rise.	Useful only where it serves the state.
Which won?	The exams, the officials and the family rules, for 2,000 years.	The unification of China in 221 BC, and much of its law.

Figure 8.4 · Two answers to the same question.
Read across each row. The last row is the one that decides the argument, and it is not a clean win for either side.

Confucius, who lived from about 551 to 479 BC, said people are basically good, or can be made good. What matters is example and education. A ruler should behave well, and people will copy him. Relationships carry duties in both directions: a father owes his son care as much as a son owes his father respect. Education should be open to anyone with ability, not only to the well-born.

The Legalists disagreed completely. People act in their own interest. Example is useless; what works is clear written law with heavy punishments and real rewards. A strong state does not need good people, only predictable ones.

Who won

Both, in a way, and this is the useful lesson.

The Legalists won first. Qin Shi Huang united China using their methods, and they worked: he built roads, standardised everything and imposed order. But the Qin dynasty was harsh, and it collapsed within fifteen years of his death.

Confucius won in the long run. From the Han dynasty onwards, Confucian ideas shaped Chinese government, and one of them had extraordinary consequences: entry to the civil service by written examination. From about 600 AD, China chose many of its officials by exam rather than by birth. It was not fully open, because years of study cost money, but compared with a Europe where office was inherited, it was a genuinely different idea about who should govern.

● AMAZING BUT TRUE

The Chinese imperial examinations ran, on and off, for more than 1 300 years, from 605 to 1905. Candidates were locked into individual cells for up to three days at a time to write. Cheating was punished with extreme severity, and one recorded method of cheating was a shirt with the entire Confucian classics written on it in miniature characters.

■ YOUR TURN

- 1 Summarise Confucius's view of people and of rulers in two sentences.
- 2 Summarise the Legalist view.
- 3 Which side won first, and what happened to them?
- 4 Explain the civil service examination and why it was unusual.



Figure 8.3 - Confucius, teacher and government adviser.

TOPIC 8.4

Four inventions

What they were, and what they changed

Chinese tradition speaks of four great inventions. What makes them worth studying is not that they were clever, but what each one did to the world afterwards.



Paper, recorded from about 105 AD, was cheap. Before it, writing went on bamboo strips, silk, clay or parchment made from animal skin: all heavy, or expensive, or both. Paper made records ordinary, and it made libraries possible.

Printing came later: carved wooden blocks from about 700 AD, then movable type in the eleventh century. One carved page could produce thousands of identical copies, so ideas could spread without being recopied by hand and corrupted a little each time.

The compass was first used for fortune telling, and by about 1000 AD for navigation. Before it, sailors mostly hugged coastlines, because losing sight of land meant losing direction under cloud. A compass makes open ocean sailing possible, and everything in the age of exploration follows from that.

Gunpowder appears around 900 AD, discovered by alchemists looking for a medicine of immortality, which it was not. Fireworks came first, then rockets, bombs and guns. It made stone castle walls obsolete, and with them a whole way of organising society.

◆ CAREFUL

"The Chinese invented gunpowder but only used it for fireworks" is a popular claim, and it is false. Chinese armies were using gunpowder bombs, fire lances and rockets in warfare from the eleventh century, centuries before European guns. Fireworks came first; weapons followed quickly.

■ YOUR TURN

- 1 Name the four great inventions and give a date for each.
- 2 For each one, write a sentence explaining what it changed.
- 3 Explain why the compass mattered more to sailors than a map did.
- 4 Correct the claim that gunpowder was only used for fireworks in China.



Figure 8.4 · More inventions from China: silk, the wheelbarrow, the hot air balloon and the seismoscope.

TOPIC 8.5

The Silk Road and the Great Wall

The road that was not a road

The Silk Road was not one road, and almost nobody travelled its whole length.

It was a network of routes running about 6 400 kilometres between China and the Mediterranean, across deserts and mountains, and goods moved along it by being sold from trader to trader. Silk leaving Chang'an might pass through a dozen pairs of hands before reaching Rome, and each pair took a share of the price. That is why silk was worth its weight in gold at the far end.



Figure 8.6 • The Silk Road.

Follow the route from Chang'an westwards. Remember that the line represents a chain of traders, not a journey anyone made.

Goods went both ways. Out of China: silk, paper, porcelain, tea. Into China: horses, glass, gold, wool, spices. But the important cargo was not goods at all.

Ideas travelled. Buddhism reached China along these routes and became one of its major religions. Paper-making spread west, reaching the Islamic world in the eighth century and Europe centuries later. Astronomy, mathematics and medicine moved in both directions.

And so did disease. The Black Death, which killed perhaps a third of Europe's population in the fourteenth century, is generally thought to have travelled west along these same routes. A network that carries silk carries everything else too.

The wall that was many walls

The Great Wall is not one wall either, and it was not built all at once. Different dynasties built different walls in different places over roughly two thousand years. What tourists visit today is mostly Ming dynasty work from the fifteenth and sixteenth centuries.

Its purpose is usually misunderstood. It was never intended to be an impenetrable barrier; a determined army could always get through or round it. What a wall does is force raiders to come through the gates, and raiders who have to use gates can be taxed, counted, controlled or fought at a place of the defender's choosing. A wall with a garrison is not a shield. It is a filter, and a signalling system: beacon towers could pass a warning along the frontier far faster than a horse.

● AMAZING BUT TRUE

The claim that the Great Wall is the only human structure visible from space is false, and it has been publicly denied by astronauts who have looked. The wall is long but narrow, roughly the width of a road, and the same colour as the ground it stands on. Motorways and cities are far easier to see.



LOOK FURTHER

University of Washington, Silk Road Seattle. Maps, objects and texts from along the route. Notice how few people travelled the whole way.

depts.washington.edu/silkroad/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● LOOK AGAIN

Look at the Silk Road map again, then at the map of the first cities in unit 6.

Both maps show the same band of the world: from the Mediterranean across to China. In unit 6 that band is where cities and writing appear first. Here it is where goods, faiths and diseases travel.

That is not a coincidence. This is the widest continuous stretch of temperate land on Earth, running east to west, so crops, animals and people could move along it without needing to adapt to a new climate. Africa and the Americas run north to south, so anything moving along them crosses climate bands and has to start again. Geography made that band the busiest corridor in human history, and it is still busy.



Figure 8.7 · The Great Wall along a ridge.



| The Andes, and terraces cut into them. Every one of those steps is a field somebody built by hand.

PART TWO • UNIT 10

10 The Inca and the Americas

An empire four thousand kilometres long, run without writing, wheels or money

■ IN THIS UNIT, YOU WILL

- ✓ Explain how altitude stacks several climates into one mountainside.
- ✓ Place the Maya and the Aztecs on the map and on the timeline.
- ✓ Explain how the Inca governed, and what a quipu did.
- ✓ Explain three Inca technologies and the problem each one solved.
- ✓ Explain the conquest, including civil war and disease.

UNIT 10 • WHAT YOU WILL LEARN

An empire in the mountains

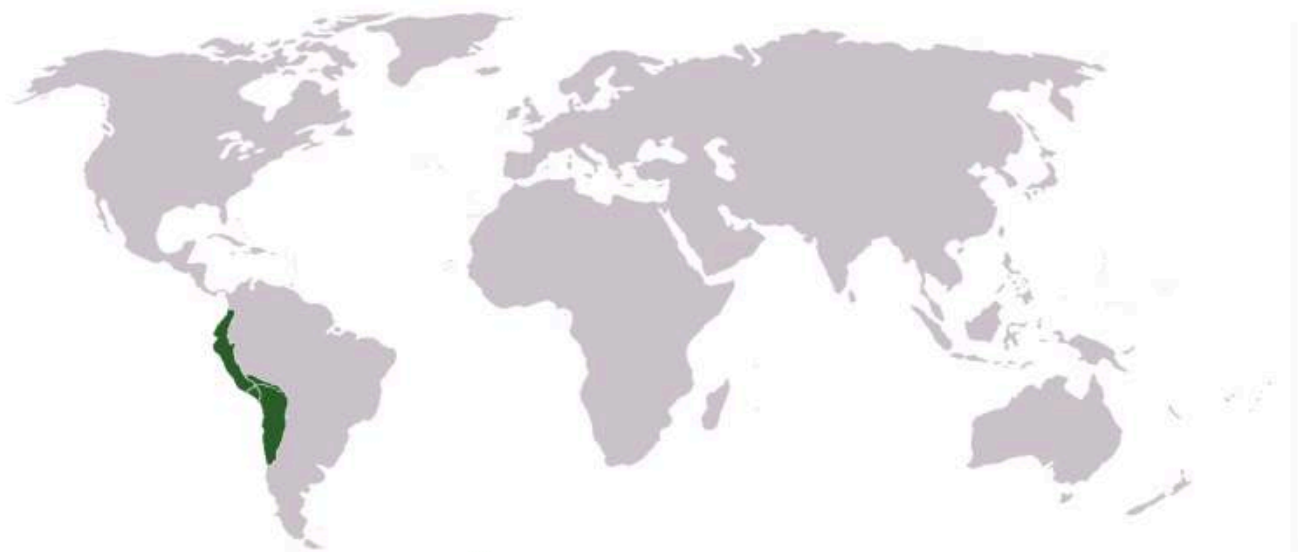


Figure 10.1 • The Inca empire, about 1530.

It is long and thin because it follows the mountains. Find Cuzco: every road in the empire started there.

You will climb the Andes and discover why one mountainside can hold four climates, meet the Maya and the Aztecs, then work out how the Inca ran an empire four thousand kilometres long with no writing, no wheels and no money. The unit ends with how that empire fell.

THE BIG QUESTION THIS UNIT ANSWERS

How do you run an empire with no writing, no wheels and no money?

THE TOPICS, IN THE ORDER YOU WILL MEET THEM

10.1 The Andes, a vertical world

10.2 Before and beside the Inca

10.3 An empire without writing or wheels

10.4 Terraces, roads and runners

10.5 Cuzco, Machu Picchu and the end

Everything is taught before it is tested, and the answers to every closed question are at the back of the book. Where a topic can be taken further, the code on that page goes to a checked English-language source. Ask your teacher before

● LOOK FIRST

Imagine you have to govern a country four thousand kilometres long, almost all of it mountain, with twelve million people speaking dozens of languages.

Now take away three things you were assuming you had. No writing. No wheeled vehicles. No money.

The Inca did it, and they did it well, in under a hundred years. This unit is about how.

TOPIC 10.1

The Andes, a vertical world

Straight up

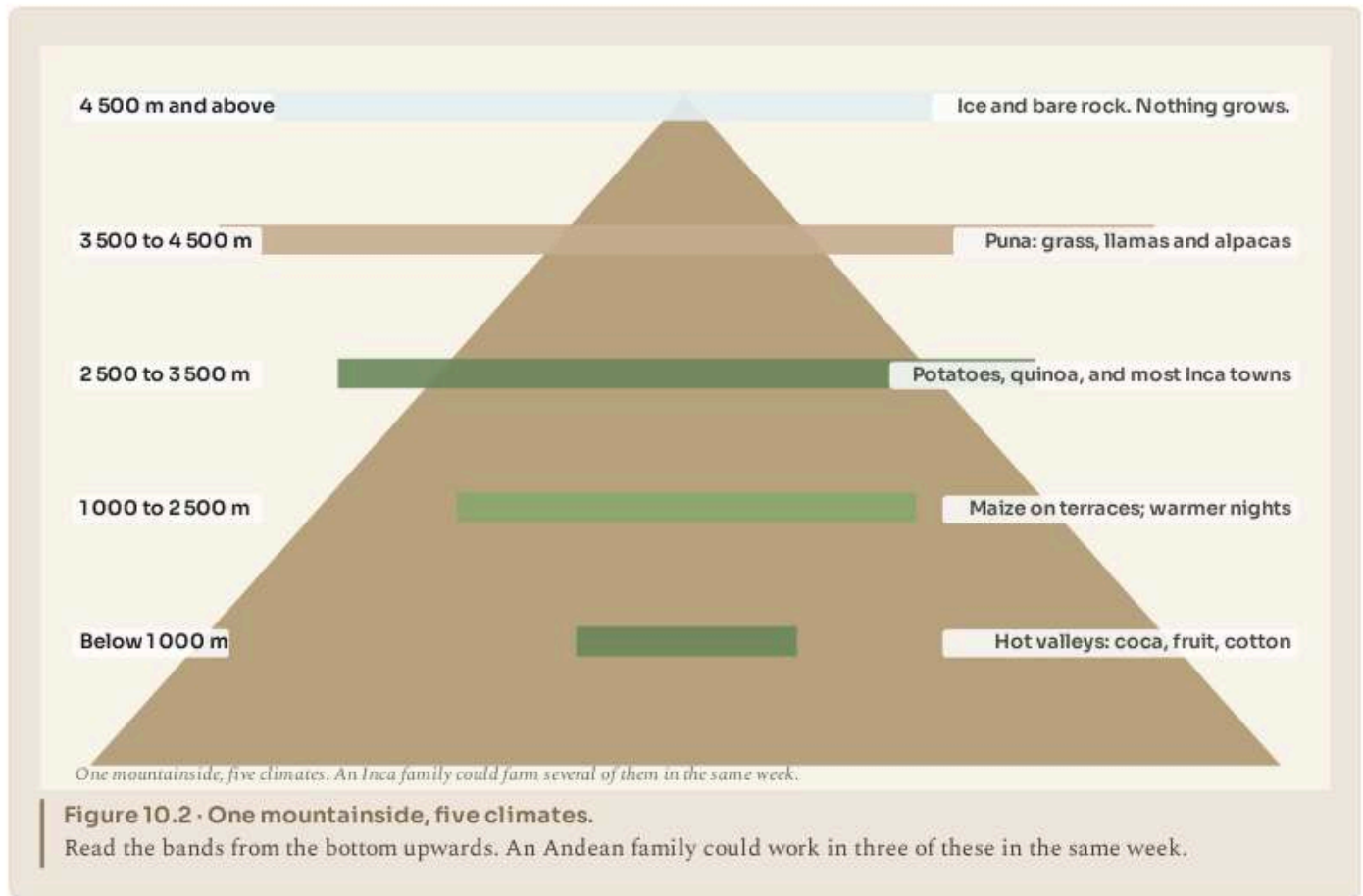
The Andes run about 7 000 kilometres down the western side of South America, the longest mountain range on any continent. They were pushed up, and are still being pushed up, where the Pacific sea floor is forced under the continent: the same process you met in unit 3.

In the middle section, in Peru and Bolivia, the range is wide and high, with a plateau called the *altiplano* sitting at around 3 800 metres. Lake Titicaca, on that plateau, is the highest large navigable lake in the world.

Being that high changes everything. There is about a third less oxygen in the air at 4 000 metres than at sea level. It is cold at night all year round. Ultraviolet light is fierce. And crops that grow at one height will not grow five hundred metres higher.

Several climates on one mountainside

The key idea of Andean geography is that climate here changes with **altitude**, not with latitude. Go up a mountainside and you pass through as many climate zones as you would by travelling thousands of kilometres north.



Andean peoples turned that into a strategy. Instead of farming one place, a community would deliberately hold land at several heights: maize lower down, potatoes higher up, herds of llamas and alpacas on the high grassland, and perhaps coca or fruit in a hot valley. A frost that destroyed one crop would not touch the others.

That is an insurance policy built out of geography, and it is one of the cleverest pieces of farming anywhere in the world.

● AMAZING BUT TRUE

Andean farmers domesticated the potato, and they did not stop at one. Several thousand varieties are grown in the Andes, bred for different heights, soils and uses. The potato later fed Europe: almost every chip, crisp and roast potato eaten anywhere in the world today descends from plants first farmed on these mountains.

■ YOUR TURN

- ① How were the Andes formed? Refer back to unit 3.
- ② Give three ways that living at 4 000 metres is difficult.
- ③ Explain what is meant by climate changing with altitude rather than latitude.
- ④ Explain the Andean strategy of farming at several heights, and why it reduces risk.



Figure 10.2 - Symbols of the Inca world: the sun god Inti, the Chakana, the rainbow of Tawantinsuyu.

TOPIC 10.2

Before and beside the Inca

Three peoples, often confused

The Inca were not the only civilisation in the Americas, they were not the first, and they were not the Aztecs. These three get mixed up constantly, so it is worth separating them clearly.



The **Maya** lived in what is now southern Mexico, Guatemala, Belize and Honduras. Their great cities, Tikal, Palenque, Copan, flourished from about 250 to 900 AD, centuries before the Inca empire existed. They built stone pyramids and observatories, worked out an extremely accurate calendar, and had a full writing system that recorded history, which no other American civilisation did. Their cities declined long before Europeans arrived; Maya people did not disappear and there are millions of them today.

The **Aztecs** were in central Mexico, at their height from about 1428 to 1521. Their capital, Tenochtitlan, was built on an island in a lake and had perhaps 200 000 people, which made it one of the largest cities in the world at the time. They farmed floating gardens called *chinampas* in the lake itself.

The **Inca** were in the Andes, roughly 3 000 kilometres to the south, and their empire was extraordinarily recent: it expanded from a single valley to its full size in under a century, from about 1438 to the 1520s.

◆ CAREFUL

The Maya, the Aztecs and the Inca were separated by thousands of kilometres and by centuries. Confusing them is like confusing the Vikings, the Romans and the Ottomans because all three were in the general direction of Europe. Learn one distinguishing fact for each: Maya writing, Aztec island city, Inca mountain roads.

■ YOUR TURN

- 1 Where and when did each of the Maya, Aztecs and Inca flourish?
- 2 Which of the three had a full writing system?
- 3 What were chinampas?
- 4 Roughly how long did the Inca take to build their empire?



Figure 10.4 - The three civilisations of the Americas: Aztec, Maya and Inca.

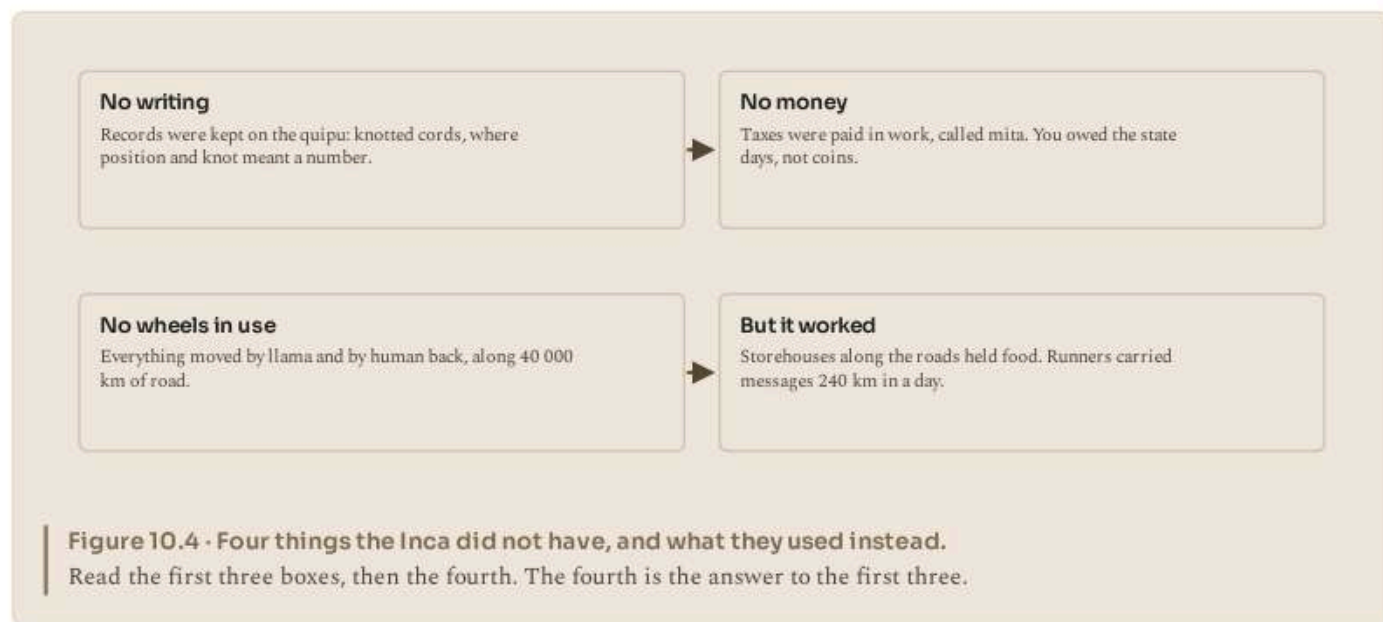
TOPIC 10.3

An empire without writing or wheels

What they did not have

The Inca empire, which they called *Tawantinsuyu*, "the four parts together", stretched about 4 000 kilometres from what is now southern Colombia to central Chile, and governed perhaps ten to twelve million people.

It did this without a writing system, without wheeled transport, without iron tools and without money.



What they had instead

The quipu. Instead of writing, the Inca recorded information on knotted cords. A main cord carried hanging strings, and the type of knot, its position along the string and the colour of the cord all carried meaning. Quipus certainly recorded numbers: populations, harvests, taxes, llamas. Whether they could also record language is still argued about by scholars. Officials called *quipucamayocs* were trained to make and read them.

The mita. Instead of taxes in money, people owed the state **labour**: a certain number of days a year building roads, terraces, temples or serving in the army. In return the state was expected to feed you in a bad year from its storehouses. It was an exchange, and when the Spanish later kept the mita and dropped the obligation to feed people, it became something much worse.

Reciprocity. Andean society ran on the *ayllu*, an extended family group that held land in common and worked each other's fields in turn. The empire was built on top of that habit rather than against it.

Resettlement. Conquered peoples who looked troublesome were moved, and loyal populations moved in. It was harsh, and it worked.

◆ CAREFUL

"No writing" does not mean "no records". The Inca ran a census, a tax system and a supply network across an empire the length of Europe. They simply did it with knots rather than letters. Judging a civilisation by whether it has the technology you happen to have is the oldest mistake in this subject.

■ YOUR TURN

- 1 What did the Inca call their empire, and what does the name mean?
- 2 Explain what a quipu was and what it certainly recorded.
- 3 Explain the mita, and what the state owed in return.
- 4 Why is it wrong to say the Inca had no records?

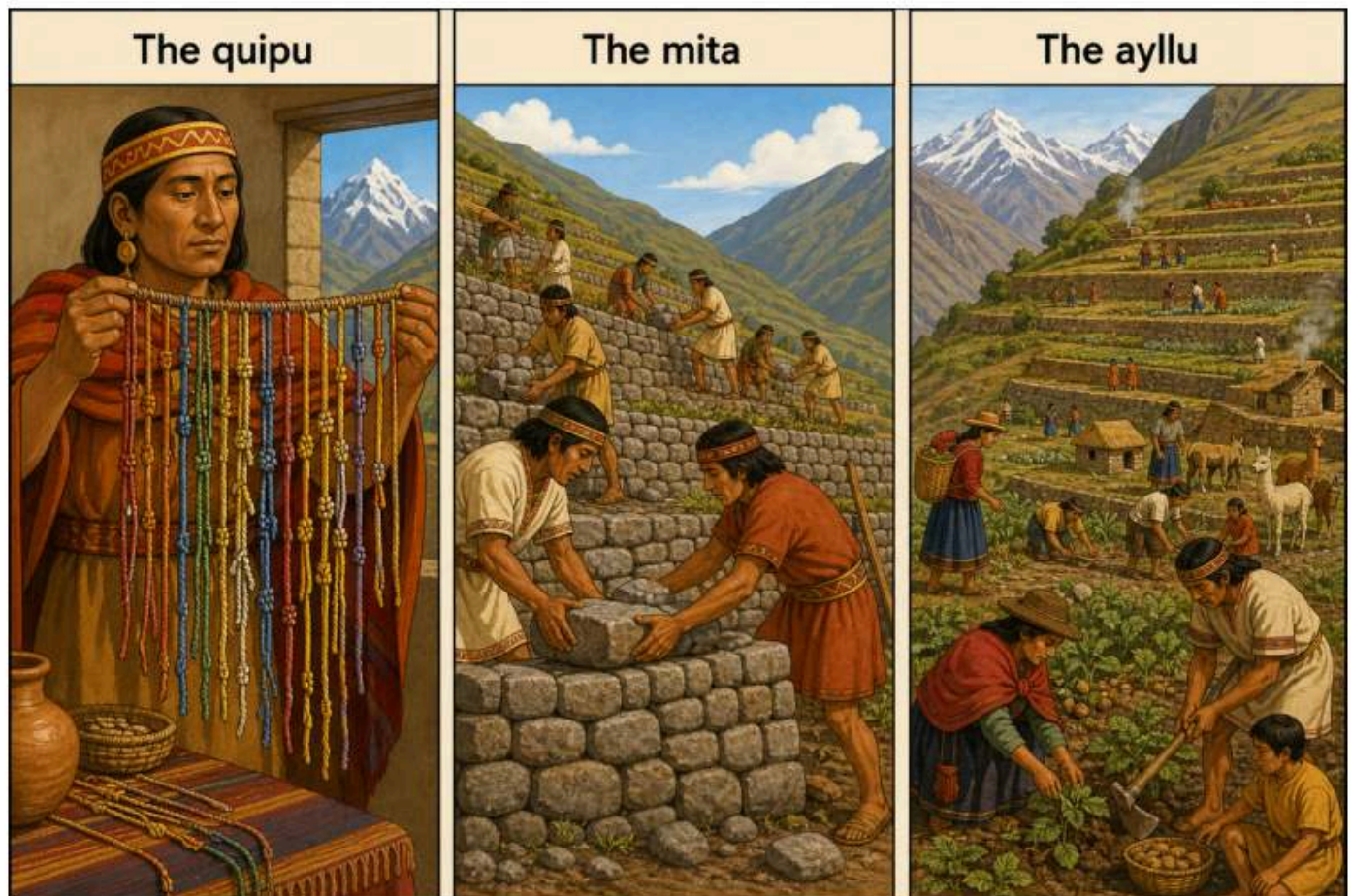


Figure 10.5 - The quipu, the mita and the ayllu: recording, working and belonging.

TOPIC 10.4

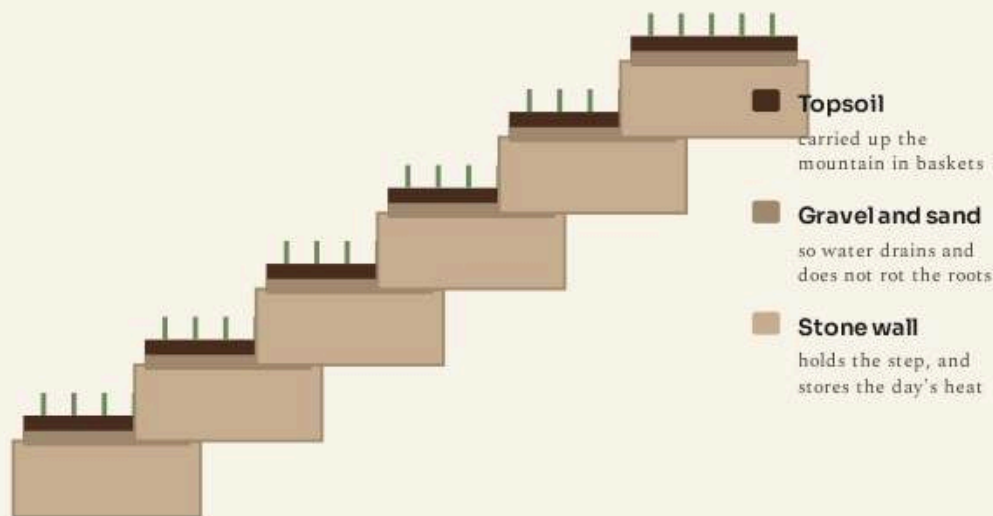
Terraces, roads and runners

Three problems, three solutions

Everything the Inca built was an answer to a specific mountain problem.

Problem: mountainsides are too steep to farm, and rain washes the soil away.

The answer was **terracing**: cutting the slope into steps held by stone walls. It is much more than a flat surface. Behind the wall the Inca layered gravel and sand for drainage, then carried topsoil up by hand. The stone wall absorbs heat during the day and releases it at night, protecting crops from frost, so a terrace can grow crops that would die on flat ground at the same height.



The wall releases heat at night, so crops on a terrace survive frosts that would kill them on flat ground.

Figure 10.5 • Inside an Inca terrace.

Look at the layers behind the wall. Only the top one is soil, and it was carried up the mountain in baskets.

Problem: an empire 4 000 km long, and no wheels or horses.

The answer was the **Qhapaq Nan**, the royal road: about 40 000 kilometres of paved roads, stairways cut into cliffs, tunnels and rope suspension bridges woven from grass and rewoven every year. Wheels are useless on a staircase, so the Inca did not want them; goods went by llama train and by human porter.

Along the roads stood *tambos*, waystations with food, and *qollqas*, storehouses holding grain, dried potato and cloth against a bad harvest.

Problem: news has to travel faster than a person can walk.

The answer was the **chasqui**, relay runners stationed a few kilometres apart. Each ran his stretch at full speed and passed the message, and any quipu, to the next. A message could cover about 240 kilometres in a day. Fresh fish is said to have reached the emperor in Cuzco from the coast, though that story may be a compliment rather than a fact.

Stone that needs no mortar

Inca stonework is famous for a reason. The finest walls are built of stones cut so precisely that they fit without any mortar at all, each block shaped to its neighbours, with many angles rather than four.

That is not decoration. Mortar cracks in earthquakes, and this is earthquake country. Interlocking blocks can shift slightly and settle back. Spanish buildings put up on top of Inca foundations have fallen down in earthquakes while the Inca walls beneath them stood.

■ YOUR TURN

- 1 Give three things a terrace does besides making flat ground.
- 2 Why did the Inca not need wheels?
- 3 What were tambos and qollqas for?
- 4 Explain the chasqui system and what it could achieve.
- 5 Explain why mortar-free stonework is a sensible choice in the Andes.

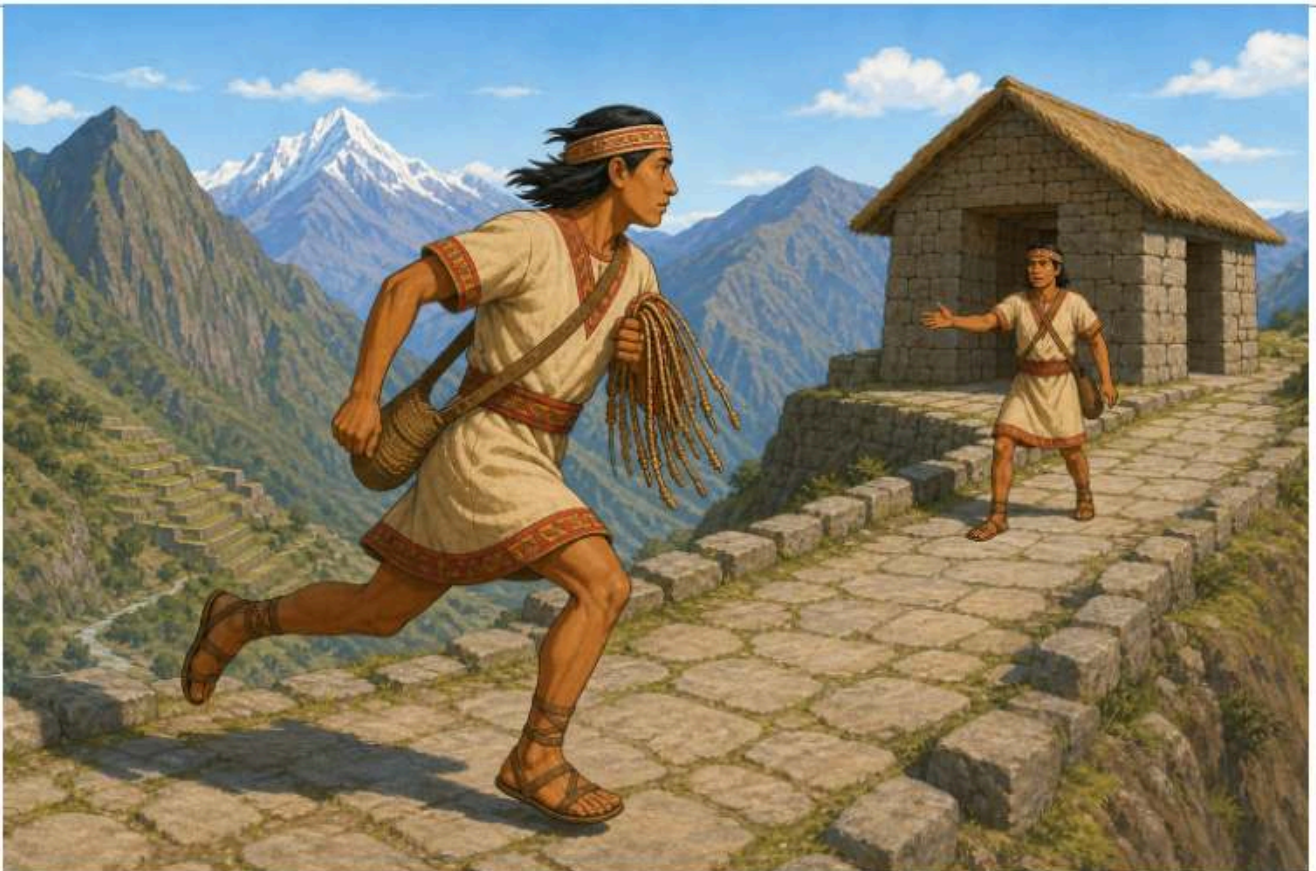


Figure 10.6 - A chasqui, the relay runner who carried messages along the Inca roads.

TOPIC 10.5

Cuzco, Machu Picchu and the end

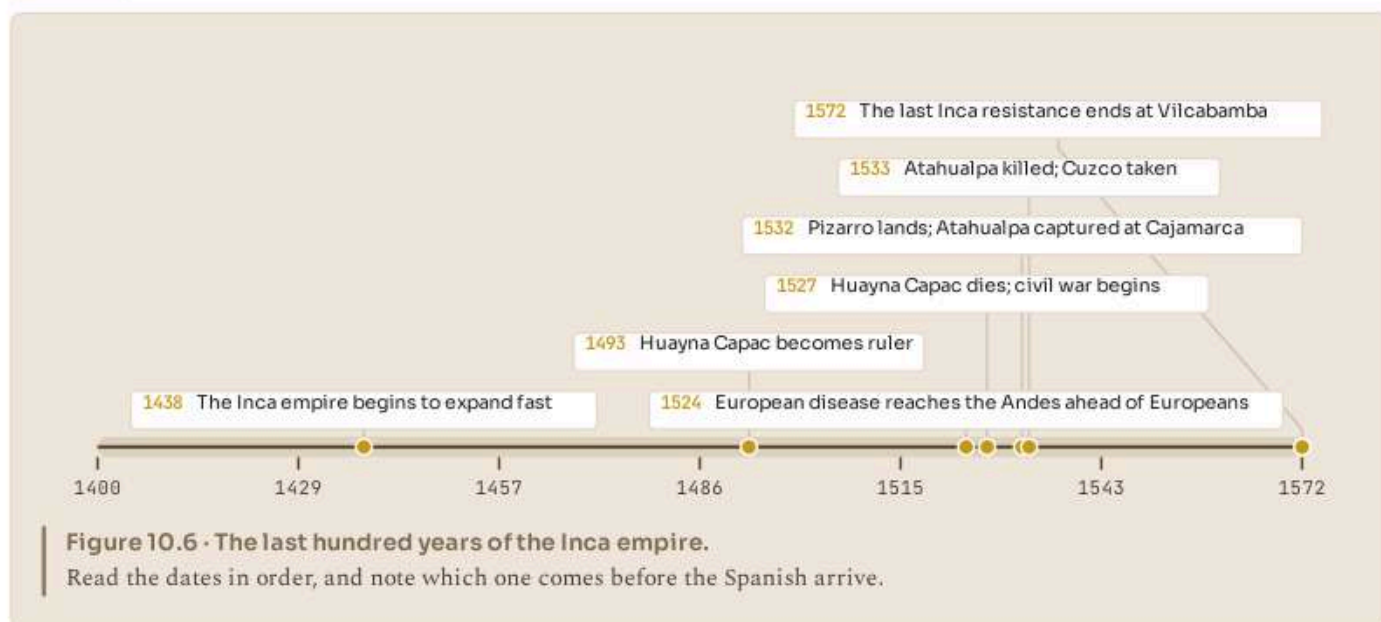
The centre of the world

Cuzco was the Inca capital, at 3 400 metres, and it was the point from which the empire's four parts were measured. Every main road began there. It held the Coricancha, the sun temple, whose walls were said to be covered in sheets of gold.

Machu Picchu sits on a ridge at 2 430 metres above the Urubamba river. It was probably built around 1450 as a royal estate for the emperor Pachacuti. It has about 200 buildings, terraces, fountains fed by a stone channel from a spring, and stonework of the highest quality.

It was never found by the Spanish, which is why it survived. It was known to local farmers all along, and it was brought to international attention in 1911 by the American explorer Hiram Bingham, who was actually looking for somewhere else.

How it fell



In 1532 Francisco Pizarro landed with about 168 men and 27 horses, and within a year the empire was in his hands. That sounds impossible, and it needs explaining honestly, because the usual explanation, that a few Europeans simply overpowered millions, is not true.

Four things happened at once.

Disease arrived first. Smallpox, brought by Europeans to the Americas earlier, spread down through the continent ahead of any European. It reached the Andes around 1524 and killed an enormous share of the population, including the emperor Huayna Capac and his named heir. Nobody in the Americas had any immunity.

Civil war followed. With the emperor and heir dead, two of his sons, Atahualpa and Huascar, fought a long and destructive war for the throne. Pizarro arrived in the middle of it, when the empire had just finished tearing itself in half.

Allies. Many peoples had been conquered by the Inca recently and did not love them. Pizarro gained thousands of local allies, and most of the fighting was done by them.

Technology and surprise. Steel, horses and guns mattered, especially at close quarters, and the capture of Atahualpa at Cajamarca decapitated a system built around one man.

Atahualpa was held for ransom, filled a room with gold, and was killed anyway in

1 Resistance continued from Vilcabamba until 1572.

■ PORTUGAL AT THIS STAGE

Here is why Portugal is not in this story, and it is worth a minute.

In 1494, before either the Aztec or the Inca empires had been seen by any European, Portugal and Spain signed the Treaty of Tordesillas, which drew a line down the Atlantic and divided the not-yet-explored world between them. Almost all of the Americas fell on the Spanish side; the eastern bulge of South America, which became Brazil, fell on the Portuguese side.

So while Pizarro was in Peru, Portuguese effort was going the other way. In 1498 Vasco da Gama reached India by sailing round Africa, and Portugal built a chain of trading posts along the African, Indian and south east Asian coasts, reaching Japan by 1543.

Two small European countries, in the same decades, took opposite directions round the world for reasons partly written into a treaty signed before anybody knew what was there.



LOOK FURTHER

Khan Academy, *Art of the Americas*. The Maya, the Aztecs and the Inca through the things they made. Find the section on Machu Picchu.

www.khanacademy.org/humanities/art-americas

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● LOOK AGAIN

Read the four causes of the conquest again, and then ask which one you would put first.

Most people, before studying it, would say guns and horses. Most historians would say disease and civil war, because by 1532 the empire had already lost a large part of its population and had just finished a war against itself.

That matters beyond this one topic. When you meet an event where a small force beats a large one, the interesting question is almost never "how good were their weapons". It is "what had already happened to the other side".

■ YOUR TURN

- 1 What was Cuzco, and why did all roads begin there?
- 2 What was Machu Picchu, and why did it survive?
- 3 Give the four reasons this book offers for the fall of the Inca empire.
- 4 Which cause do you think was most important? Justify your answer with evidence from the timeline.
- 5 Explain why Portugal was not involved in the conquest of Peru.

■ WHAT YOU CAN NOW DO

- ✓ Explain how altitude stacks several climates onto one mountainside.
- ✓ Tell the Maya, the Aztecs and the Inca apart, on the map and on the timeline.
- ✓ Explain how the Inca governed without writing, wheels or money.
- ✓ Explain three Inca technologies and the problem each solved.
- ✓ Explain the conquest, and put its causes in order of importance.

— WORDS TO KEEP

altiplano, altitude, terrace, quipu, mita, ayllu, chasqui, tambo, reciprocity

UNIT 10 AT A GLANCE

The Inca and the Americas

How do you run an empire with no writing, no wheels and no money?

THE TOPICS YOU HAVE COVERED

- 10.1 The Andes, a vertical world
- 10.2 Before and beside the Inca
- 10.3 An empire without writing or wheels
- 10.4 Terraces, roads and runners
- 10.5 Cuzco, Machu Picchu and the end

WHAT YOU CAN NOW DO

- ✓ Explain how altitude stacks several climates onto one mountainside.
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- ✓ Explain how the Inca governed without writing, wheels or money.
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- ✓ Explain the conquest, and put its causes in order of importance.

THE WORDS THIS UNIT TAUGHT

altiplano

altitude

terrace

quipu

mita

ayllu

chasqui

tambo

reciprocity

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to

Answers

Every closed question in this book has an answer here. Open questions have acceptance criteria instead: what a good answer has to contain, rather than the exact words it has to use.

Use these after you have tried, not instead of trying. If your answer differs from the one printed here, the useful question is where the two of you took different turnings, not who was right.

Unit 1 · How the Universe began

1.1 A STORY WITH A BEGINNING

- 1 Because a conclusion can rest on evidence rather than on a witness. Accept any answer that names evidence and explains that several independent lines pointing the same way make a strong case.
- 2 Eight minutes and twenty seconds, because that is how long light takes to travel from the Sun to the Earth. Nothing can arrive faster.
- 3 Open. A good answer names one clue from the figure and gives a reason connected to what that clue rules out.
- 4 Acceptance criteria: uses the word evidence; mentions at least one of the three clues; does not claim the theory is proved beyond all doubt.

1.2 THE BIG BANG

- 1 The Universe is smaller than a full stop; the first hydrogen forms; the fog clears.
- 2 A firework throws pieces outwards through space that was already there. The raisin bread has the space itself getting bigger, which is what actually happens, and it explains why the far raisins move away fastest.
- 3 Because space expanded everywhere at once, so every point sees the others moving away and no point is the middle.
- 4 13 800 000 000 years.

1.3 HOW GALAXIES WERE MADE

- 1 Gravity.
- 2 A denser patch pulls harder, so it gathers more gas, which makes it denser still, which makes it pull harder again.
- 3 A star is a single ball of gas hot enough to shine. A galaxy is an enormous collection of stars, gas and dust held together by gravity.
- 4 About two thousand million.

1.4 THE MILKY WAY, OUR HOME

- 1 A flat disc with spiral arms and a bulge in the middle, about 100 000 light

years across.

- 2 About halfway out, in one of the spiral arms. It is not in the centre.
- 3 Because we are inside the disc, so looking along it we see huge numbers of distant stars blurred together into a band.
- 4 Light pollution from outdoor lighting has made the night sky too bright in most places where people live.

1.5 BLACK HOLES

- 1 Nothing has been removed. It is matter packed extremely tightly, so it is the opposite of an absence.
- 2 The boundary past which nothing, not even light, can escape.
- 3 Wrong. Gravity depends on mass and distance, not on density at a distance. A black hole with the Sun's mass would leave the Earth's orbit unchanged.
- 4 From the collapse of the core of a very large star at the end of its life, in a supernova.
- 5 About 26 000 light years.

Unit 2 · The Sun and its family

2.1 THE SUN, OUR OWN STAR

- 1 Because it is far closer: about 150 million km, against 270 000 times that for the next nearest star.
- 2 Any two of: plants would stop growing, because photosynthesis needs sunlight; wind would stop, because it is driven by uneven heating; rain would stop, because evaporation needs heat; temperatures would fall far below freezing.
- 3 Burning is a chemical reaction needing oxygen. Fusion squeezes small atoms into bigger ones and needs no oxygen, which is why it works in space and lasts far longer.
- 4 More than 99 per cent.

2.2 THE ROCKY PLANETS

- 1 Mercury, Venus, Earth, Mars.
- 2 It has almost no atmosphere, so there is nothing to hold the heat once the Sun sets.
- 3 Its thick carbon dioxide atmosphere traps heat and stops it escaping, so heat builds up. Mercury has no such blanket.
- 4 Dry river valleys on the surface, and ice at the poles.
- 5 Open. A good answer picks one row and explains its consequences rather than simply restating it.

2.3 THE GIANTS, AND THE EDGE OF THE SYSTEM

- 1 Jupiter, Saturn, Uranus, Neptune.
- 2 Jupiter: the largest, with a storm wider than Earth. Saturn: rings made of

billions of pieces of ice and rock. Uranus: tipped on its side, so each pole gets 42 years of light then 42 of dark. Neptune: winds over 2 000 km/h.

- 3 Uranus was not moving exactly as predicted. Astronomers calculated where an unseen planet would have to be to explain the difference, pointed a telescope there and found it.
- 4 It must orbit the Sun, be round, and have cleared other objects from its orbital path.
- 5 It shares its region of space with many similar icy bodies, so it fails the third condition.

2.4 EARTH, THE PLANET THAT GOT EVERYTHING RIGHT

- 1 Liquid water; an atmosphere; a magnetic field; time with a steady star.
- 2 Below a certain temperature water freezes and above it boils, and living chemistry needs a liquid to happen in.
- 3 The magnetic field deflected the solar wind. Without it, the solar wind stripped most of the atmosphere away over time.
- 4 Oxygen is reactive and would combine with other substances and disappear. Living things, mainly plants and plankton, keep producing it.

2.5 THE MOON

- 1 From a collision between the young Earth and an object about the size of Mars. Evidence: Moon rocks are chemically very like Earth's outer layers, and the Moon has only a very small iron core.
- 2 Half is always lit. What changes is how much of the lit half is turned towards us as the Moon goes round.
- 3 Because there is a tidal bulge on the side facing the Moon and another on the opposite side, so a coast passes through two bulges each day.
- 4 There is no permanently dark side. There is a far side, which we never see from Earth but which receives just as much sunlight.
- 5 Tides will be unusually large, because the Sun's pull and the Moon's pull add together when they are in line.

Unit 3 · The restless Earth

3.1 INSIDE THE EARTH

- 1 Crust, mantle, outer core, inner core.
- 2 The mantle, nearly 2 900 km deep.
- 3 The swirling liquid iron of the outer core. It turns a compass and deflects much of the Sun's radiation.
- 4 Waves travel at different speeds through different materials and some cannot pass through liquid, so comparing arrival times at stations worldwide shows what the waves passed through.
- 5 The pressure on it is so great that the iron stays solid despite the temperature.

3.2 RODINIA, PANGAEA AND THE WANDERING CONTINENTS

- 1 That the coastlines of South America and Africa fit together. Evidence: matching fossils of land animals on both sides, matching rock formations, and glacier marks in places that are now tropical.
- 2 Because he could not explain what force could move continents.
- 3 Rodinia; Pangaea; Pangaea breaking up; today.
- 4 $2.5 \text{ cm} \times 1\,000 = 2\,500 \text{ cm} = 25 \text{ m}$ wider.

3.3 PLATES THAT PUSH, PULL AND SLIDE

- 1 About fifteen large ones, plus many smaller.
- 2 Pulling apart makes new crust, mid-ocean ridges and rift valleys; pushing together makes mountains, trenches and volcanoes; sliding past makes earthquakes.
- 3 Because India is still pushing north into Asia at about 5 cm a year, and neither plate will sink, so the rock keeps crumpling upwards.
- 4 A continent is land. A plate is a piece of the Earth's outer shell, and most plates carry both continental crust and ocean floor, so plate edges are not coastlines.
- 5 Because no rock is being pushed down and melted, so there is no molten rock rising to the surface.

3.4 WHEN THE EARTH MOVES

- 1 Friction locks two sides of a fault while the plates keep moving. Strain builds until the rock breaks and both sides lurch, releasing energy as waves.
- 2 Because most deaths are caused by buildings collapsing, and building standards differ. Accept any answer naming construction quality or preparedness.
- 3 A horseshoe of volcanoes and earthquakes round the edge of the Pacific. It is that shape because it follows the edges of the Pacific plate.
- 4 Any two of: very fertile volcanic soil; geothermal energy; tourism; building stone; family and cultural ties to the place.
- 5 About $30 \times 30 = 900$ times.

3.5 THE CONTINENTS TODAY

- 1 Any correct route round the cycle, for example: magma cools to igneous rock; weathering breaks it into sediment; burial and pressure make sedimentary rock; heat and pressure make metamorphic rock; deeper still it melts back to magma.
- 2 Because plates are still moving, so the current arrangement is temporary. Accept any answer using a named future change.
- 3 Open. Acceptance criteria: identifies that both figures are explained by plate movement, and distinguishes the two timescales.

Unit 4 • The five oceans

4.1 MEET THE FIVE OCEANS

- 1 Pacific, Atlantic, Indian, Southern, Arctic.
- 2 The Southern Ocean, defined by the Antarctic Circumpolar Current.
- 3 An ocean is one of the five great bodies of water; a sea is smaller and usually partly enclosed by land. Example: the Atlantic Ocean; the Mediterranean Sea.
- 4 165.2 divided by 14.1, which is about 11.7 times.

4.2 WHAT THE OCEAN IS REALLY LIKE

- 1 About 3.5 per cent. It is dissolved out of rock by slightly acidic rain and carried to the sea by rivers, and it stays behind when water evaporates.
- 2 Sunlight zone, 0 to 200 m; twilight zone, 200 to 1 000 m; midnight zone, 1 000 to 4 000 m; the deep, 4 000 m to 11 034 m.
- 3 Because plants need sunlight, and useful amounts of sunlight only reach the top 200 m.
- 4 Light made by a living thing. Used for finding food, attracting mates and startling predators.

4.3 THE GREAT CONVEYOR

- 1 Wind, which drags the surface water along; and density, because cold salty water is heavier and sinks.
- 2 Cold water is denser than warm water and salty water is denser than fresh, so cold salty water is heaviest of all.
- 3 About a thousand years.
- 4 Because ocean currents and winds carry heat away from the tropics towards the poles, so the heat does not simply build up.

4.4 THE OCEAN MAKES THE WEATHER

- 1 Because Lisbon is on a coast warmed by the Atlantic and by the warm current crossing it, while Beijing is far inland, where land cools rapidly in winter.
- 2 Water heats and cools more slowly than land, so coasts are milder; and currents carry heat from one part of the world to another.
- 3 Air above cold water holds little moisture, so it brings little rain to the land it crosses.
- 4 Cold water is rich in nutrients, which supports large populations of fish.

4.5 THE OCEAN AND US

- 1 Any three of: food, transport for trade, oil and gas, offshore wind energy.
- 2 About 80 per cent.
- 3 Because plastic does not rot. It only breaks into smaller pieces, which enter the food chain and spread everywhere.
- 4 Good: it absorbs about a quarter of our carbon dioxide, slowing the warming of the air. Bad: dissolved carbon dioxide makes the sea more acidic, which makes shell-building harder for corals and shellfish.
- 5 Open. Acceptance criteria: any two reasonable consequences, such as fishing

policy, marine protection, scientific research or how the country negotiates internationally.

Unit 5 · Seven continents

5.1 HOW TO READ A MAP

- 1 Title, what it shows; key, what the symbols mean; scale, real distance; direction arrow, which way is north; grid, an address for every place.
- 2 Open. Lisbon is about 38.7 degrees north, 9.1 degrees west; accept any reasonable local figure.
- 3 Because a sphere cannot be flattened without stretching, so every projection must distort shape, area, distance or direction.
- 4 Because the common classroom projection stretches areas far from the equator. Africa is in fact about fourteen times larger than Greenland.

5.2 AFRICA

- 1 Because the equator crosses it near the middle, so the bands of climate are repeated north and south.
- 2 Rivers drop off the edge of the plateau in waterfalls and rapids near the coast, so boats could not sail far inland.
- 3 Any three with a matching animal, for example rainforest and gorilla, savannah and elephant, desert and camel.
- 4 54. No continent has more.

5.3 ASIA

- 1 About six in every ten people alive.
- 2 Any three of: they block cold air moving south, keeping northern India warmer; they feed the great rivers; they are still rising; they made land travel between India and China extremely difficult.
- 3 A seasonal reversal of wind. In summer land heats faster than sea, air rises over land and moist sea air is drawn in, bringing rain; in winter the flow reverses and is dry.
- 4 India, capital New Delhi; China, capital Beijing.

5.4 EUROPE

- 1 Because Europe and Asia are one continuous landmass, so any dividing line is chosen rather than natural. The Urals are the usual convention.
- 2 Nowhere in western Europe is far from the sea, which made trade, fishing and later overseas voyages far easier.
- 3 Because it is further from the Atlantic, so the moderating effect of the ocean is weaker.
- 4 Lisbon, Paris, Berlin, Warsaw.

5.5 NORTH AMERICA

- 1 The Rocky Mountains in the west and the Appalachians in the east.
- 2 Because the mountains run north to south, there is nothing to stop cold Arctic air and warm Gulf air meeting in the middle, which produces violent weather.
- 3 About one fifth of the world's fresh surface water.
- 4 Because Central America narrows to about 50 km at that point, so a canal there saves a 13 000 km voyage round South America.

5.6 SOUTH AMERICA

- 1 The Andes, about 7 000 km long, raised where the Pacific sea floor is forced under the continent.
- 2 The largest river by volume, and the largest drainage basin and rainforest.
- 3 Because the Humboldt Current brings cold water up the coast, and air above cold water carries little moisture.
- 4 Any four correct pairs, for example Brazil and Brasilia, Argentina and Buenos Aires, Peru and Lima, Colombia and Bogota.

5.7 OCEANIA

- 1 Because Australia has been separated from other landmasses for tens of millions of years, so its animals evolved in isolation.
- 2 Any two: Australia is old, flat and dry, New Zealand is mountainous and wet; Australia is not on a plate boundary in the same way, New Zealand is and has volcanoes and earthquakes.
- 3 Because they are only a metre or two above sea level, so even a small rise floods them and spoils their fresh water.
- 4 A coral reef about 2 300 km long off north east Australia, and the largest structure on Earth built by living things.

5.8 ANTARCTICA

- 1 Desert, because it receives very little precipitation; freezer, because it holds about 90 per cent of the world's ice; library, because its ice cores record the atmosphere going back more than 800 000 years.
- 2 About 70 per cent.
- 3 Snow traps bubbles of the air of the year it fell. Drilling down and sampling those bubbles gives a direct record of past atmospheres.
- 4 It sets the continent aside for peaceful scientific use, bans military activity and suspends territorial claims. It is unusual because no other continent is governed by agreement rather than by countries.

Unit 6 · The first people

6.1 OUT OF AFRICA

- 1 About 300 000 years, and it appeared in Africa.
- 2 Fossils, DNA, and tools and objects.
- 3 Because populations that left Africa were relatively small, so they carried

only part of the variety that existed there, and that reduction shows up in their descendants.

- 4 Some humans left Africa; Africa has been continuously inhabited ever since.

6.2 LIVING THROUGH THE ICE AGE

- 1 About 120 metres lower.
- 2 Any two: Britain and mainland Europe; Asia and North America at the Bering land bridge; islands of south east Asia.
- 3 Any four of: identifying edible and poisonous plants; tracking; making fire; making stone tools; sewing fitted clothing; knowing seasonal movements.
- 4 Because it is open-air rock carving rather than painting inside caves, which is rare and much more fragile.

6.3 THE GREAT CHANGE

- 1 About 11 500 years ago, in south west Asia.
- 2 Any three, for example China and rice, Central America and maize, the Andes and potatoes, Africa and sorghum, New Guinea and taro.
- 3 Better: far more food per area; larger populations; permanent settlements.
Worse: less varied diet; more disease; longer working hours; shorter, less healthy skeletons.
- 4 Stored grain can be taken, so somebody has to control it, defend it and record it. That produces rulers, soldiers, taxes and writing.

6.4 THE FIRST CITIES

- 1 In Mesopotamia, between the Tigris and Euphrates, by about 3500 BC.
- 2 Irrigation canals need many people to build and maintain them and rules about who gets water when, which requires organisation and authority.
- 3 Accounts: lists of grain, sheep, beer and oil, and who owed what. Surprising because we expect writing to begin with stories or religion.
- 4 Prehistory is the time before written records; history is the time we can read about from the people who lived it.
- 5 Open. Acceptance criteria: recognises that large-scale organisation does not require cities, writing or kings.

Unit 7 · Egypt, the gift of the Nile

7.1 THE RIVER THAT MADE A COUNTRY

- 1 North. Rivers flow downhill, and the highlands are in the south; north being at the top of a map is a convention, not a slope.
- 2 Kemet, the black land, was the fertile strip beside the river. Deshret, the red land, was the desert beyond it.
- 3 Fed, by renewing the soil with silt; watered, in a land with almost no rain; carried, because current and wind run in opposite directions; protected, by desert, sea and cataracts.
- 4 Because Egypt is named for the river: upstream is Upper Egypt, and the river

flows from south to north.

7.2 PHARAOH AND THE PYRAMID OF PEOPLE

- 1 A god on Earth, and the link between the gods and the people, responsible for keeping the world in order.
- 2 If harvests failed or the country was invaded, people concluded the pharaoh had failed in that duty, and the dynasty could be overthrown.
- 3 Pharaoh; officials and priests; scribes; craftworkers and soldiers; farmers.
- 4 Because during the flood the fields were under water and could not be farmed.
- 5 The excavated workers' town at Giza, with bakeries, housing, medical care and tombs. Accept the evidence of healed, properly set broken bones.

7.3 GODS, TEMPLES AND THE AFTERLIFE

- 1 Any four, for example Ra, the sun; Osiris, the dead and rebirth; Isis, magic and protection; Anubis, care of the dead; Thoth, writing.
- 2 Truth, balance, order and justice: the way the world was supposed to be, and the pharaoh's duty to uphold.
- 3 Because they believed a person needed their body, their name and their possessions to continue after death.
- 4 Because tombs and temples were built of stone in the dry desert, while houses were built of mud brick beside the river and have washed away.

7.4 HIEROGLYPHS AND THE SCRIBE

- 1 Stand for a sound; stand for a whole word or idea; act as a silent determinative showing what kind of thing is meant.
- 2 The oval drawn round a royal name. It was useful because scholars could guess the royal names from the Greek text and test their sound values.
- 3 A stone carrying the same decree in hieroglyphs, demotic and Greek. Since Greek could be read, it worked as an answer key.
- 4 That hieroglyphs were not purely picture-writing: many signs recorded sounds.
- 5 Because Coptic descends from ancient Egyptian, so when he sounded out signs he recognised real words.

7.5 BUILDING THE PYRAMIDS

- 1 Around 2560 BC, for the pharaoh Khufu. It was 146.6 m tall.
- 2 About 3 800 years.
- 3 Wetting sand to the right dampness roughly halves the friction, so a sledge needs about half the pulling force.
- 4 That slaves built them: refuted by the excavated workers' town, pay in bread and beer, medical care and burials. That outside help was needed: refuted by the quarries, tools, ramps and records, and it is an insult to the builders.
- 5 Organisation: feeding, housing, paying and rotating tens of thousands of workers and delivering the right stone on time for twenty years.

Unit 8 · Ancient China

8.1 TWO GREAT RIVERS

- 1 The Huang He or Yellow River, with millet and wheat; the Chang Jiang or Yangtze, with rice.
- 2 Fine yellow soil blown into place by wind. It is easy to dig and very fertile, which is why farming began there.
- 3 Because it carries very large amounts of silt and drops it on the bed, and the banks people build to contain it raise the whole channel further.
- 4 Because dykes, dredging and canals require organisation across whole regions, which no village could manage alone, so a strong central government was a practical necessity.

8.2 THREE THOUSAND YEARS OF DYNASTIES

- 1 A dynasty takes power and restores order; the country prospers; over generations government becomes distant and costly; disasters are handled badly; rebellion follows; a new dynasty takes over.
- 2 Shang: the first Chinese writing. Qin: unified China and standardised writing, coins, laws and cart axles. Han: four centuries, opened the Silk Road. Song: printing, paper money, gunpowder weapons and the compass spread.
- 3 The belief that Heaven grants a just ruler the right to govern and withdraws it from a bad one, which makes rebellion legitimate.
- 4 Similarity: authority came from beyond the human world. Difference: the pharaoh was a god, while the emperor was a man holding a licence that could be withdrawn.

8.3 CONFUCIUS AND HOW TO RUN A COUNTRY

- 1 People are basically good, or can be made good by example and education. A ruler should behave well and be worth following.
- 2 People act in their own interest, so only clear written laws with heavy punishments and real rewards will make them behave.
- 3 The Legalists. Qin Shi Huang united China using their methods, but the Qin dynasty was harsh and collapsed within fifteen years of his death.
- 4 Officials were chosen by written examination rather than by birth. It was unusual because elsewhere office was usually inherited.

8.4 FOUR INVENTIONS

- 1 Paper, about 105 AD; printing, from about 700 AD; the compass, about 1000 AD; gunpowder, about 900 AD.
- 2 Paper made records and books cheap. Printing allowed thousands of identical copies. The compass allowed sailing out of sight of land. Gunpowder ended the age of the castle wall.
- 3 Because a map tells you where things are but not which way you are facing. A compass gives direction even under cloud and out of sight of land.
- 4 Chinese armies used gunpowder bombs, fire lances and rockets from the

eleventh century, well before European guns. Fireworks came first, but weapons followed quickly.

8.5 THE SILK ROAD AND THE GREAT WALL

- 1 Because it was not one road and it did not carry only silk. It was a network of routes along which goods passed from trader to trader.
- 2 West: silk, paper, porcelain, tea. East: horses, glass, gold, wool, spices.
- 3 Ideas, such as Buddhism and paper-making; and diseases, such as the plague.
- 4 To force raiders to come through gates, where they could be taxed, counted or fought at a place of the defenders' choosing; and to pass warnings quickly along the frontier by beacon.
- 5 Because Asia's main axis runs east to west, so crops, animals and people could move along it without crossing climate bands.

Unit 10 · The Inca and the Americas

10.1 THE ANDES, A VERTICAL WORLD

- 1 Where the Pacific sea floor is forced under the South American plate, which crumples and raises the rock: a pushing-together boundary.
- 2 Any three of: about a third less oxygen; cold nights all year; strong ultraviolet light; crops that will not grow above a certain height.
- 3 Going up a mountainside passes through as many climate zones as travelling thousands of kilometres towards a pole.
- 4 Communities held land at several heights and grew different crops at each. A frost or failure at one height would not destroy the others, so risk was spread.

10.2 BEFORE AND BESIDE THE INCA

- 1 Maya: southern Mexico and Central America, about 250 to 900 AD. Aztecs:

central Mexico, about 1428 to 1521. Inca: the Andes, about 1438 to 1533.

- 2 The Maya.
- 3 Floating gardens built on the lake around Tenochtitlan and farmed by the Aztecs.
- 4 Under a hundred years.

10.3 AN EMPIRE WITHOUT WRITING OR WHEELS

- 1 Tawantinsuyu, meaning "the four parts together".
- 2 A set of knotted cords. It certainly recorded numbers, such as populations, harvests, taxes and herds.
- 3 Tax paid in days of work rather than money or goods. In return the state was expected to feed people from its storehouses in a bad year.
- 4 Because they kept a census, a tax system and a supply network across an enormous empire, using knots instead of letters.

10.4 TERRACES, ROADS AND RUNNERS

- 1 It stops soil washing away; the layered gravel and sand drain the water; the stone wall stores heat and releases it at night, protecting crops from frost.
- 2 Because their roads included stairways, steep slopes and rope bridges, where wheels are useless. Goods went by llama and by porter.
- 3 Tambos were waystations with food and shelter; qollqas were storehouses holding grain, dried potato and cloth against a bad harvest.
- 4 Relay runners stationed a few kilometres apart, each running his stretch at full speed. A message could travel about 240 km in a day.
- 5 Because mortar cracks in earthquakes, while interlocking blocks can shift slightly and settle back.

10.5 CUZCO, MACHU PICCHU AND THE END

- 1 The Inca capital and the point from which the four parts of the empire were measured, so every main road began there.
- 2 A royal estate built for Pachacuti around 1450, on a ridge at 2 430 m. It survived because the Spanish never found it.
- 3 Disease arriving ahead of Europeans; the civil war between Atahualpa and Huascar; local allies who resented Inca rule; and European technology and the surprise capture of Atahualpa.
- 4 Open. Acceptance criteria: names one cause, uses at least one date from the timeline as evidence, and explains why that cause was decisive.
- 5 Because the Treaty of Tordesillas of 1494 placed almost all of the Americas on the Spanish side of the dividing line, and Portuguese effort went east round Africa to India and Asia instead.

Every word this book teaches, defined as it is used here, with the unit that introduces it. 66 entries.

altiplano	The high plateau of the central Andes, around 3 800 m above sea level. Unit 10
altitude	Height above sea level. In the Andes it changes climate more than latitude does. Unit 10
asteroid belt	The region between Mars and Jupiter holding many rocky and metal fragments. Unit 2
atmosphere	The layer of gas held around a planet by its gravity. Unit 2
ayllu	An Andean extended family group that held land together and worked each other's fields. Unit 10
bioluminescence	Light made by a living thing, common in the deep sea. Unit 4
biome	A large region defined by its climate and the living things suited to it, such as desert or rainforest. Unit 5
cartouche	The oval drawn around a royal name in hieroglyphs. It was the key to deciphering the script. Unit 7
cataract	A stretch of rocky rapids on the Nile, which boats could not easily pass. Unit 7
chasqui	An Inca relay runner who carried messages along the royal roads. Unit 10
civil service	The officials who run a government. China chose many of them by written examination from about 600 AD. Unit 8
continent	One of the seven great land areas of the Earth. Unit 5
core	The Earth's centre: a liquid iron outer core and a solid iron inner core. Unit 3
crust	The Earth's thin outer layer of solid rock, between about 5 and 70 km thick. Unit 3
cuneiform	The wedge-shaped writing pressed into clay in Mesopotamia, the earliest known writing. Unit 6
current	A steady flow of water within the ocean, driven by wind at the surface and by density in the deep. Unit 4
determinative	A silent hieroglyph added to the end of a word to show what kind of thing is meant. Unit 7
dwarf planet	A round object orbiting the Sun that has not cleared other objects from its path. Pluto is one. Unit 2
dynasty	A sequence of rulers from the same family. Unit 7
event horizon	The boundary around a black hole, past which nothing, not even light, can escape. Unit 1
evidence	Something you can point at that supports a conclusion. In this subject a claim without evidence is only an opinion. Unit 1

exclusive economic zone	The area of sea, out to 200 nautical miles, in which a country controls fishing and resources. Unit 4
expansion	The stretching of space itself, which is what carries galaxies away from each other. Not an evulsion of things flying through space. Unit 1
fault	A crack in the Earth's crust where the rock on either side can move. Unit 3
fusion	Squeezing small atoms together into bigger ones, releasing energy. It is why stars shine, and it is not burning. Unit 2
galaxy	An enormous collection of stars, gas and dust held together by gravity. Ours is the Milky Way. Unit 1
garrison	The soldiers stationed at a fort or along a wall. Unit 8
gravity	The pull that every piece of matter has on every other piece. It gets weaker with distance and stronger with mass. Unit 1
hieroglyph	A sign in the Egyptian writing system, which could stand for a sound, a word or a category. Unit 7
hunter-gatherer	Someone who lives by hunting animals and gathering wild plants rather than by farming. Unit 4
ice age	A long, cold period when ice sheets cover large parts of the Earth. The last cold peak was about 20,000 years ago. Unit 4
ice sheet	A layer of ice covering a large land area, as in Antarctica and Greenland. Unit 5
igneous	Rock formed when molten rock cools and hardens. Unit 3
irrigation	Bringing water to fields by canal or ditch so crops can grow where rain is not enough. Unit 4
land bridge	Land exposed when sea level falls, joining places that are separate today. Unit 6
latitude	How far north or south of the equator a place is, measured in degrees from 0 to 90. Unit 4
Legalism	The Chinese school of thought holding that people obey clear written laws and firm punishment, not good example. Unit 8
light year	A distance, not a time: how far light travels in one year, about 9.5 million million kilometres. Unit 1
loess	Fine yellow soil blown into place by wind. It gives the Yellow River its name and its fertility. Unit 8
longitude	How far east or west a place is, measured in degrees from 0 at Greenwich to 180 in the Pacific. Unit 5
maat	The Egyptian idea of truth, balance, order and justice, which the pharaoh had to uphold. Unit 7
magnetic field	The invisible field made by the Earth's swirling liquid iron core, which turns a compass and shields us from the Sun's radiation. Unit 2
magnitude	The measure of an earthquake's size. Each step up is about thirty times more energy released. Unit 3
Mandate of Heaven	The Chinese idea that Heaven gives a just ruler the right to govern, and withdraws it from a bad one. Unit 8
terrace	A step cut into a hillside and held by a stone wall, so that steep ground can be farmed. Unit 4
tide	The regular rise and fall of the sea, caused mainly by the Moon's gravity. Unit 2

mantle	The thick layer beneath the crust, nearly 2 900 km deep, made of rock that flows very slowly. Unit 3
megalith	A very large stone used in a monument or tomb, as in the Almendres Cromlech near Evora. Unit 6
metamorphic	Rock that has been changed by heat and pressure without melting. Unit 3
mita	The Inca system of paying tax in days of work rather than in money or goods. Unit 10
monsoon	A seasonal reversal of the wind that brings months of heavy rain, then months of dry weather. Unit 4
movable type	Printing with separate reusable characters rather than a whole carved page. Unit 8
mummification	Preserving a body by removing the organs and drying it with natron salt before wrapping it. Unit 7
overfishing	Taking fish faster than the population can replace itself. Unit 4
pharaoh	The ruler of ancient Egypt, believed to be a god on Earth. Unit 7
phase	How much of the Moon's sunlit half we can see from Earth at a given time. Unit 2
plateau	A large area of high, fairly flat land. Unit 5
prehistory	The time before written records. Everything after them is history. Unit 6
quipu	A set of knotted cords used by the Inca to record numbers and information instead of writing. Unit 10
reciprocity	The Andean habit of giving help in the expectation that it will be returned, on which the empire was built. Unit 10
salinity	How much salt is dissolved in water. Sea water is about 3.5 per cent salt. Unit 4
savannah	Tropical grassland with scattered trees, between rainforest and desert. Unit 5
sedimentary	Rock formed from fragments that settled, were buried and were squeezed together. Unit 3
silt	Fine, fertile material carried by a river and left behind when it floods. Unit 7
supernova	The enormous explosion of a very large star at the end of its life. Most black holes are what is left behind. Unit 1
surplus	Food grown beyond what the growers eat. It can be stored, taken and taxed, which is how governments begin. Unit 6
tambo	A waystation on an Inca road, holding food and supplies for travellers and officials. Unit 10
tectonic plate	One of about fifteen large pieces the Earth's outer shell is broken into, which move on the mantle below. Unit 3
treaty	A written agreement between countries. The Antarctic Treaty sets a whole continent

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4. Khan Academy, Art of the Americas. www.khanacademy.org/humanities/art-americas
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7. United States Geological Survey, Earthquake Hazards. www.usgs.gov/programs/earthquake-hazards
8. United States Geological Survey, Volcano Hazards Programme. volcanoes.usgs.gov/
9. NOAA Ocean Exploration. oceanexplorer.noaa.gov/
10. NOAA, Ocean facts. oceanservice.noaa.gov/facts/
11. Oceanario de Lisboa. www.oceanario.pt/en/
12. British Antarctic Survey. www.bas.ac.uk/about/antarctica/
13. NASA Earth Observatory. earthobservatory.nasa.gov/
14. Coa Valley Archaeological Park, Portugal. arte-coa.pt/en/
15. Smithsonian Institution, Human Origins. humanorigins.si.edu/
16. Institute for the Study of Ancient Cultures, University of Chicago. isac.uchicago.edu/
17. Columbia University, Asia for Educators. afe.easia.columbia.edu/
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19. Perseus Digital Library, Tufts University. www.perseus.tufts.edu/hopper/

For teachers

What this book assumes, how the year is built, and what it deliberately does not do.

What this book assumes

It assumes no prior humanities knowledge beyond a primary programme, and no mathematics beyond division and percentages. It assumes English as the language of instruction, and it assumes that a significant number of pupils are reading the subject in an additional language, which is why the body type is set a full point above the Year 7 floor, the measure is short, and panel names are one or two plain words.

How the year is structured

Nine units in two parts, running strictly forward in time.

Part one is physical geography and it is deliberately front-loaded: the Universe, the Solar System, the Earth's structure and movement, the oceans, and the continents. Part two is world history in chronological order: the first people, Egypt, China and the Inca.

The two parts are not independent. Part two repeatedly depends on part one: Egypt is a river topic, China is a river and loess topic, the Inca are an altitude topic. Teaching part two first breaks those links.

Where Portugal fits

Portugal has no chapter in this book, and that is a decision rather than an omission. The pupils are international, and Portuguese history taught as a separate block invites them to file it as somebody else's subject.

Instead there are ten short **Portugal at this stage** panels, one in each unit, each placed inside the world period it belongs to and each between 80 and 150 words. A pupil reading about the Great Pyramid meets, on the same spread, the walls of Zambujal, begun in the same century. The intended effect is that Portugal is normal rather than special: one place among many, on one timeline.

What changed from the first edition

The first edition was an eighteen page, two unit sample built on a different plan, with Unit 1 as "Geography and you". It is kept in _superseded_2026-08-07/ and nothing has been overwritten.

The scope is wider in both strands, on the argument set out in the curriculum recommendation filed in the project hub: geography now begins with the Universe rather than with map skills, and history now covers four continents rather than the usual one or two. The Input document's chapters on the Inca, China and Egypt are all carried forward, reordered chronologically and rewritten from scratch. Its chapter on the European Union and the United Kingdom is not carried forward as a chapter; its map skills survive as topic 5.1 and its European geography as topic 5.4.

Assessment

Every topic ends with a numbered **Your turn** set, easiest question first. Closed questions have answers at the back; open questions have acceptance criteria, which is what a good answer must contain rather than the words it must use. Marking to acceptance criteria is slower than marking to a key and it is the only honest way to mark an argument.

Each unit ends with **What you can now do**, which doubles as a self-check and as the source for the knowledge organiser pages that fill what would otherwise be blank versos.

What this book does not do

It does not teach the science of climate change, ecosystems or natural hazards in depth. Those are taught in far greater scientific detail in the Science programme, and duplicating them here would waste pages this subject needs.

It does not pretend that every question has a settled answer. Where scholars disagree, for instance about whether quipus recorded language, the book says so.

It does not treat any civilisation as the normal one against which others are measured. That is the single reason the geography strand comes first: a pupil who knows where the Andes, the Nile and the Huang He are does not need to be told that civilisation happened in more than one place.

A note on the sources

Every date, figure and statistic in this book was checked against a source named in the Sources section before it was printed. Where a figure is approximate or disputed, the text says so in words a Year 7 pupil can read. Pupils are told, in the front matter, that they may check the book and should report what they find.

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Learn to read the world.

This Year 7 Student Book teaches geography and history the way real geographers and historians work: by looking closely at evidence. You will meet great explorers, learn the six questions that unlock any place, and study how a real Portuguese town is changing.

Written for Prime School pupils, in clear British English, with real places and real sources you can check.

GEOGRAPHY

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